

BFS Traversal

Step 1: Start at A

Legend

Complete

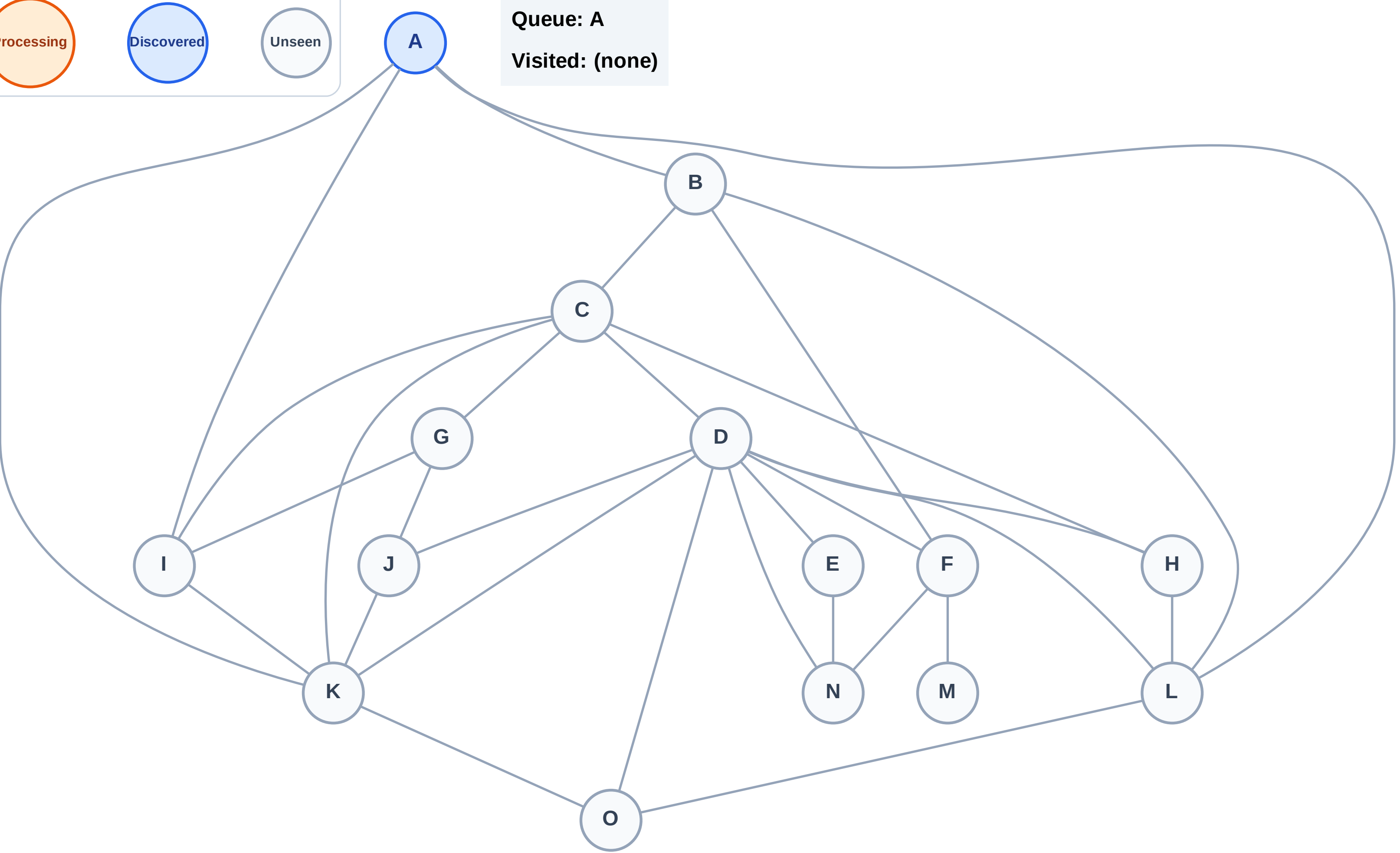
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

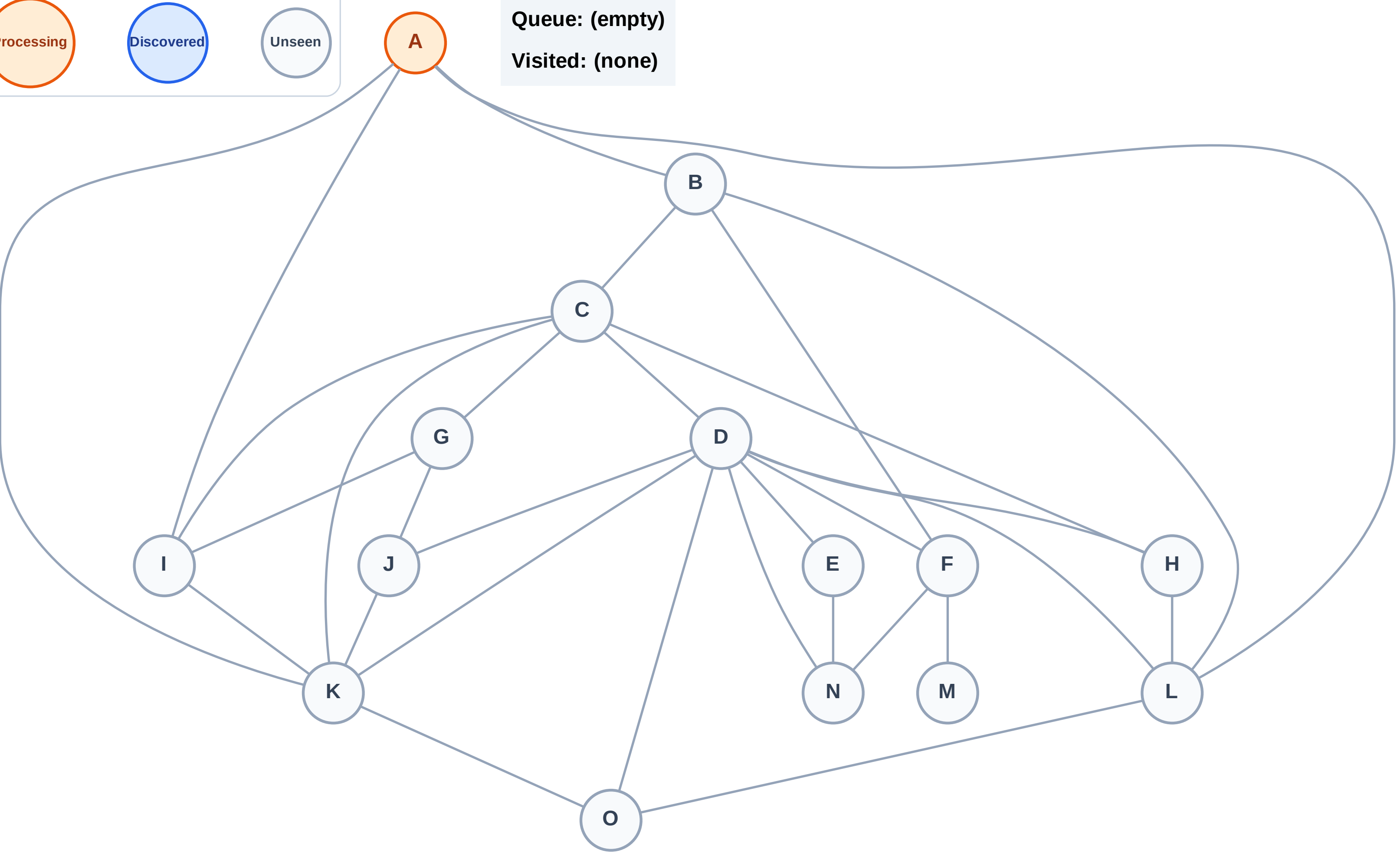
Complete

Processing

Discovered

Unseen

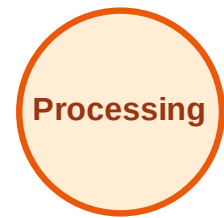
Queue: (empty)
Visited: (none)



BFS Traversal

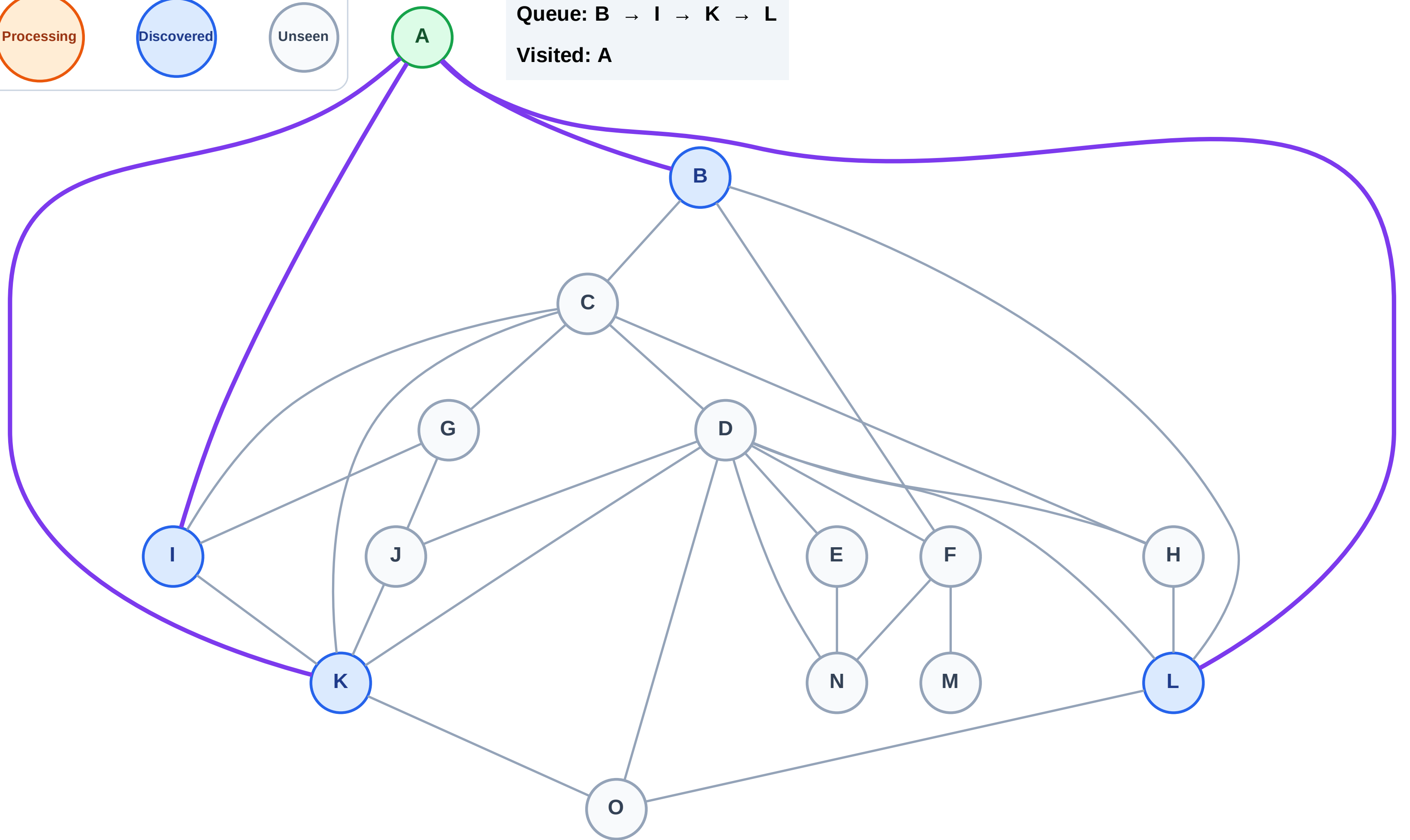
Step 3: Discover B, I, K, L from A

Legend



Queue: B → I → K → L

Visited: A



BFS Traversal

Step 4: Process node B

Legend

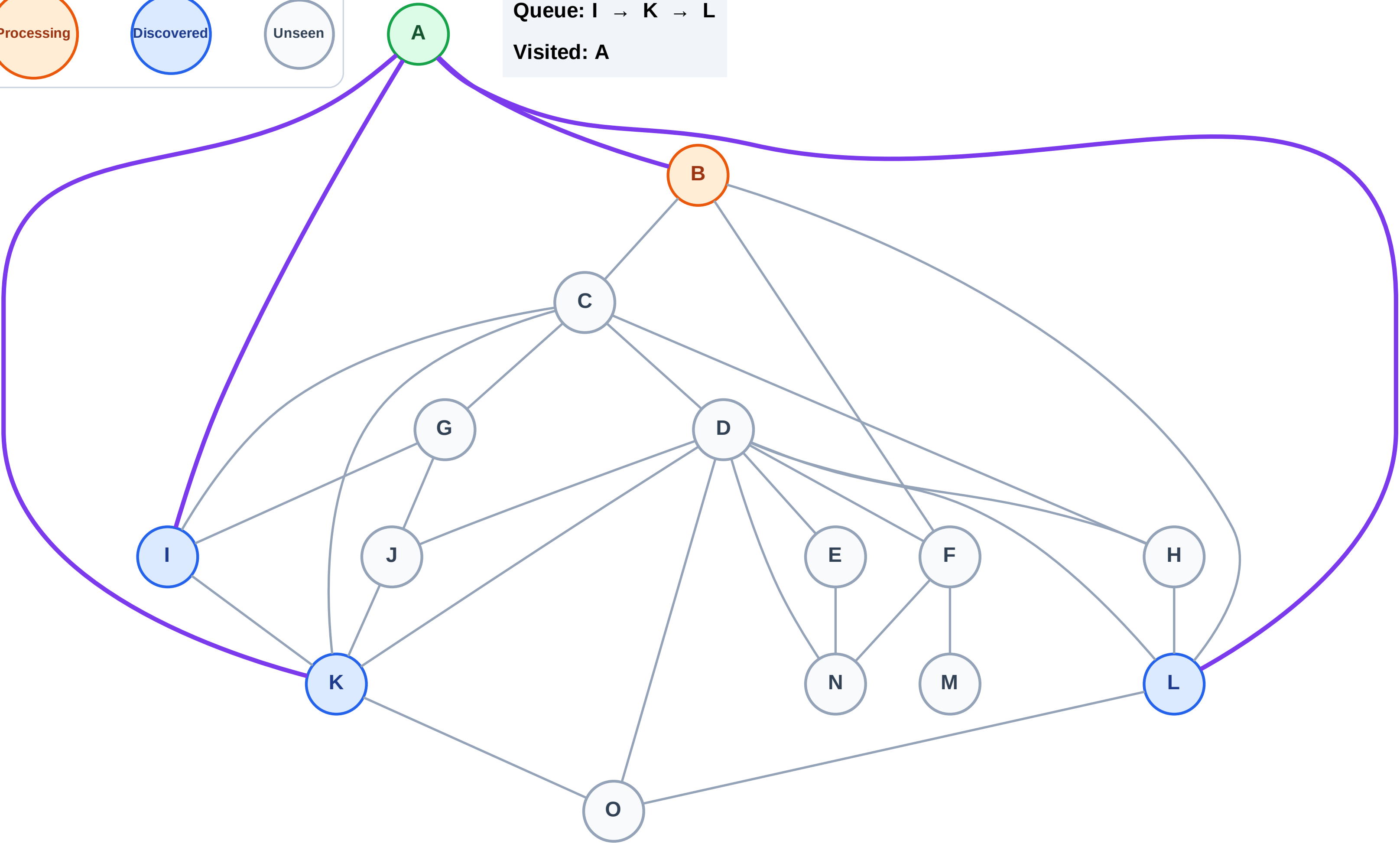
Complete

Processing

Discovered

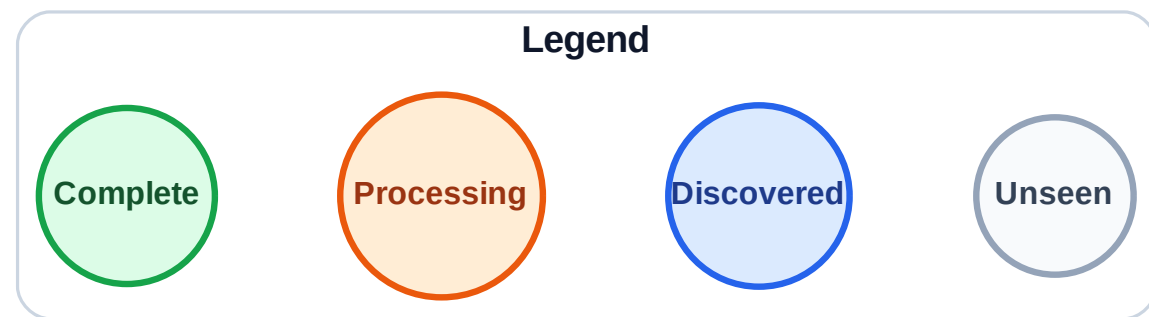
Unseen

Queue: I → K → L
Visited: A

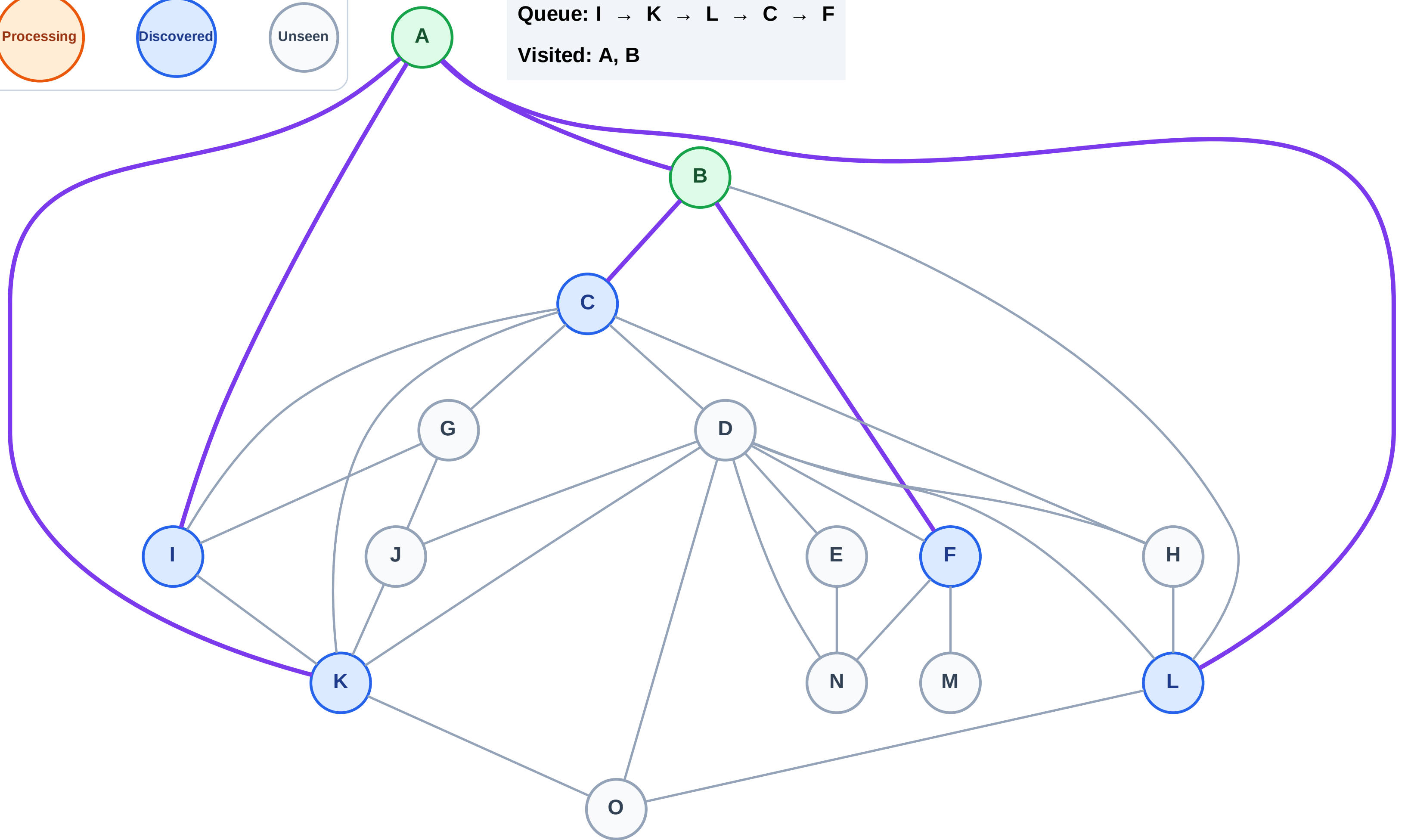


BFS Traversal

Step 5: Discover C, F from B



Queue: I → K → L → C → F
Visited: A, B



BFS Traversal

Step 6: Process node I

Legend

Complete

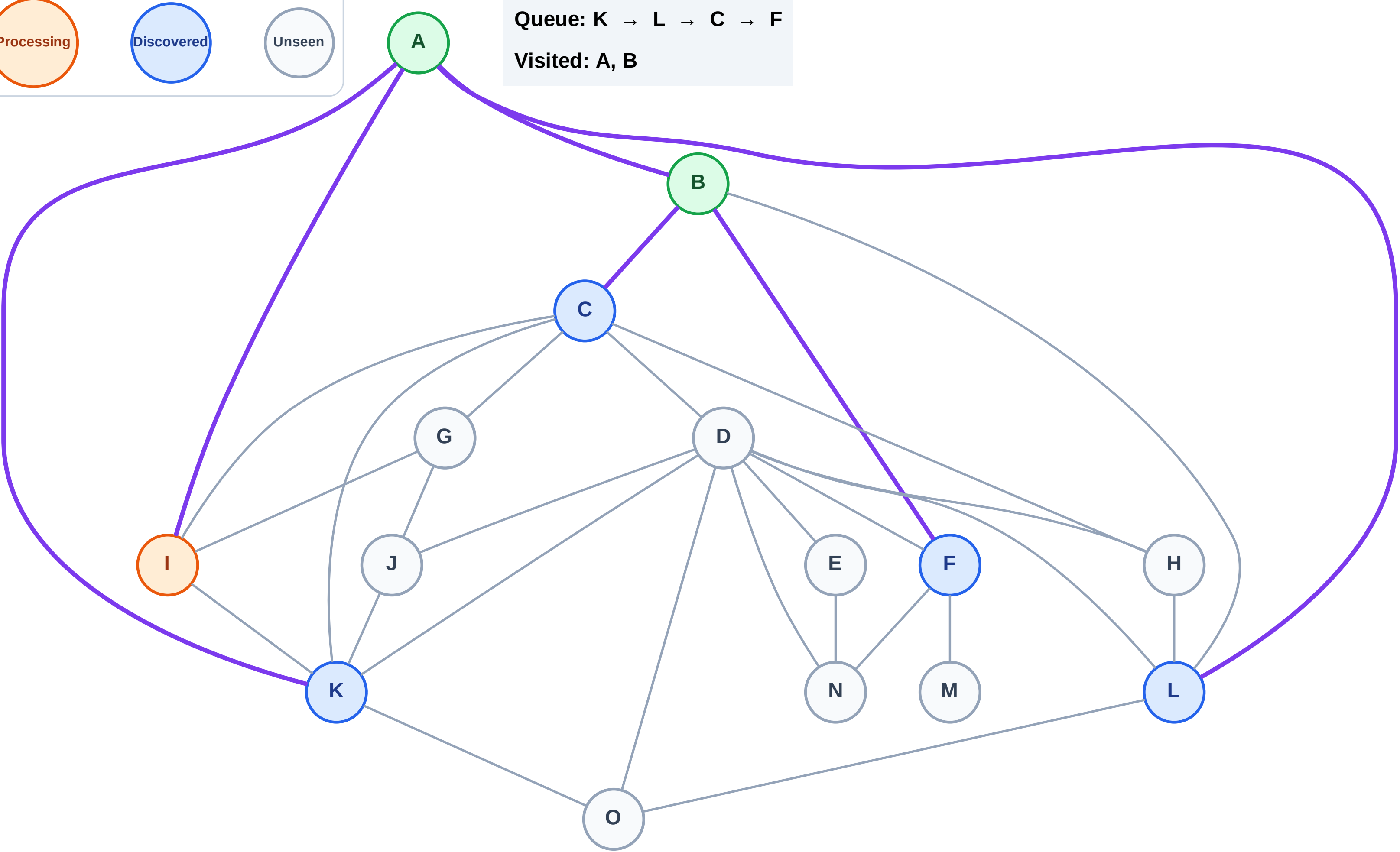
Processing

Discovered

Unseen

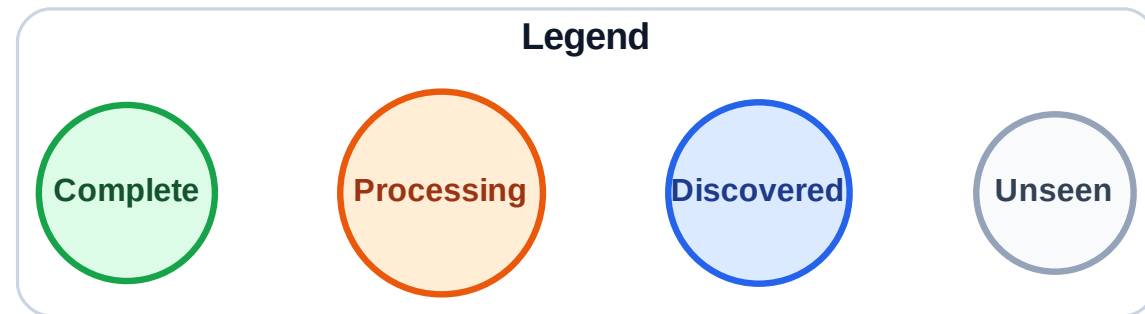
Queue: K → L → C → F

Visited: A, B



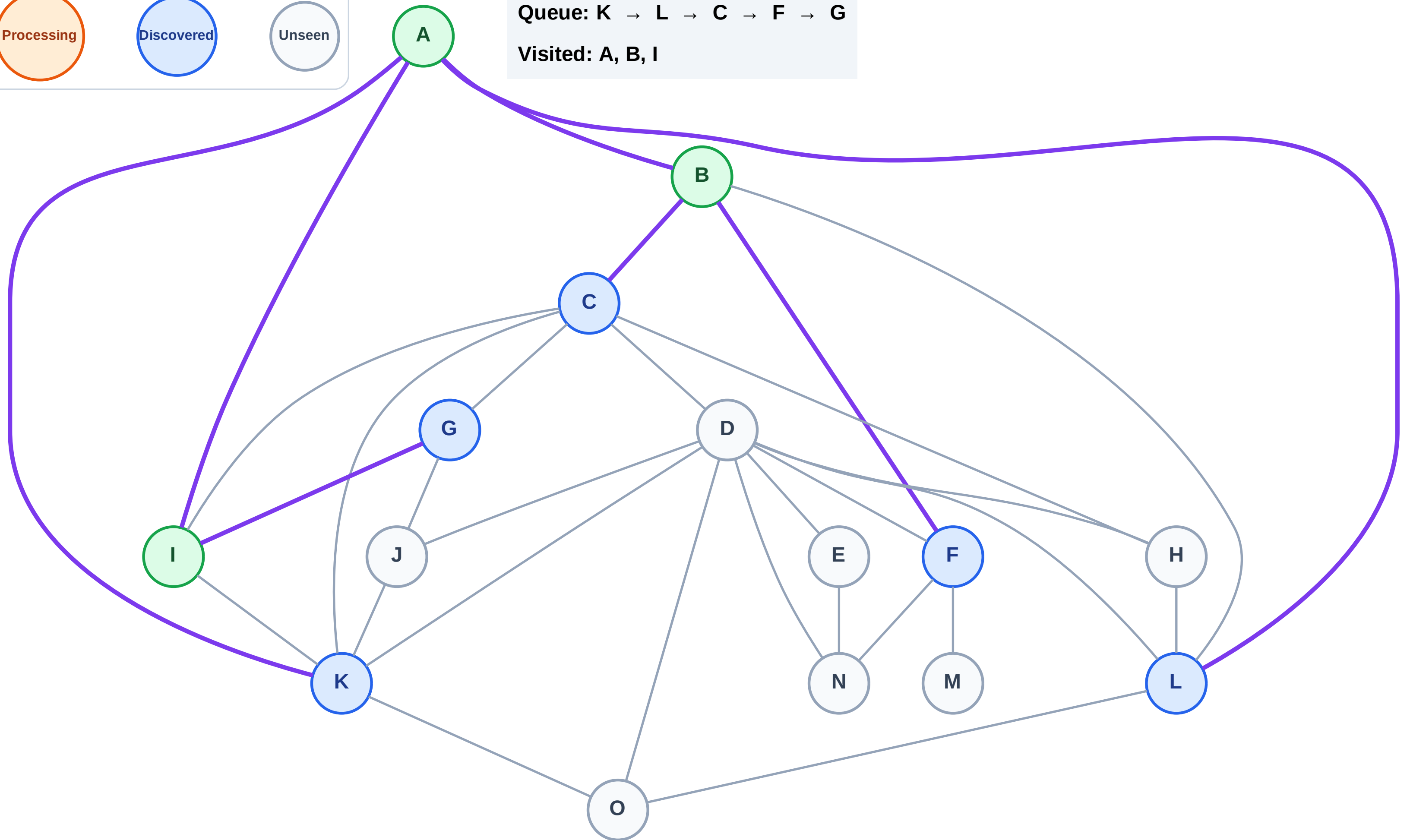
BFS Traversal

Step 7: Discover G from I



Queue: K → L → C → F → G

Visited: A, B, I



BFS Traversal

Step 8: Process node K

Legend

Complete

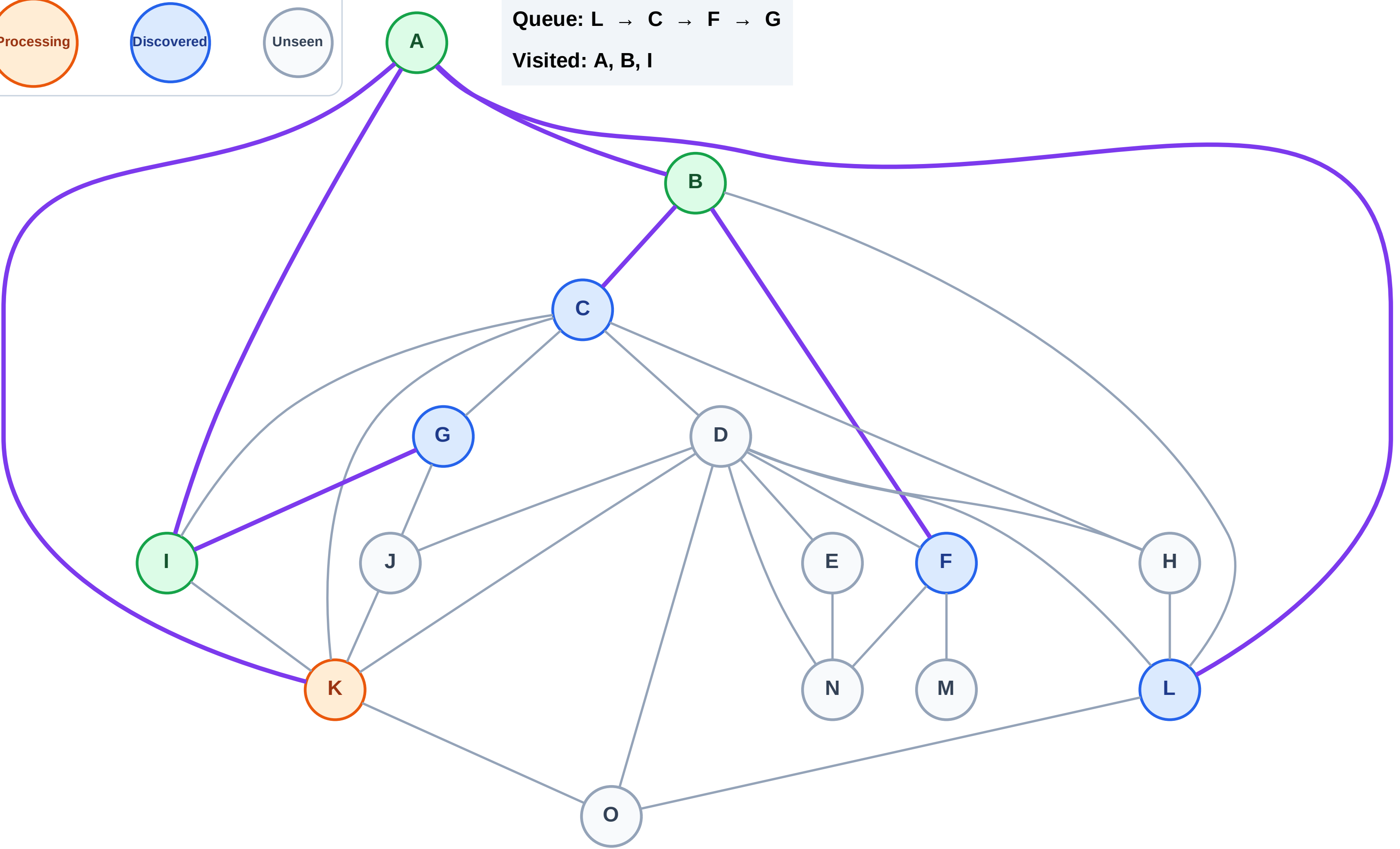
Processing

Discovered

Unseen

Queue: L → C → F → G

Visited: A, B, I



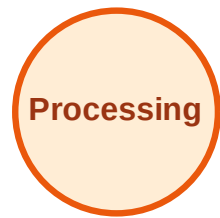
BFS Traversal

Step 9: Discover D, J, O from K

Legend



Complete



Processing



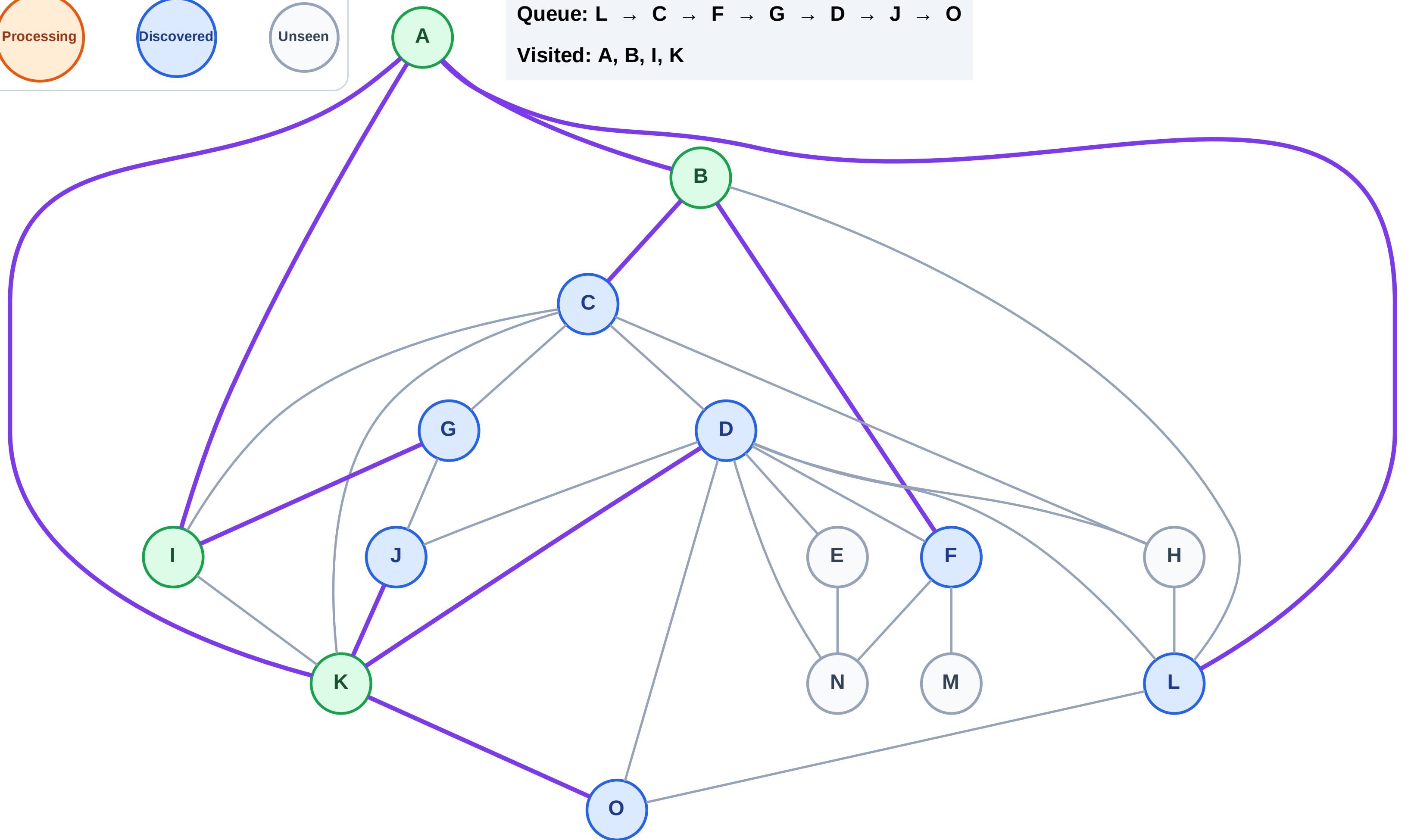
Discovered



Unseen

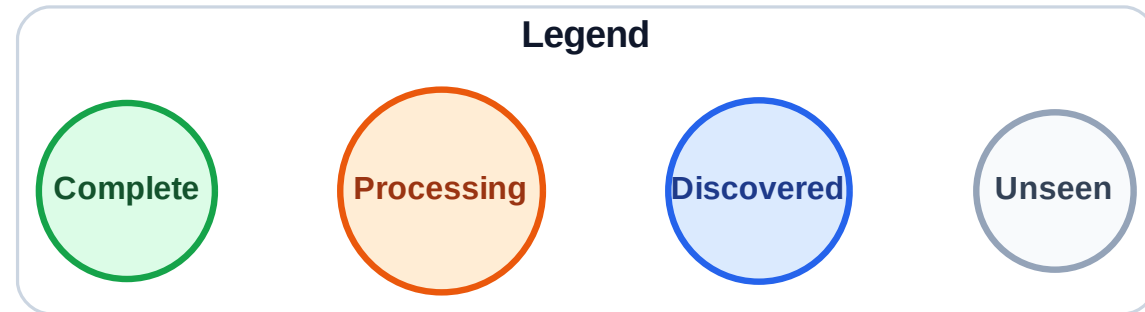
Queue: L → C → F → G → D → J → O

Visited: A, B, I, K



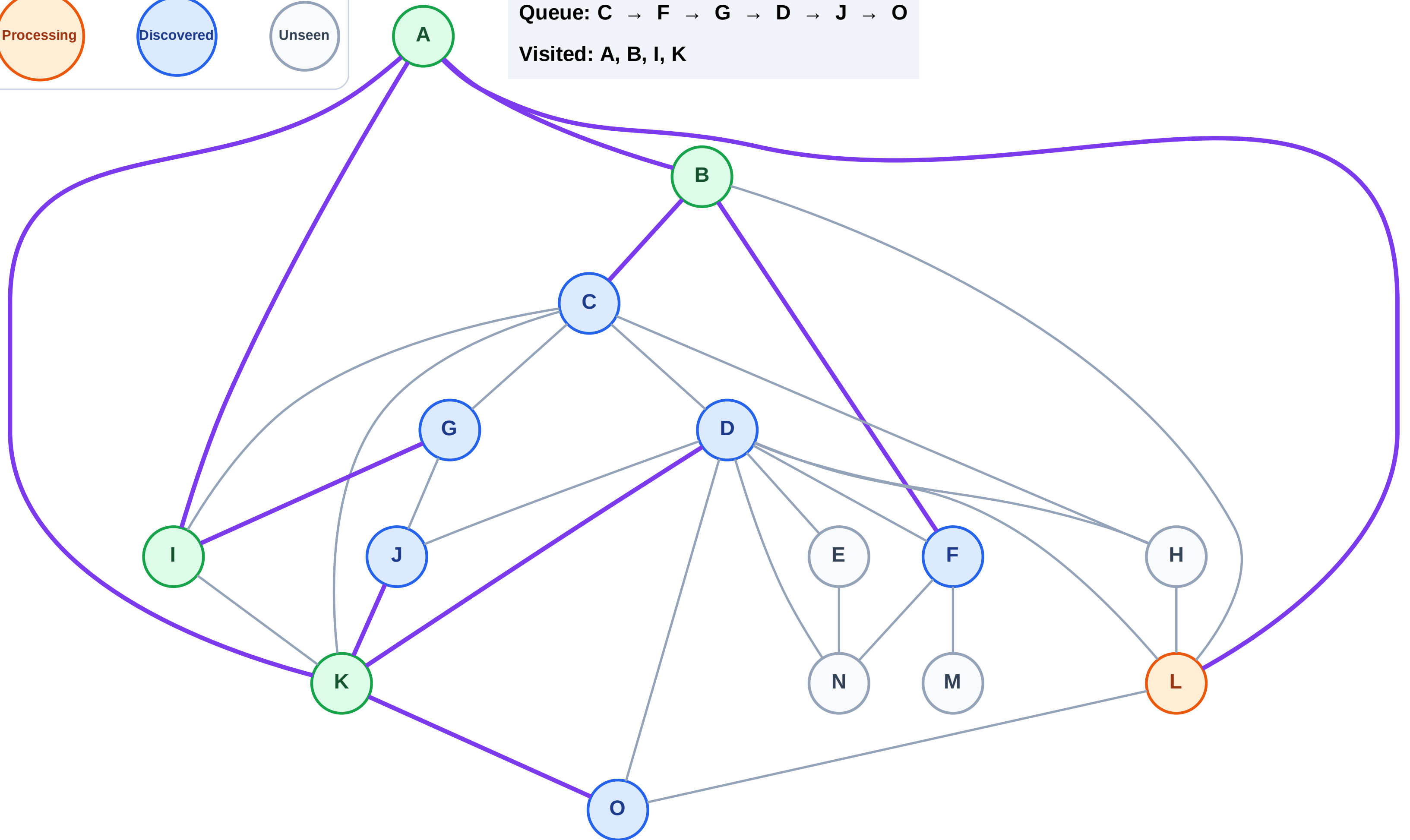
BFS Traversal

Step 10: Process node L



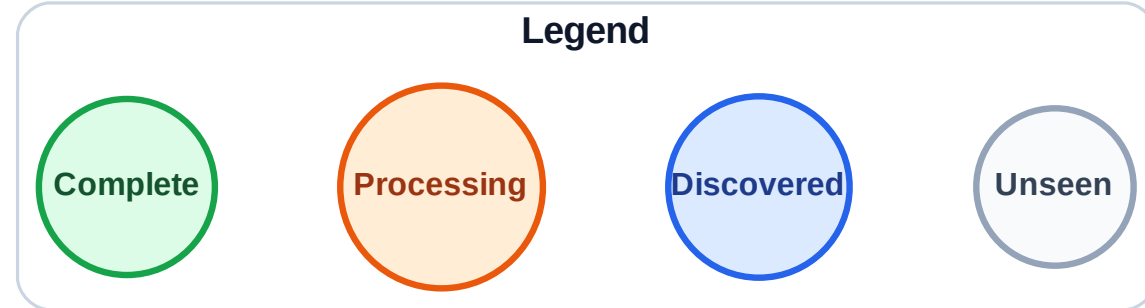
Queue: C → F → G → D → J → O

Visited: A, B, I, K



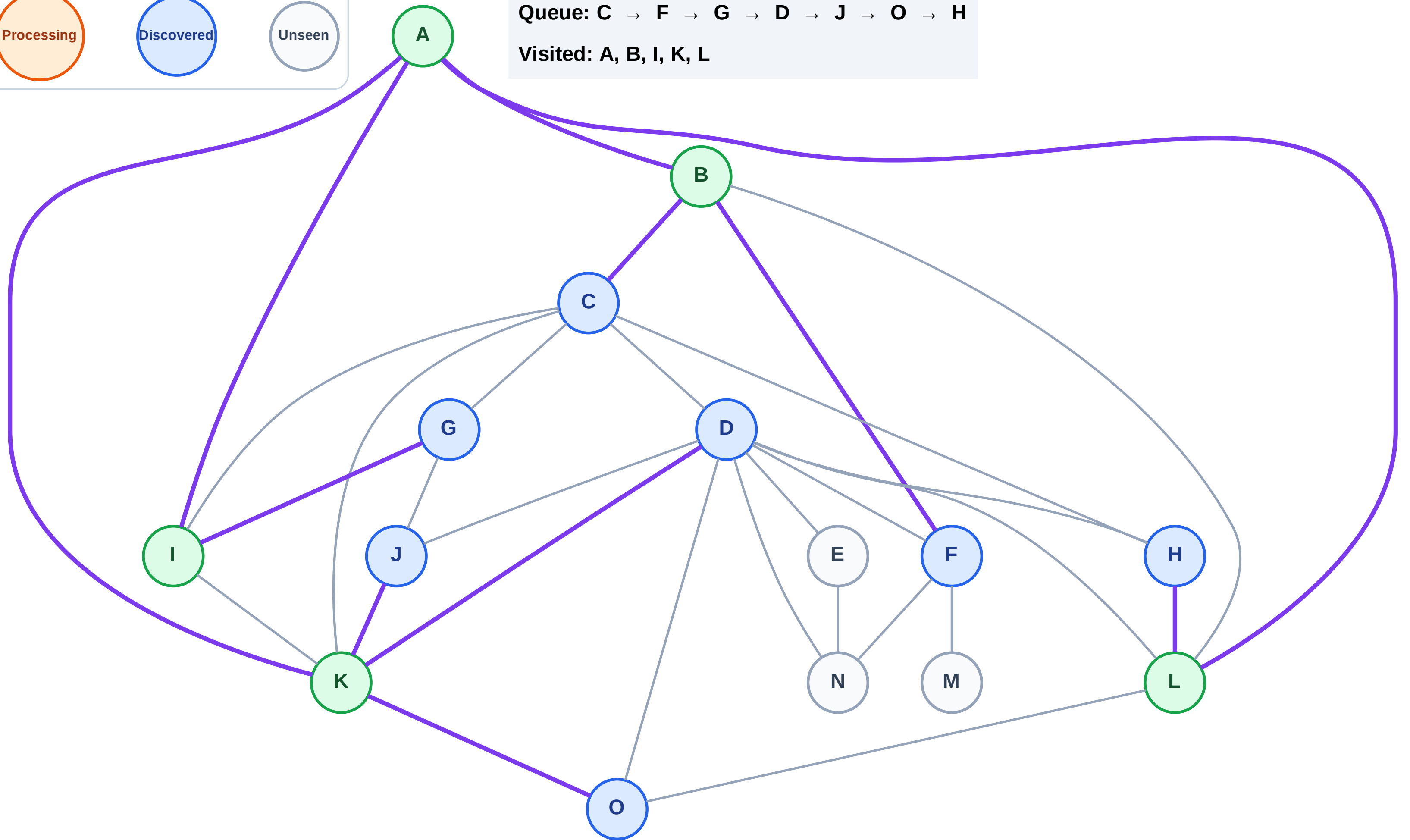
BFS Traversal

Step 11: Discover H from L



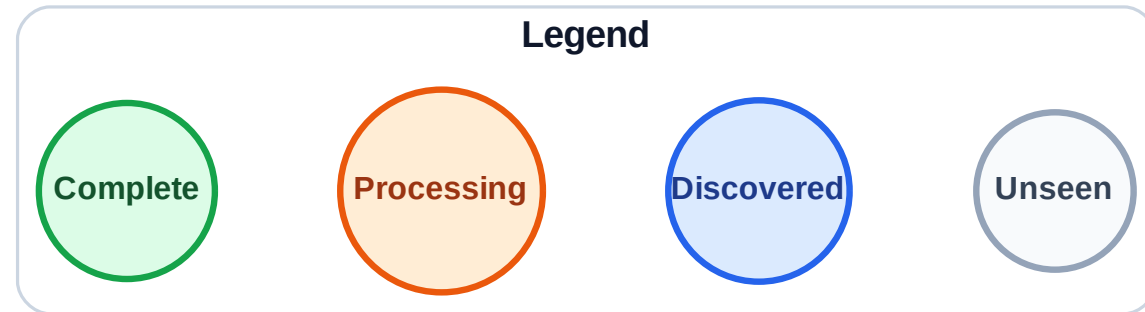
Queue: C → F → G → D → J → O → H

Visited: A, B, I, K, L



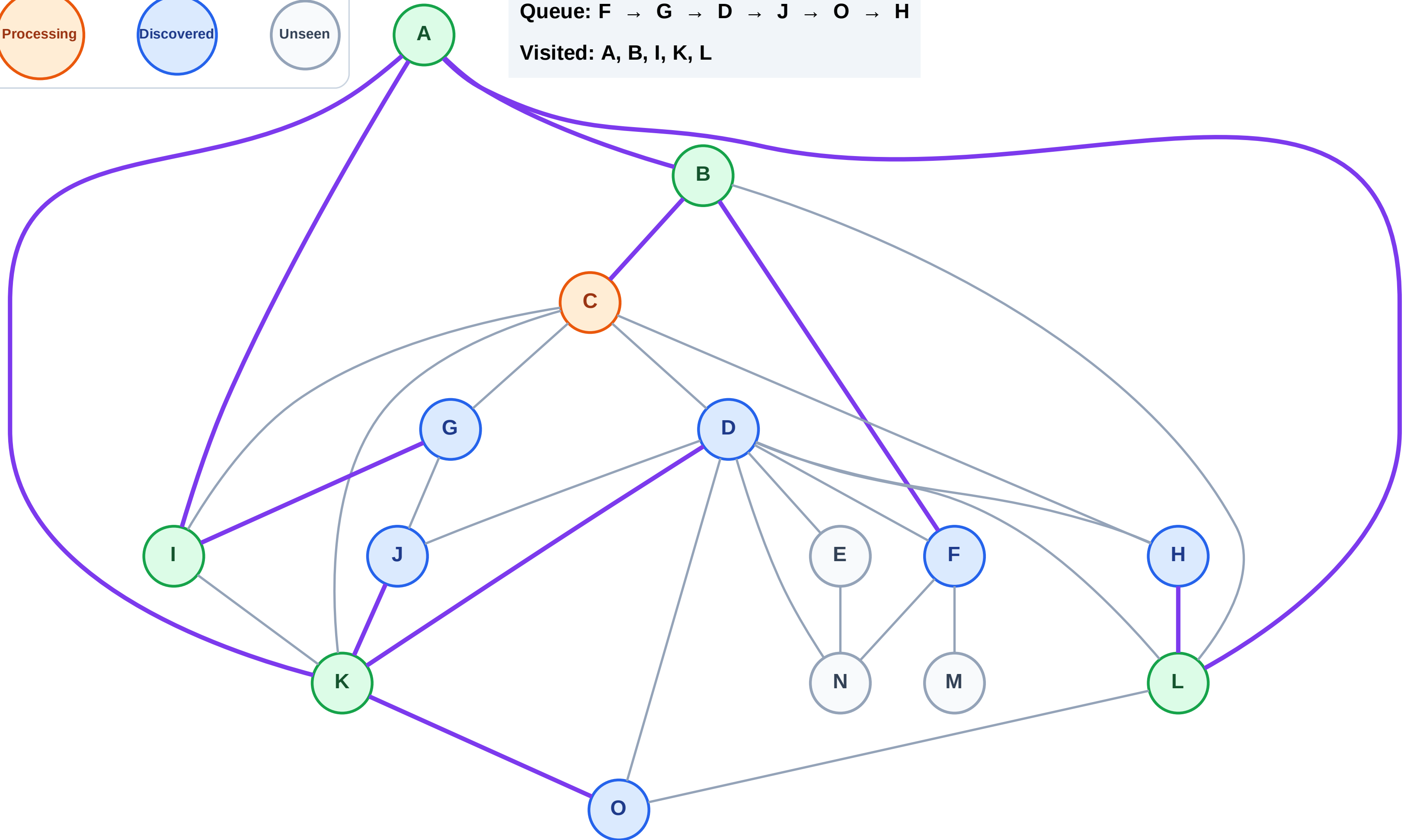
BFS Traversal

Step 12: Process node C



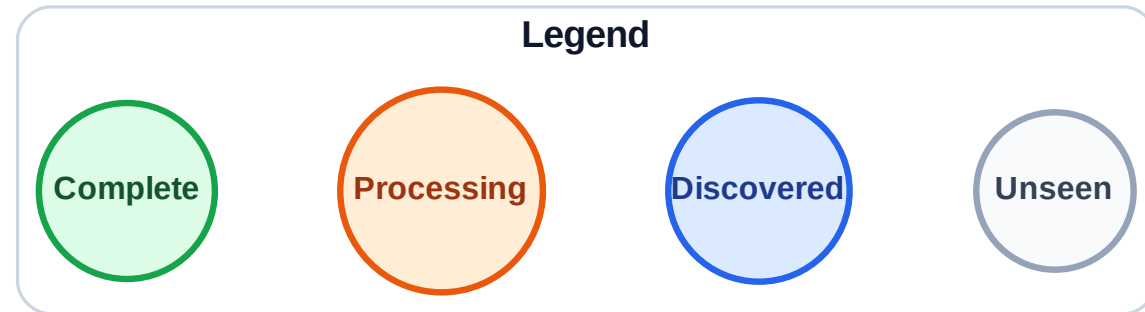
Queue: F → G → D → J → O → H

Visited: A, B, I, K, L



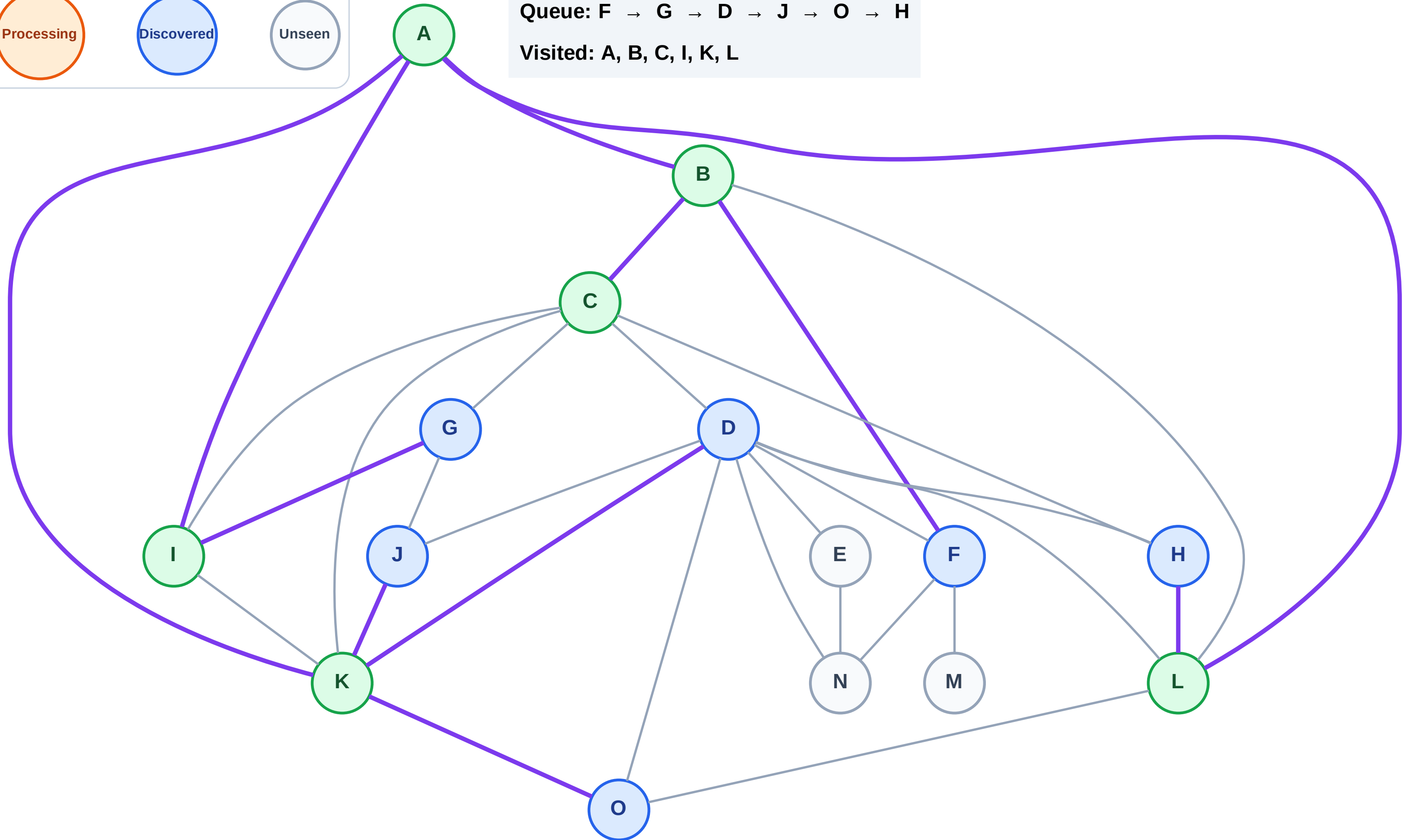
BFS Traversal

Step 13: No new neighbors from C



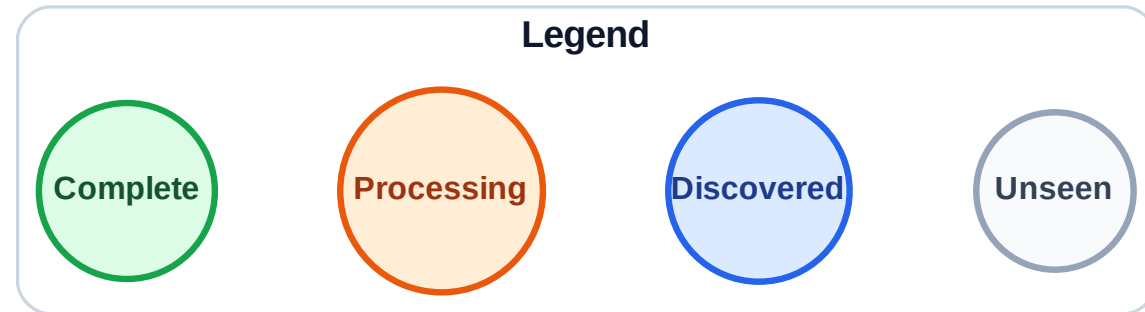
Queue: F → G → D → J → O → H

Visited: A, B, C, I, K, L



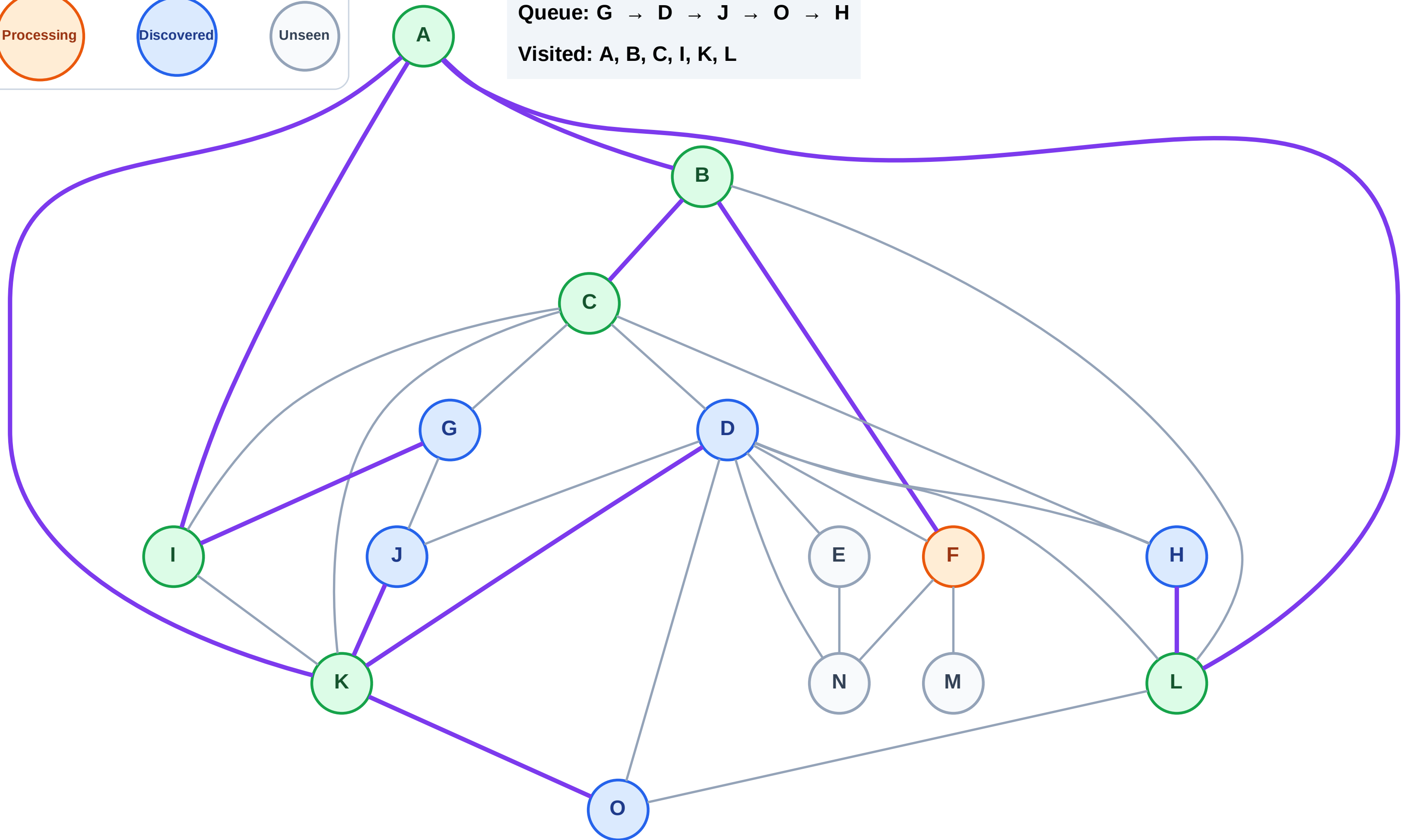
BFS Traversal

Step 14: Process node F



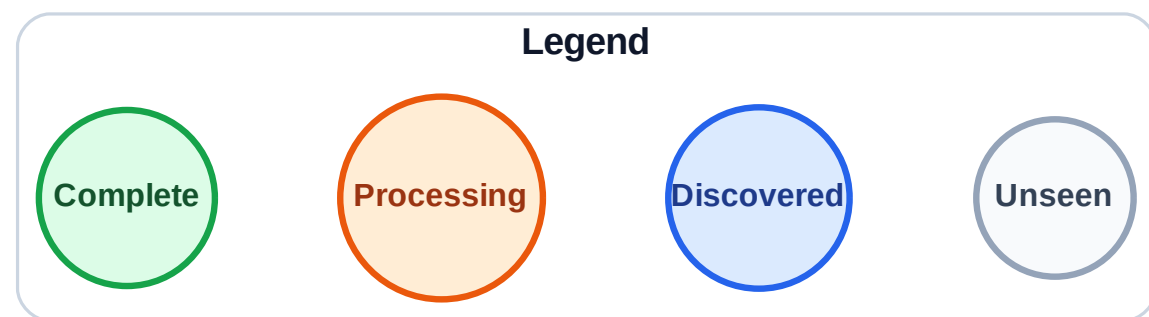
Queue: G → D → J → O → H

Visited: A, B, C, I, K, L



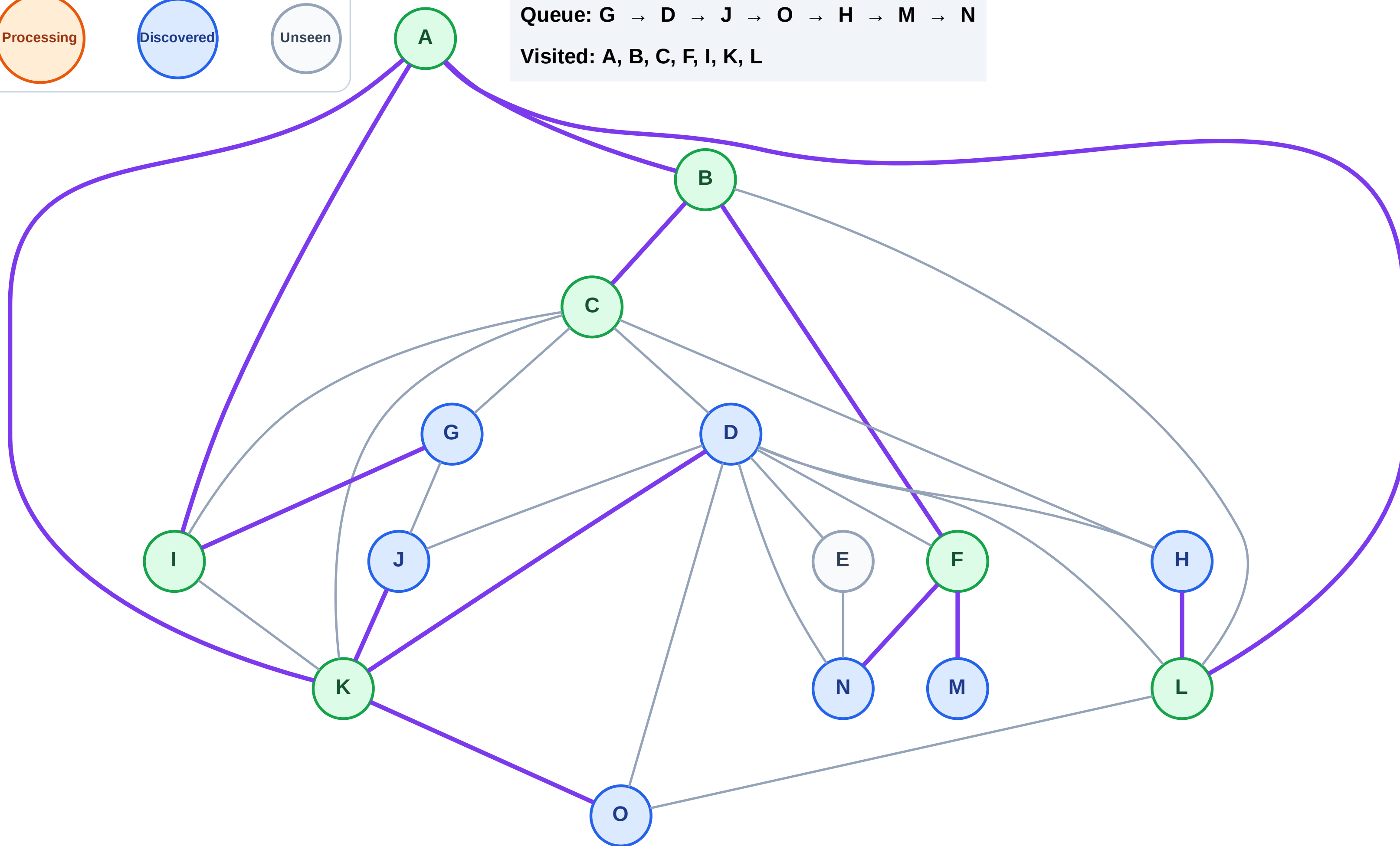
BFS Traversal

Step 15: Discover M, N from F



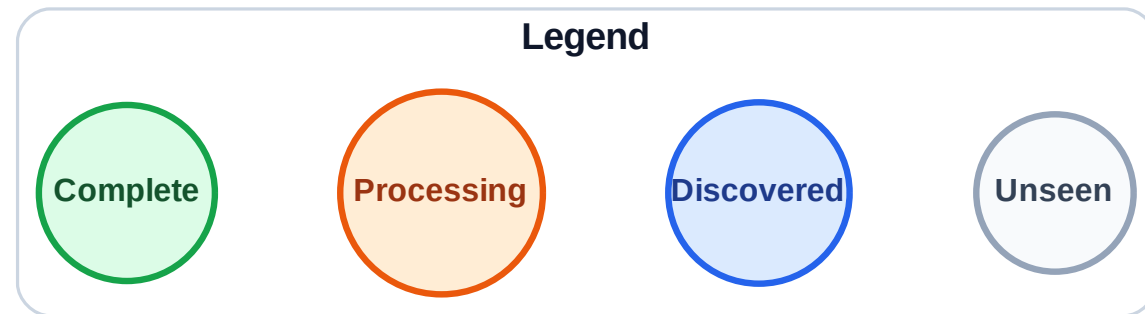
Queue: G → D → J → O → H → M → N

Visited: A, B, C, F, I, K, L



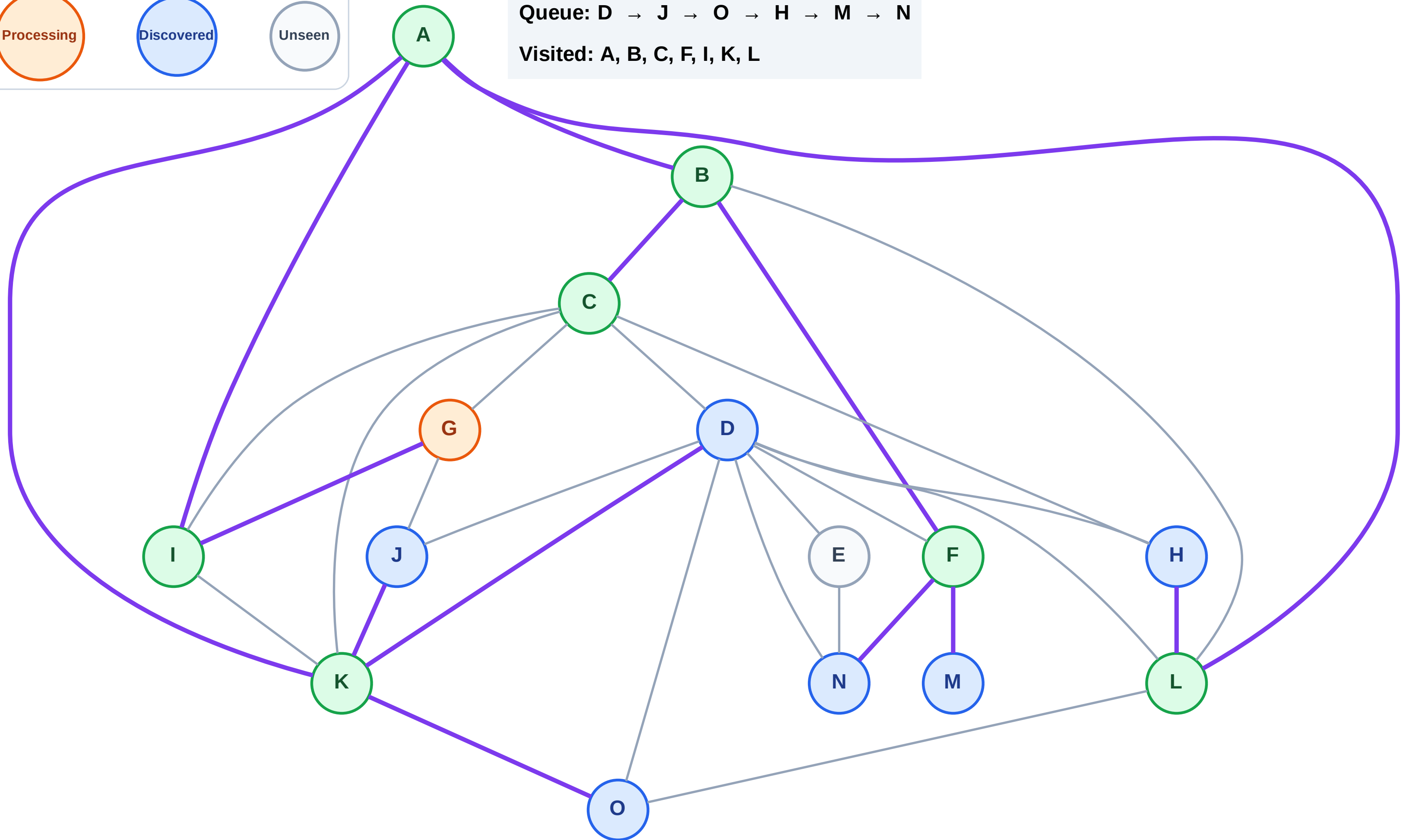
BFS Traversal

Step 16: Process node G



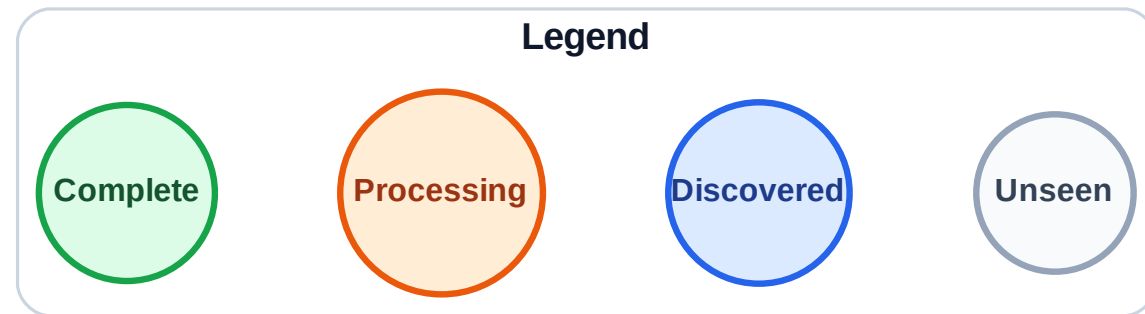
Queue: D → J → O → H → M → N

Visited: A, B, C, F, I, K, L



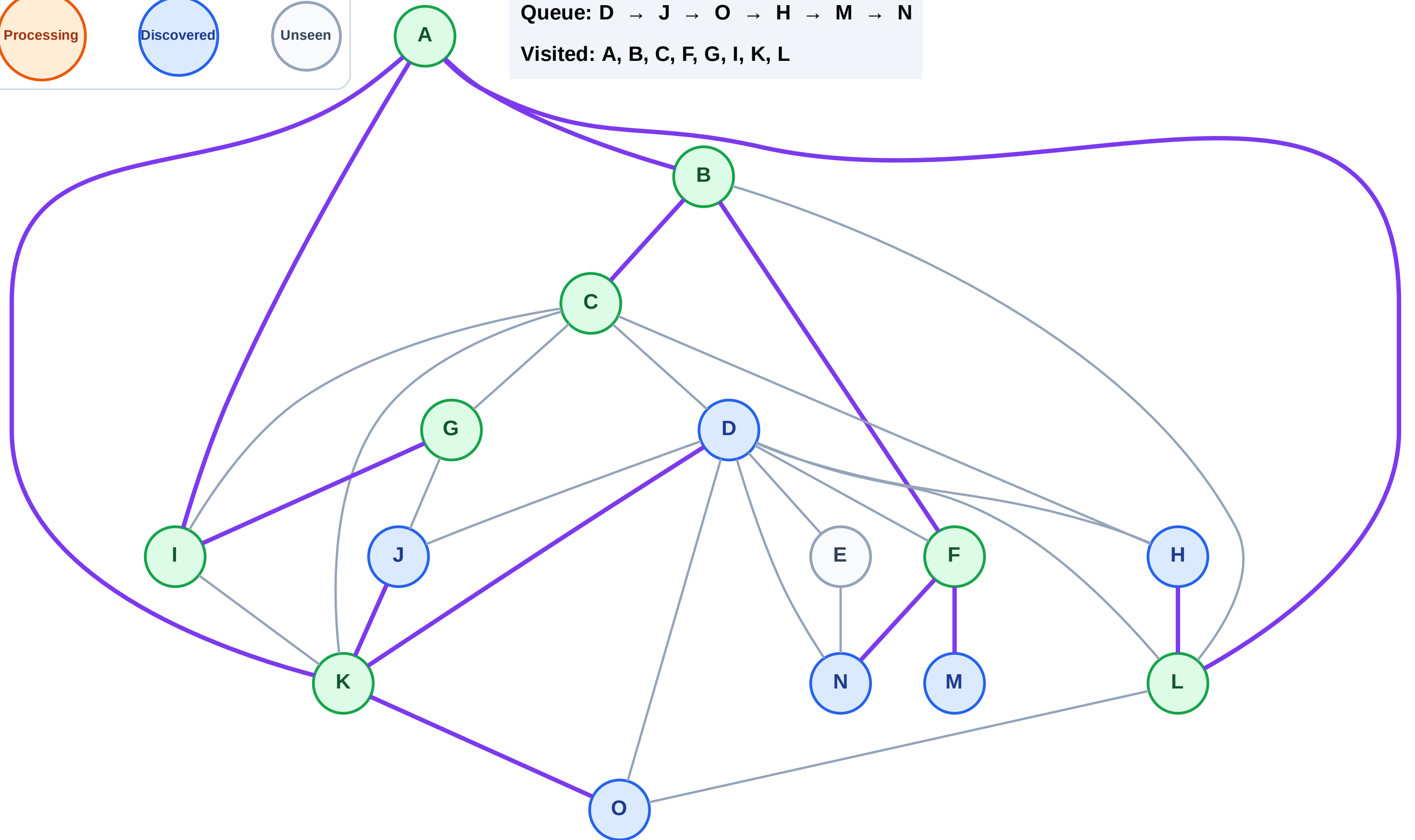
BFS Traversal

Step 17: No new neighbors from G



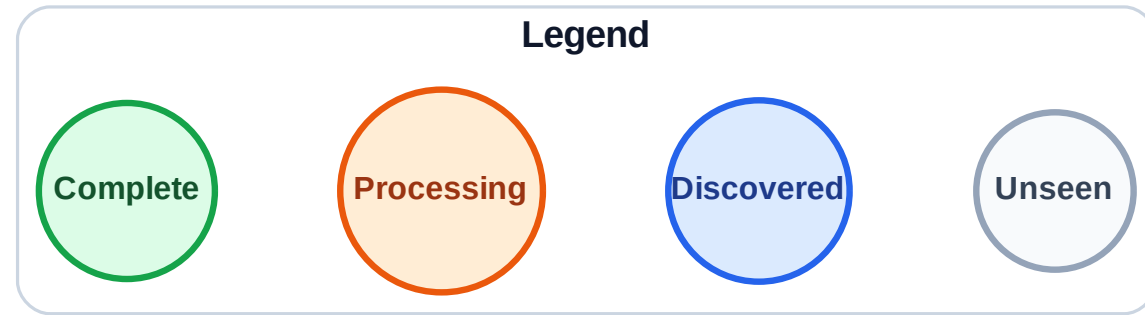
Queue: D → J → O → H → M → N

Visited: A, B, C, F, G, I, K, L



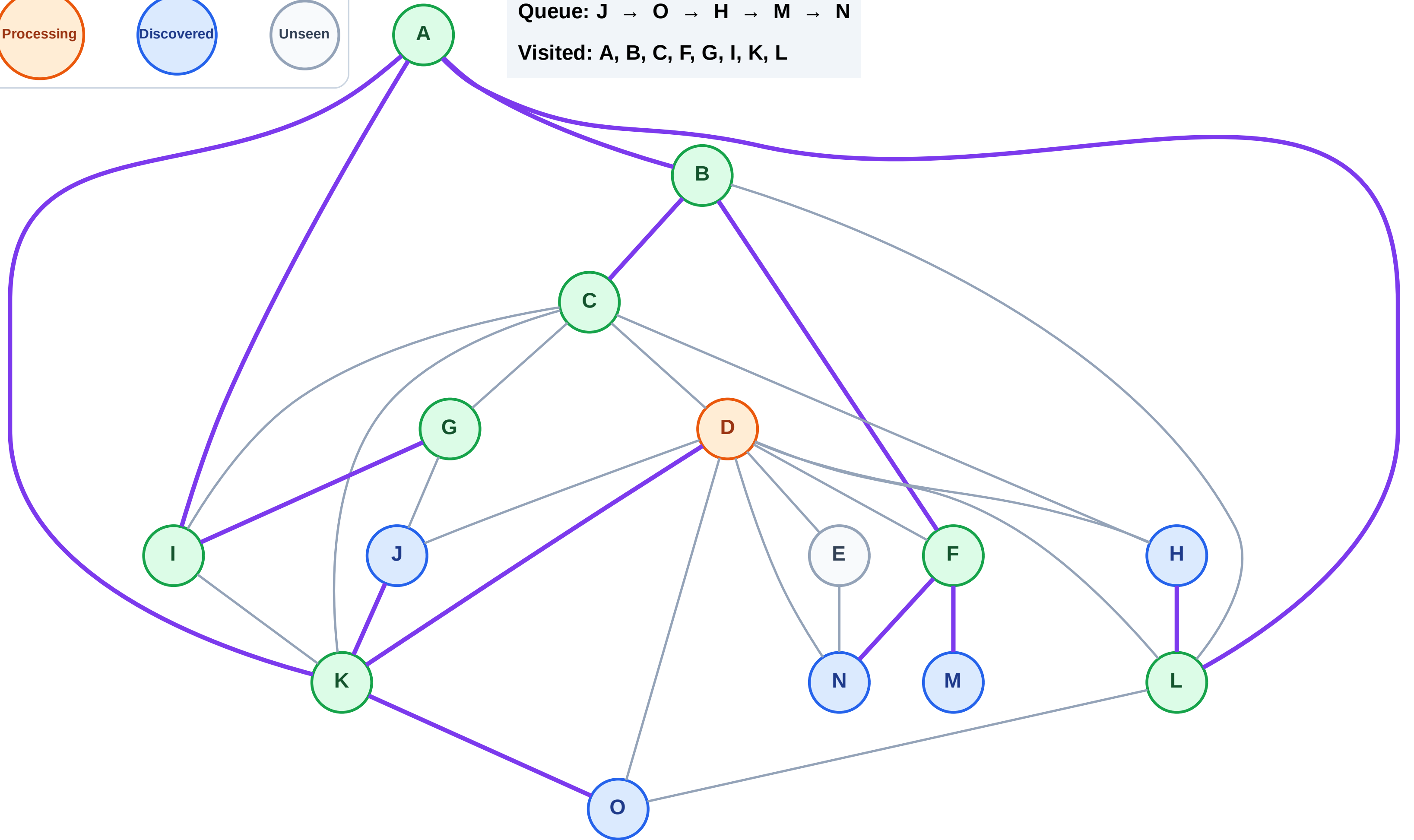
BFS Traversal

Step 18: Process node D



Queue: J → O → H → M → N

Visited: A, B, C, F, G, I, K, L



BFS Traversal

Step 19: Discover E from D

Queue: J → O → H → M → N → E
Visited: A, B, C, D, F, G, I, K, L

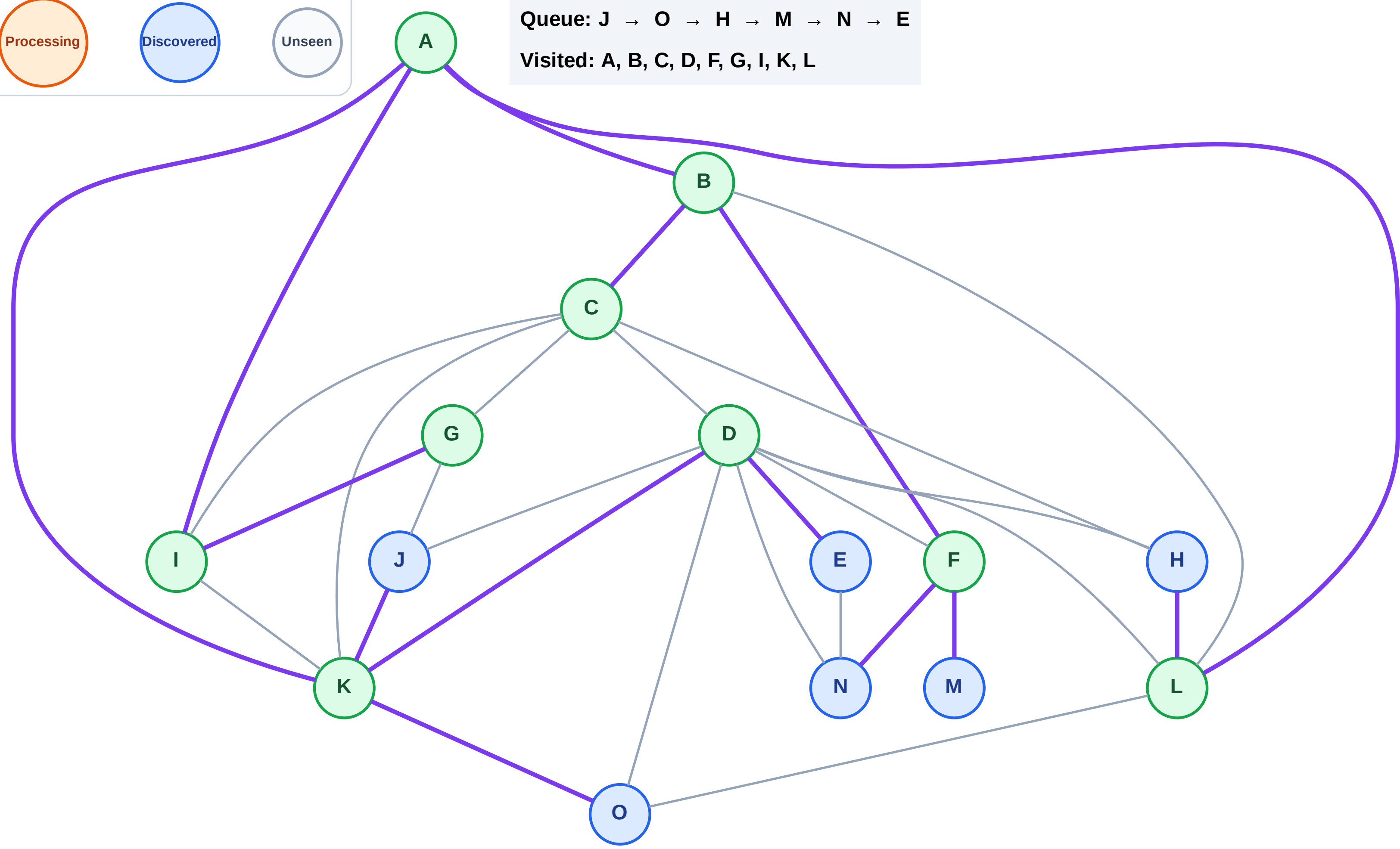
Legend

Complete

Processing

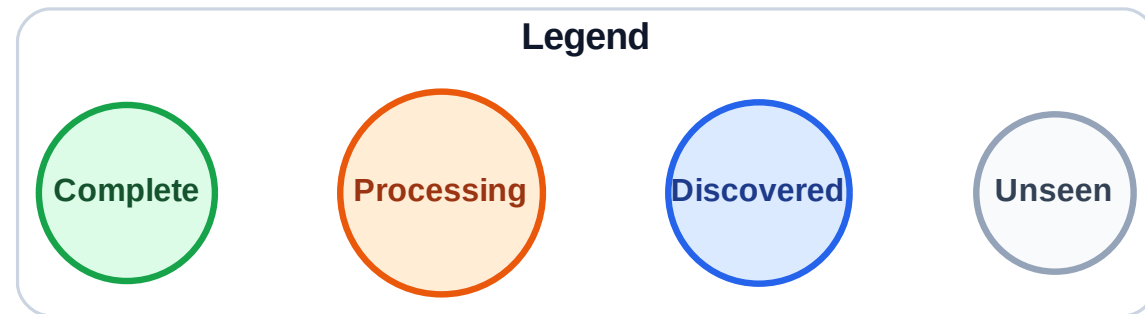
Discovered

Unseen



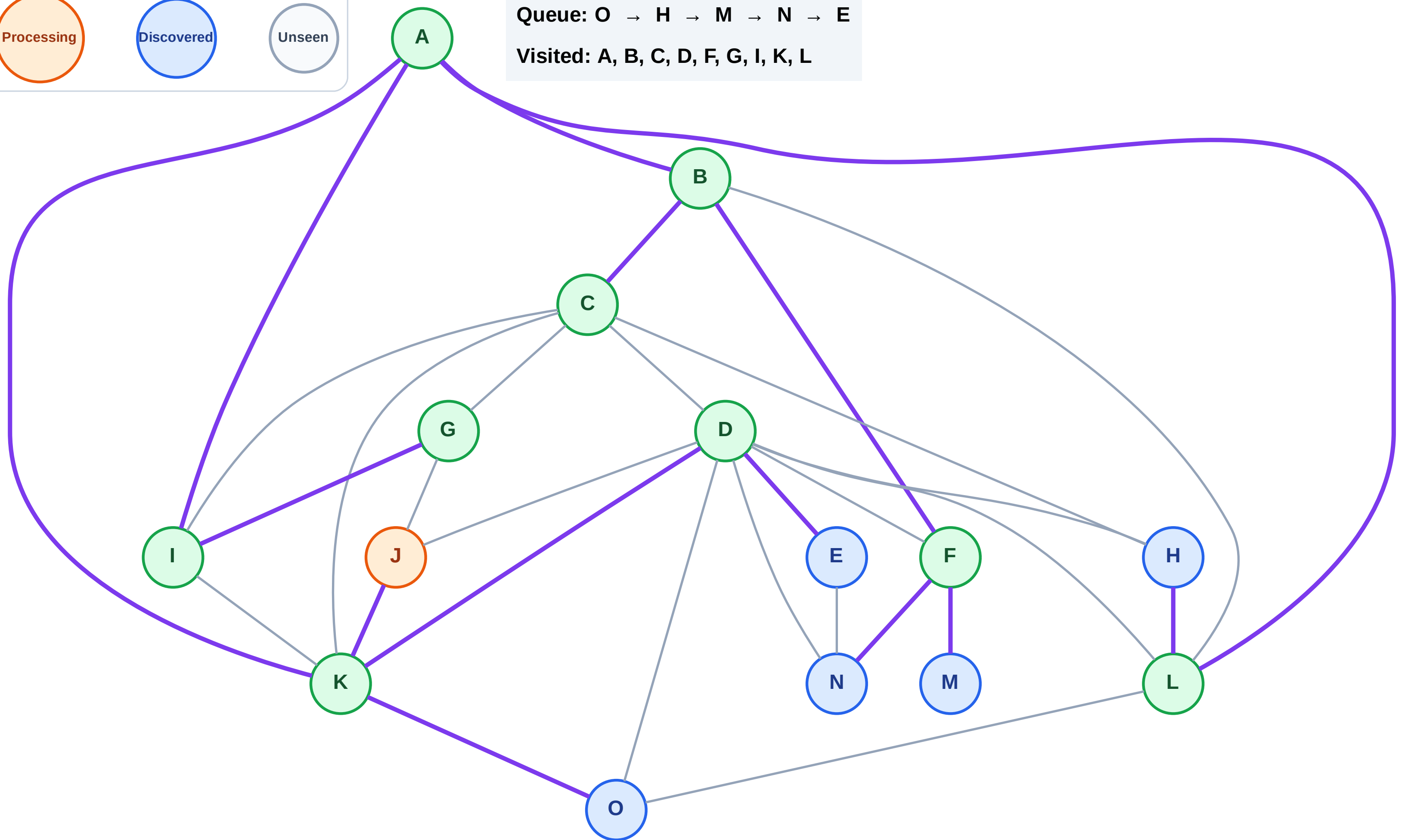
BFS Traversal

Step 20: Process node J



Queue: O → H → M → N → E

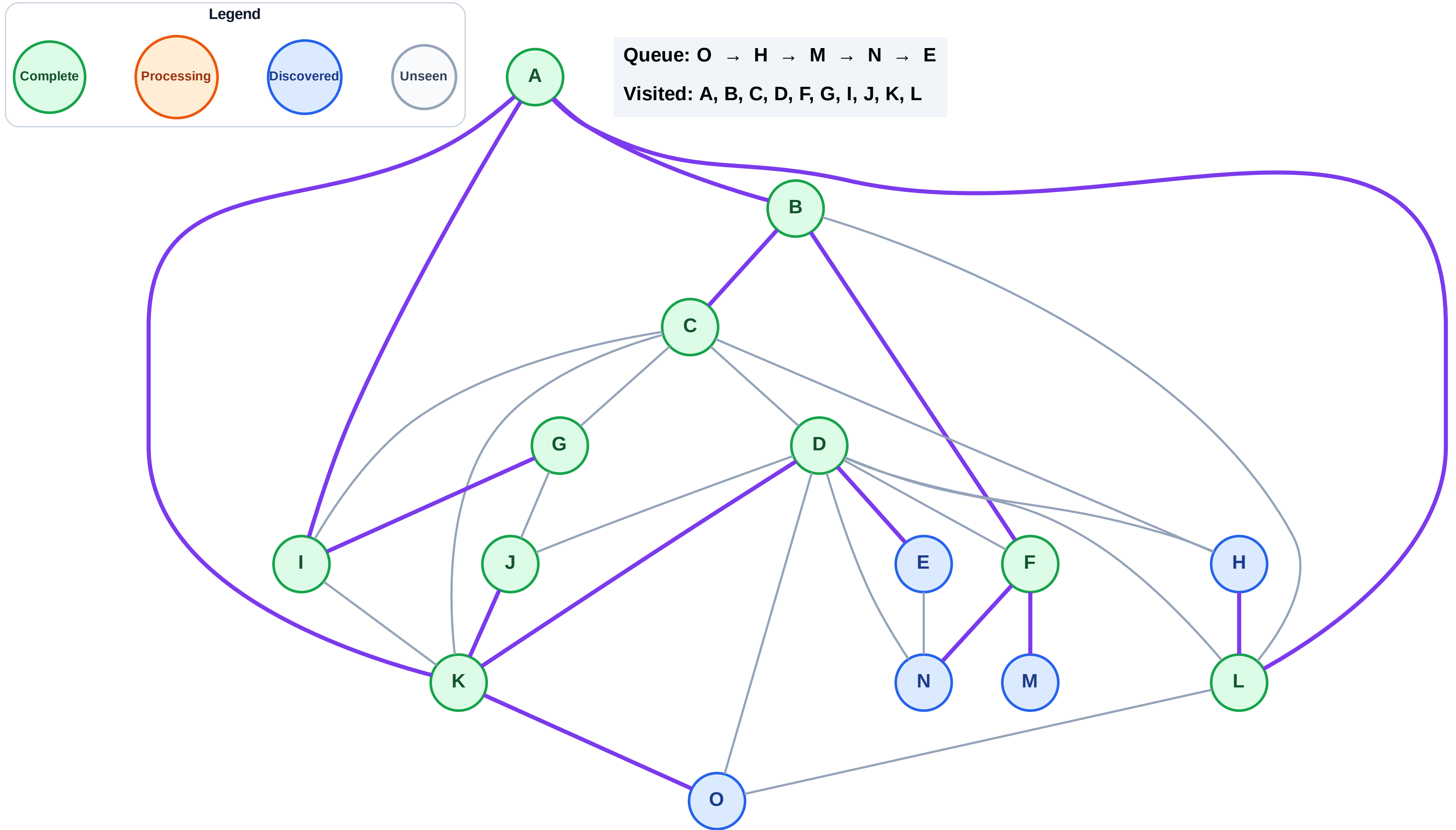
Visited: A, B, C, D, F, G, I, K, L



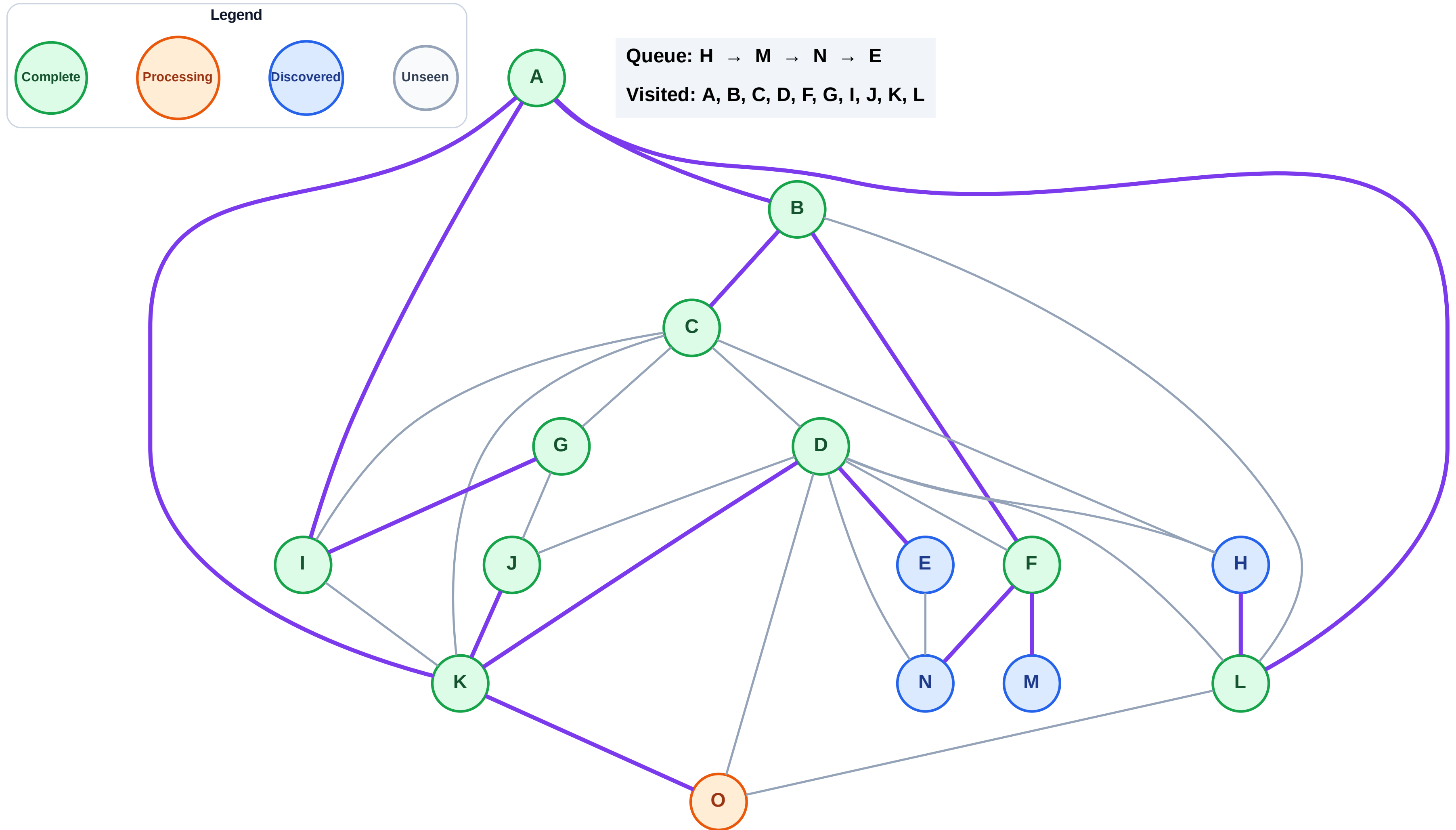
Step 21: No new neighbors from J

Queue: O → H → M → N → E

Queue: O → H → M → N → E

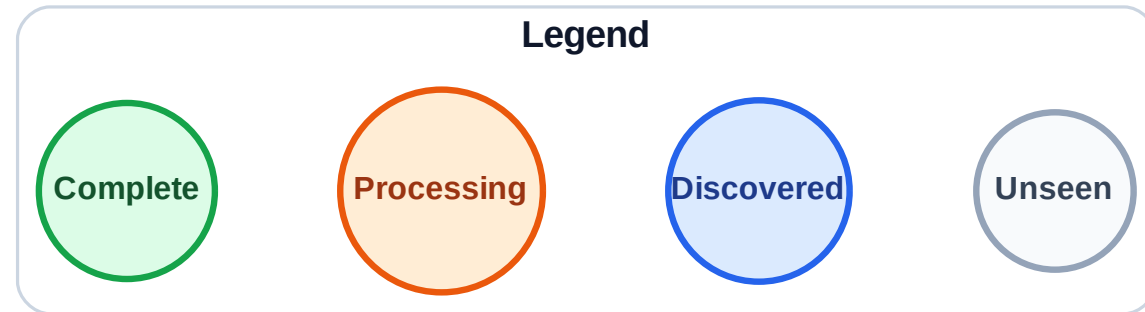


Step 22: Process node O



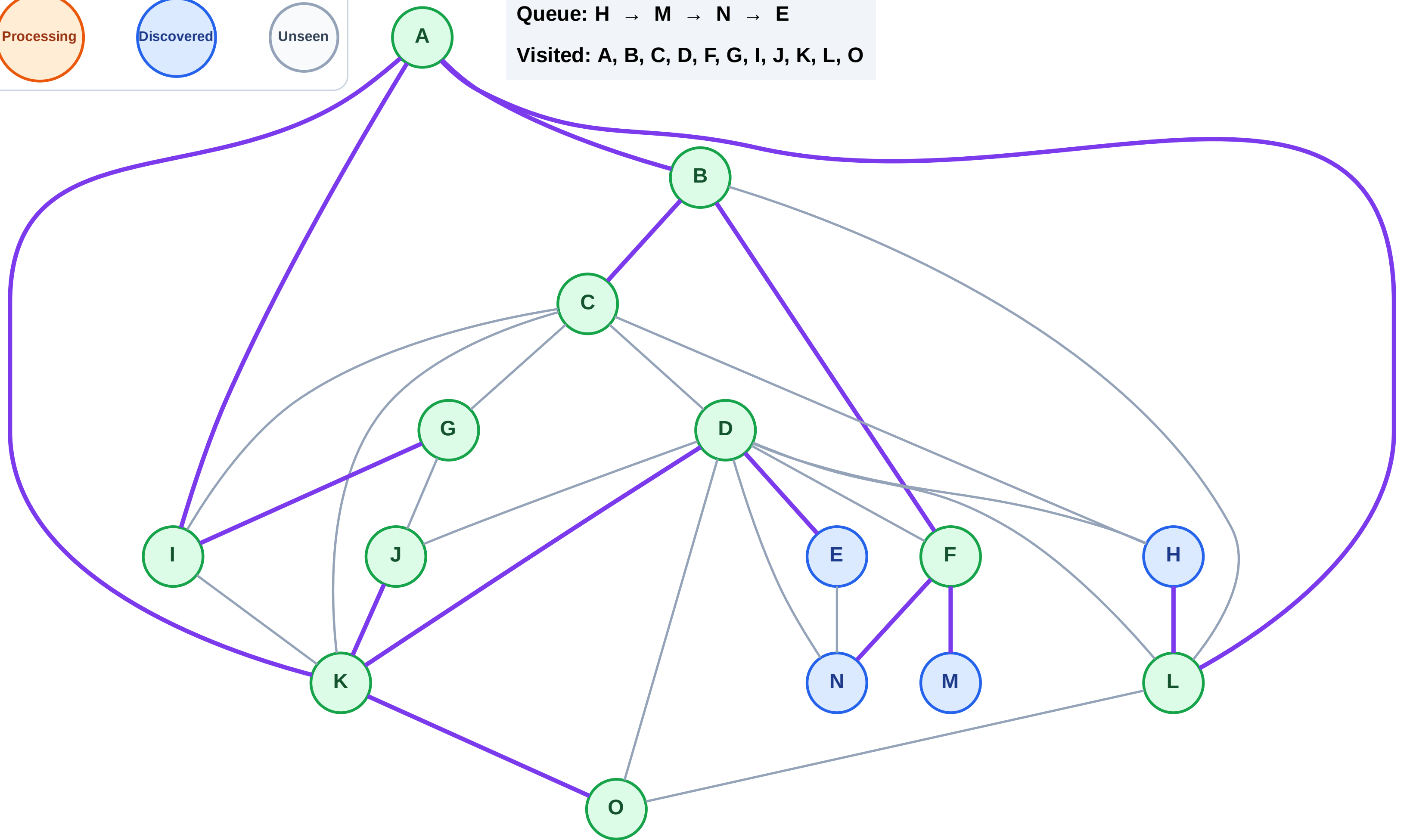
BFS Traversal

Step 23: No new neighbors from O



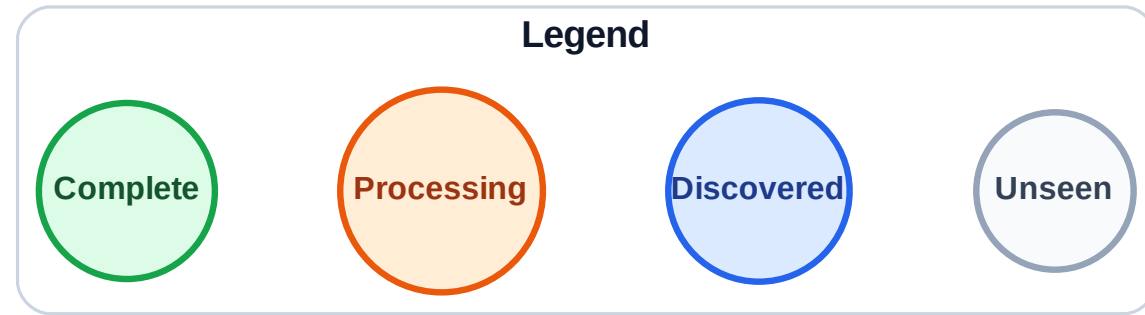
Queue: H → M → N → E

Visited: A, B, C, D, F, G, I, J, K, L, O



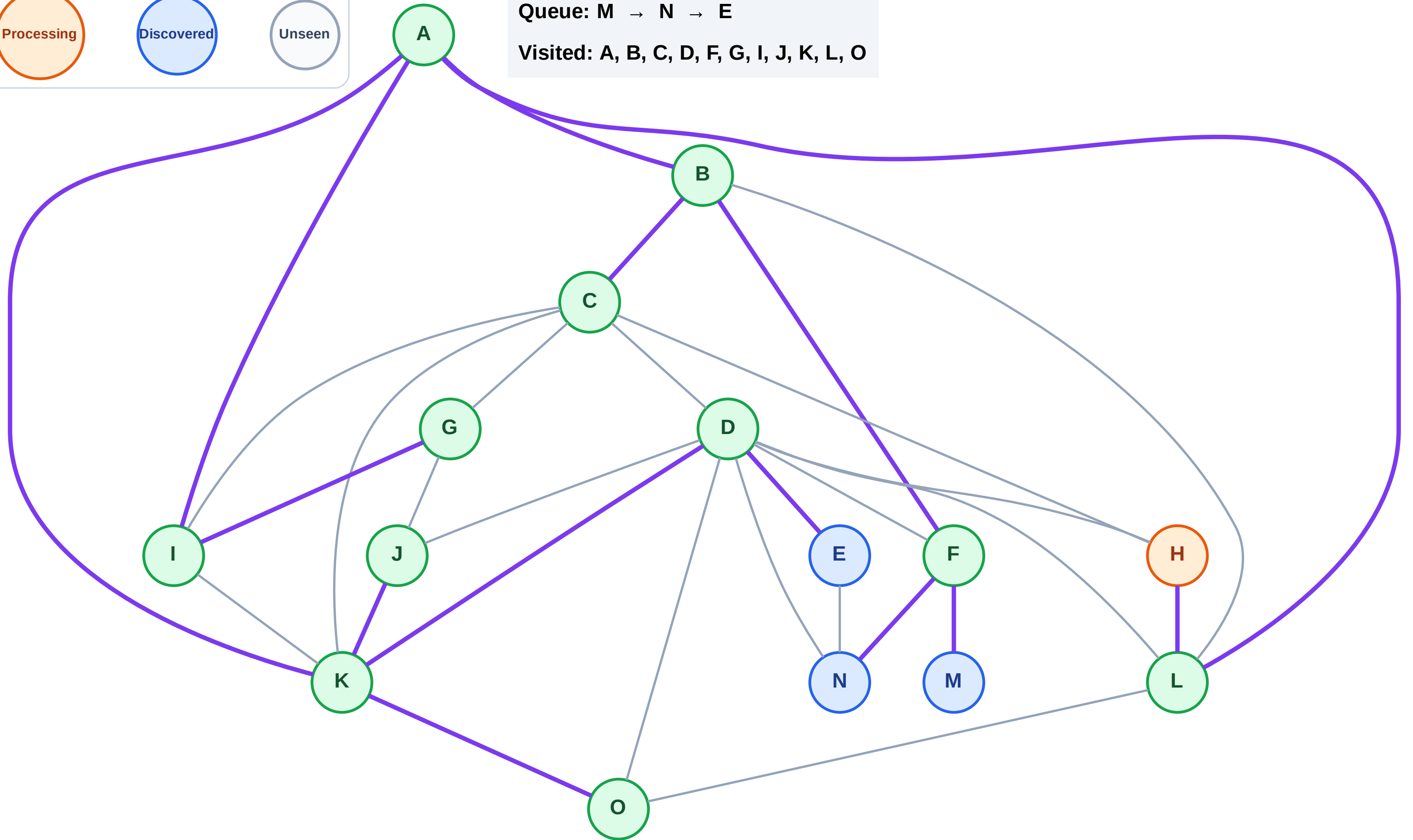
BFS Traversal

Step 24: Process node H



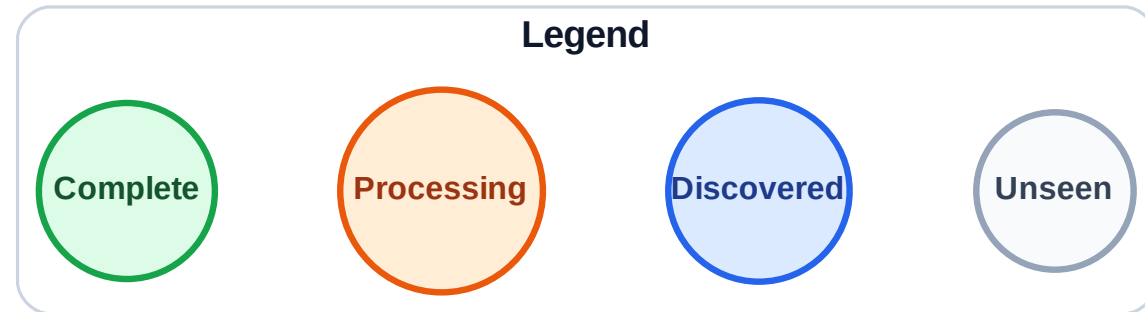
Queue: M → N → E

Visited: A, B, C, D, F, G, I, J, K, L, O



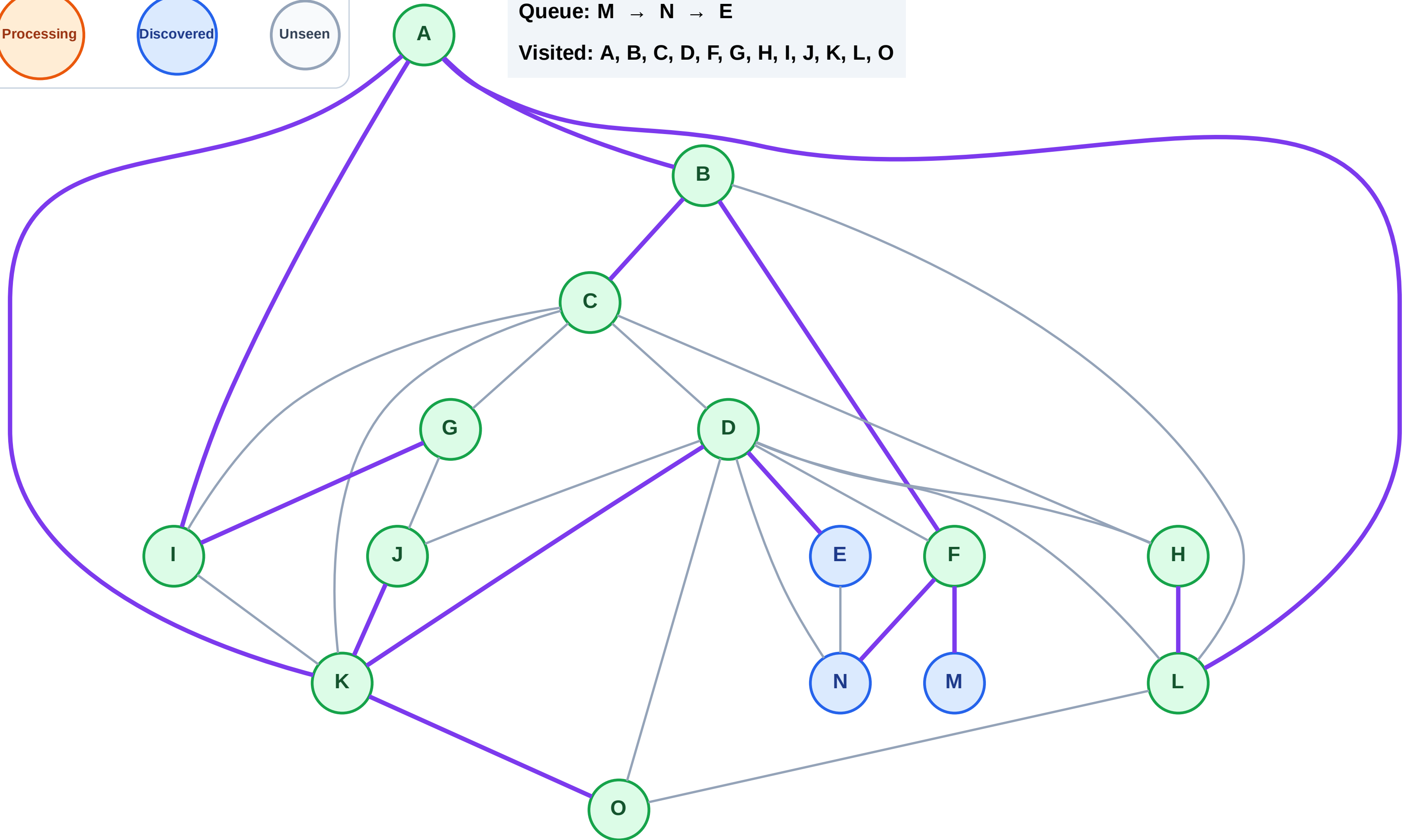
BFS Traversal

Step 25: No new neighbors from H



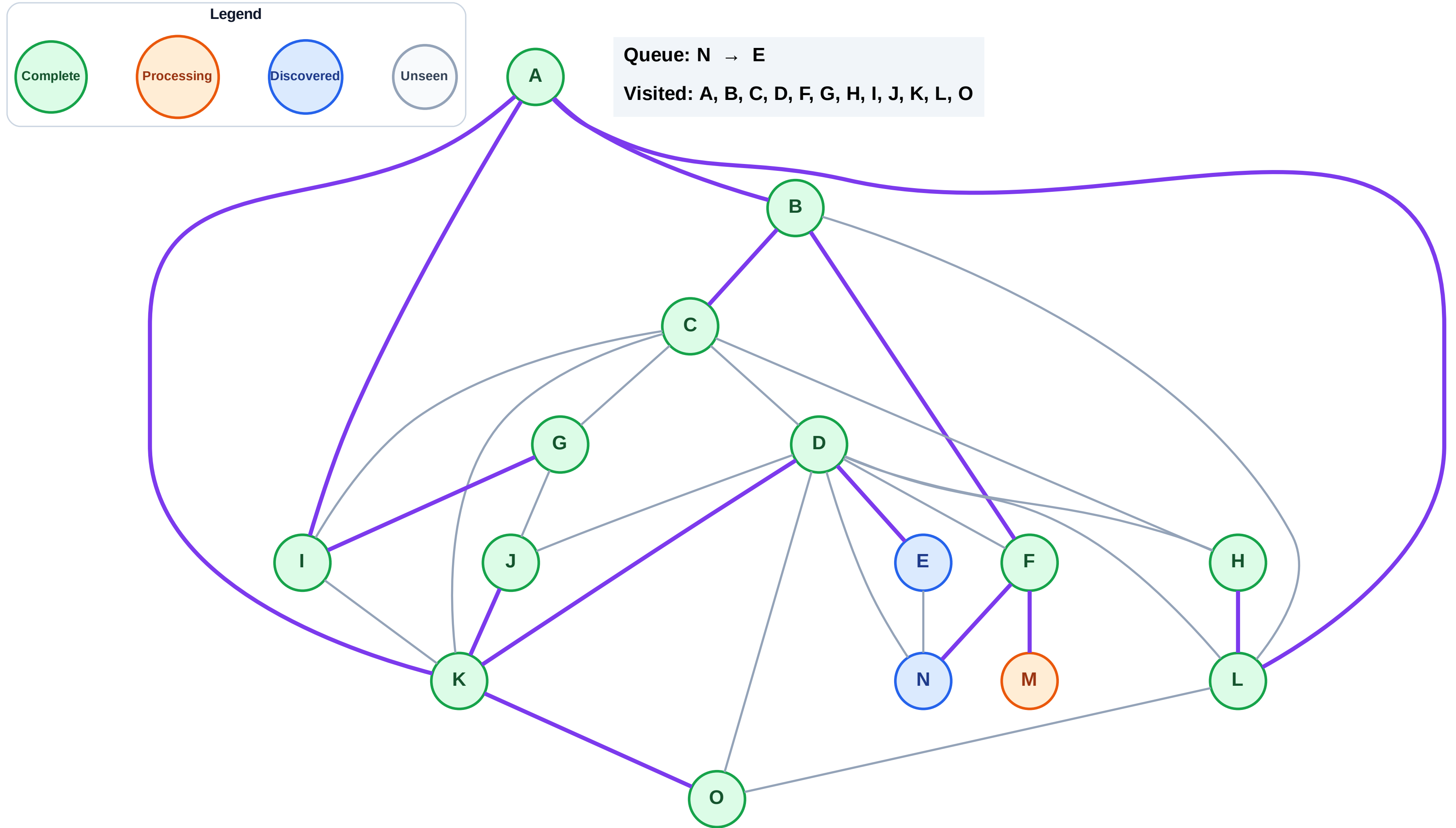
Queue: M → N → E

Visited: A, B, C, D, F, G, H, I, J, K, L, O



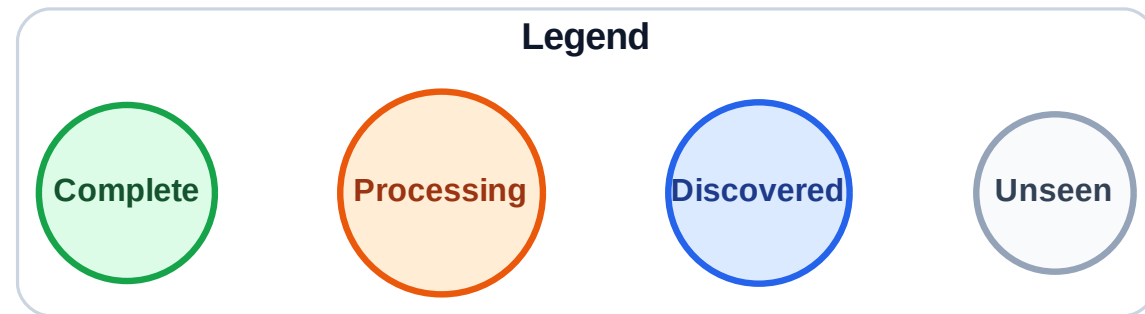
Step 26: Process node M

Visited: A B C



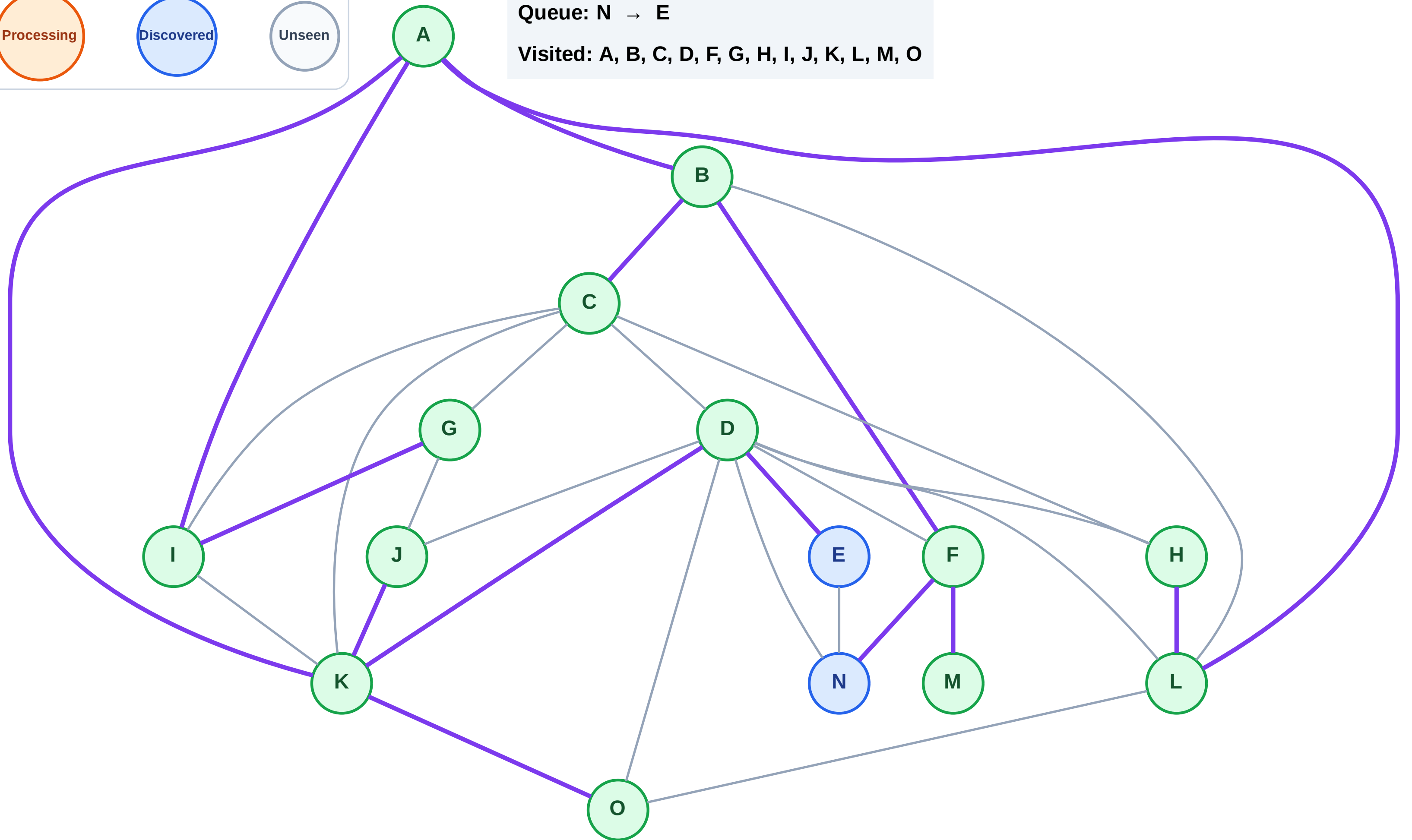
BFS Traversal

Step 27: No new neighbors from M



Queue: N → E

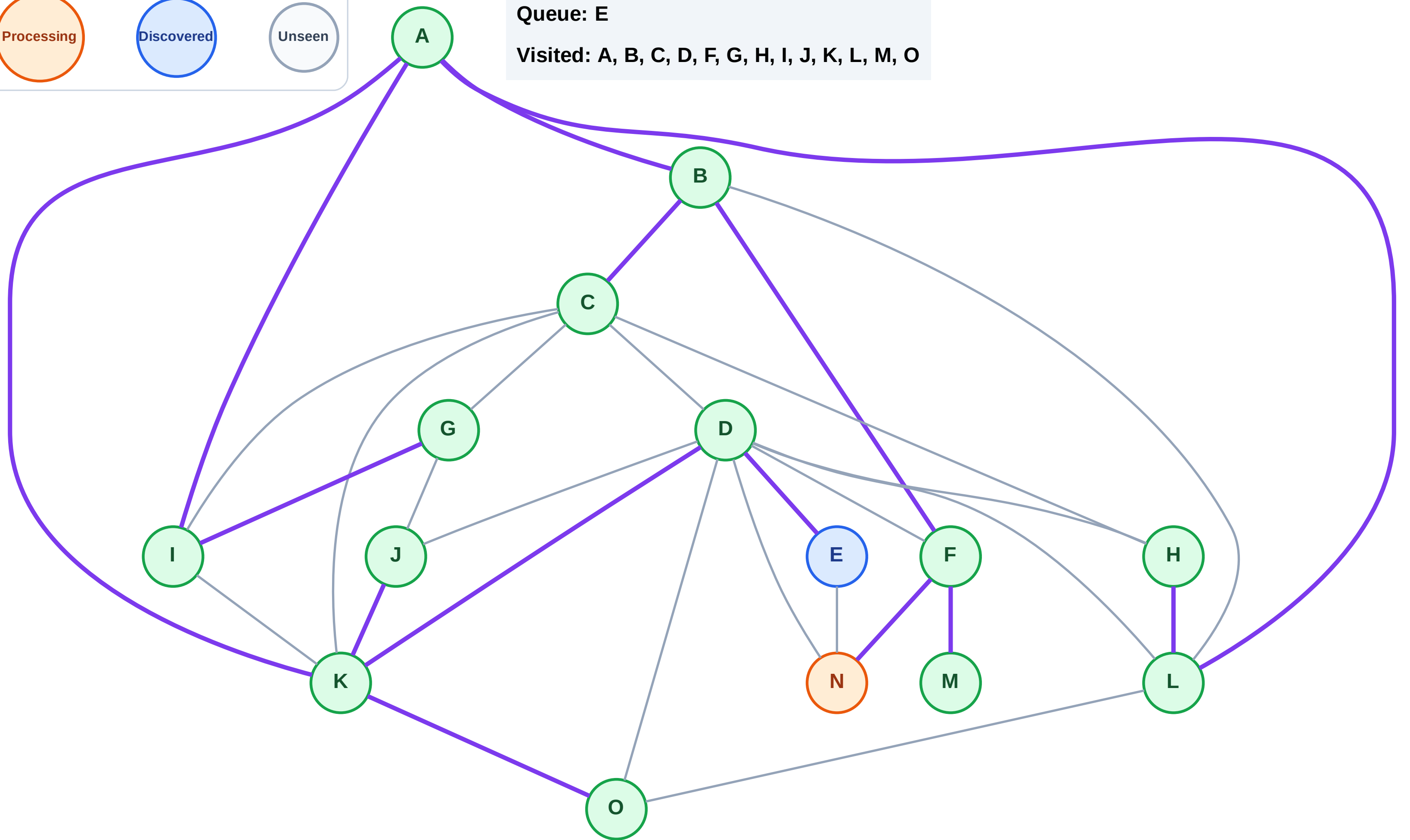
Visited: A, B, C, D, F, G, H, I, J, K, L, M, O



Step 28: Process node N

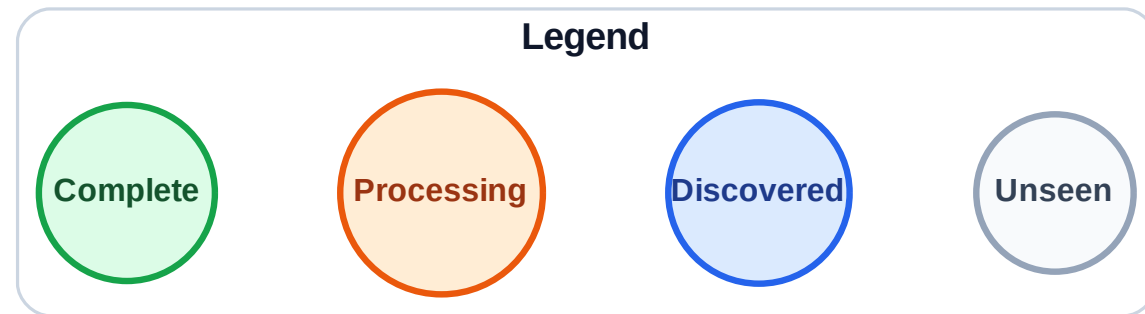


Visited: A



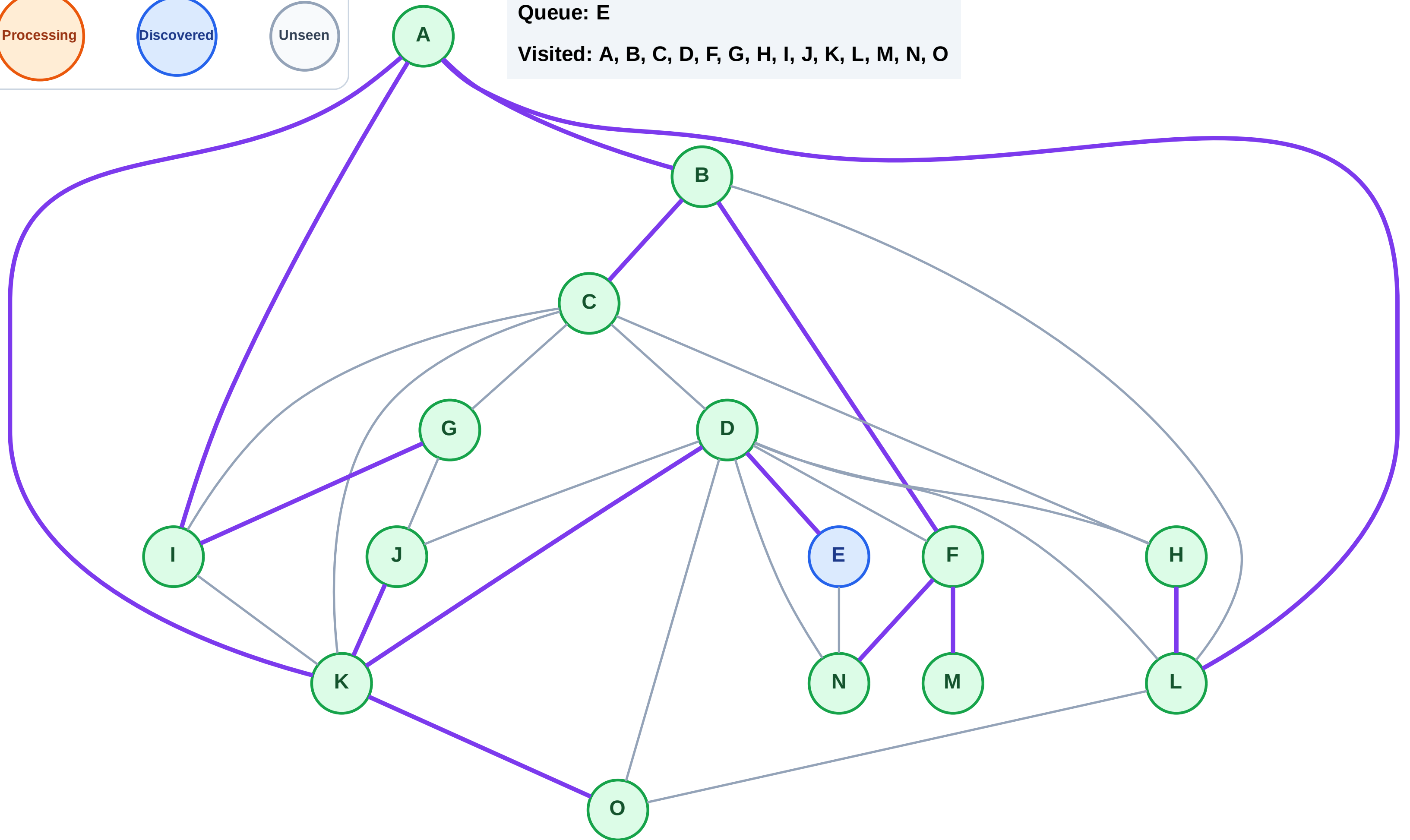
BFS Traversal

Step 29: No new neighbors from N



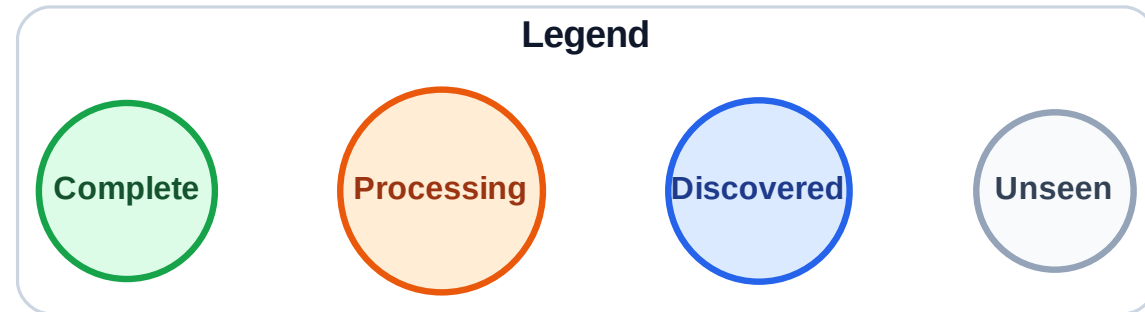
Queue: E

Visited: A, B, C, D, F, G, H, I, J, K, L, M, N, O



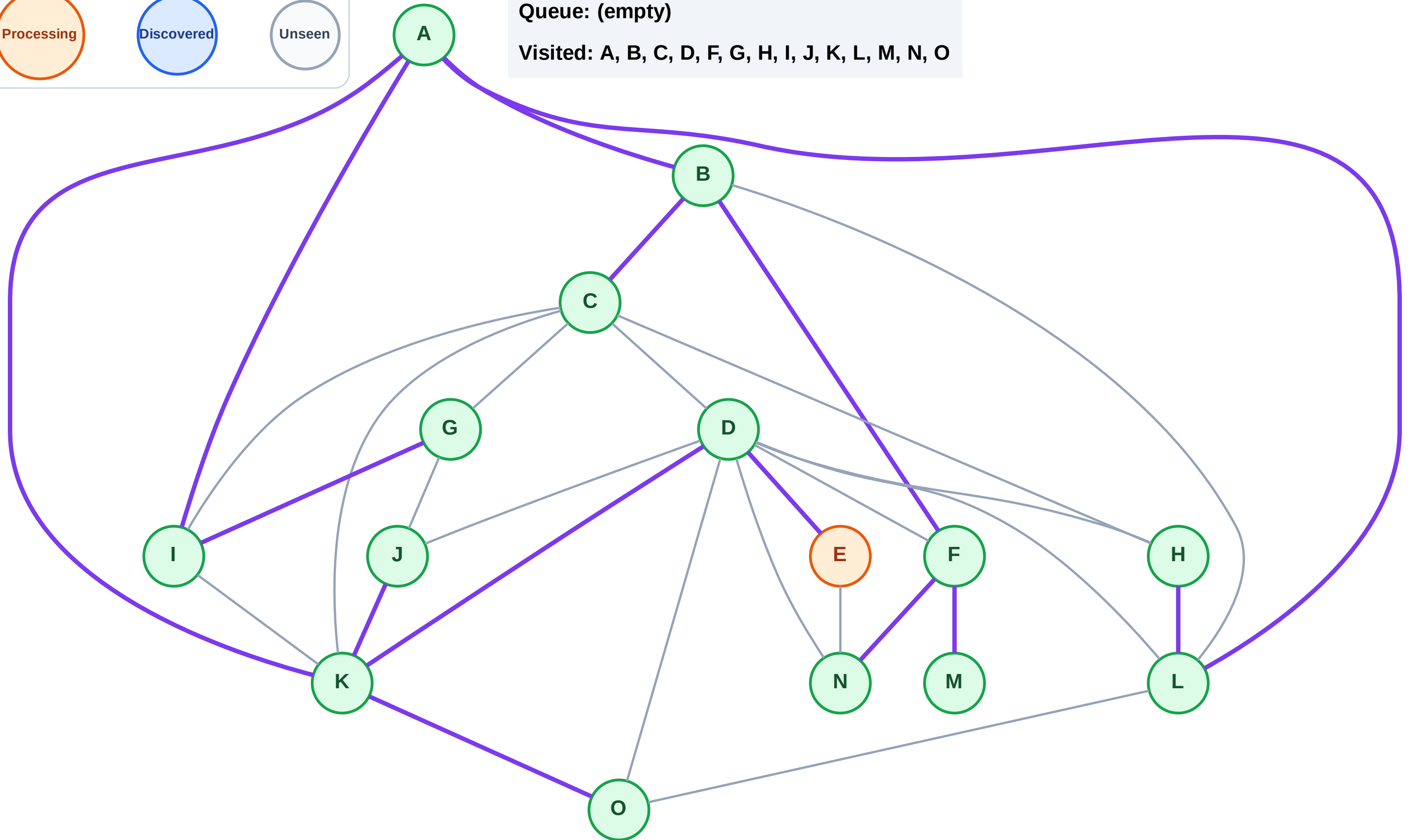
BFS Traversal

Step 30: Process node E



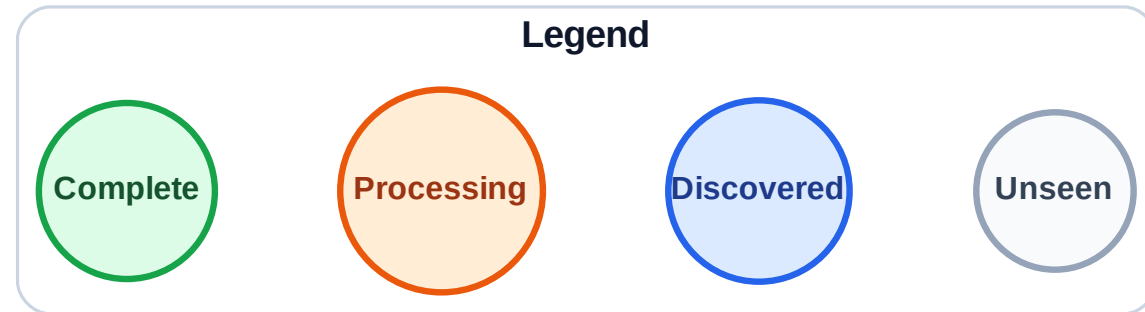
Queue: (empty)

Visited: A, B, C, D, F, G, H, I, J, K, L, M, N, O



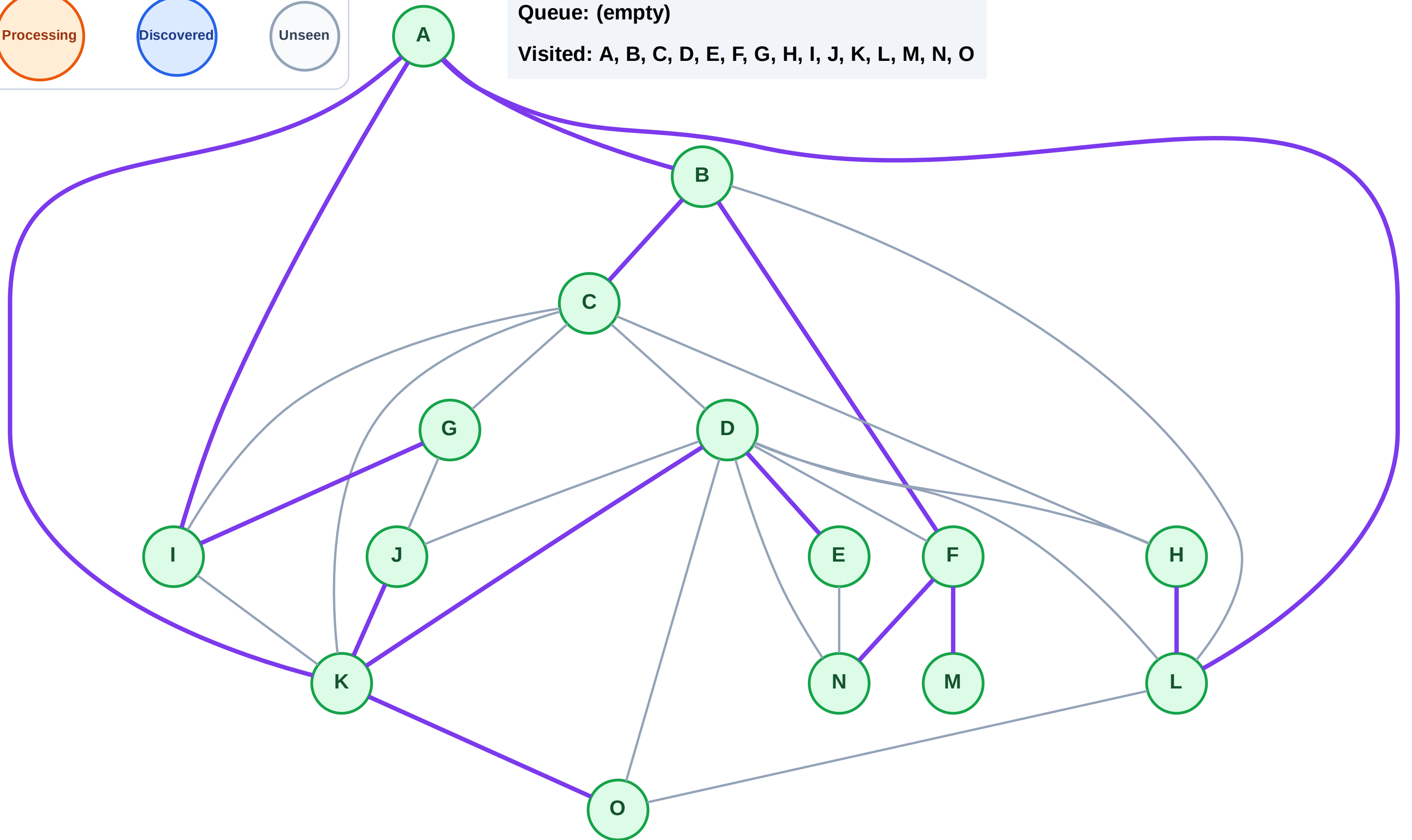
BFS Traversal

Step 31: No new neighbors from E



Queue: (empty)

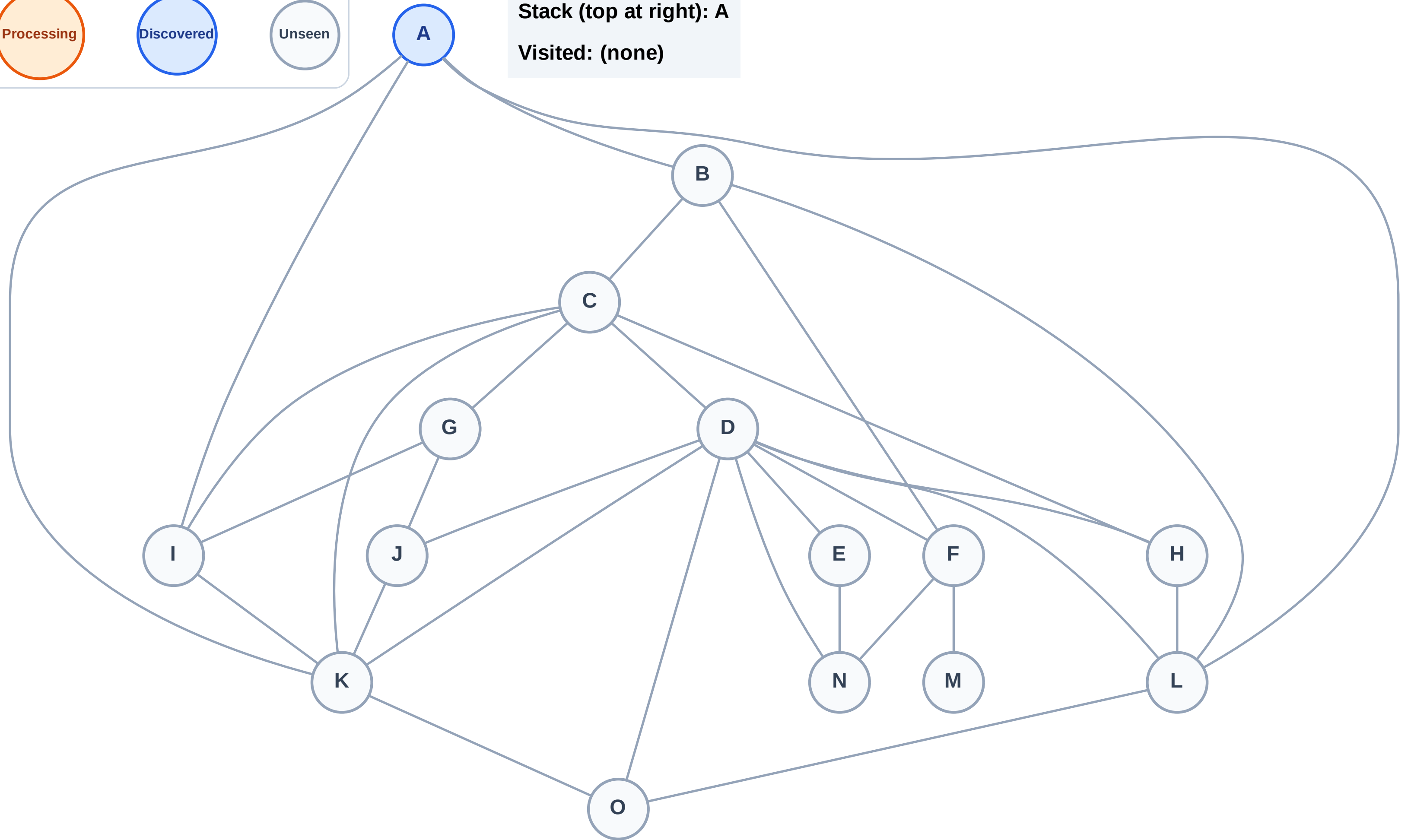
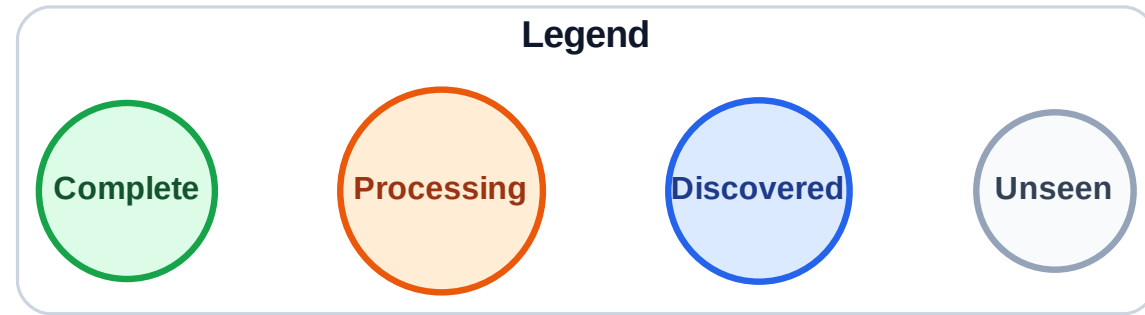
Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N, O



Step 1: Start at A

Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

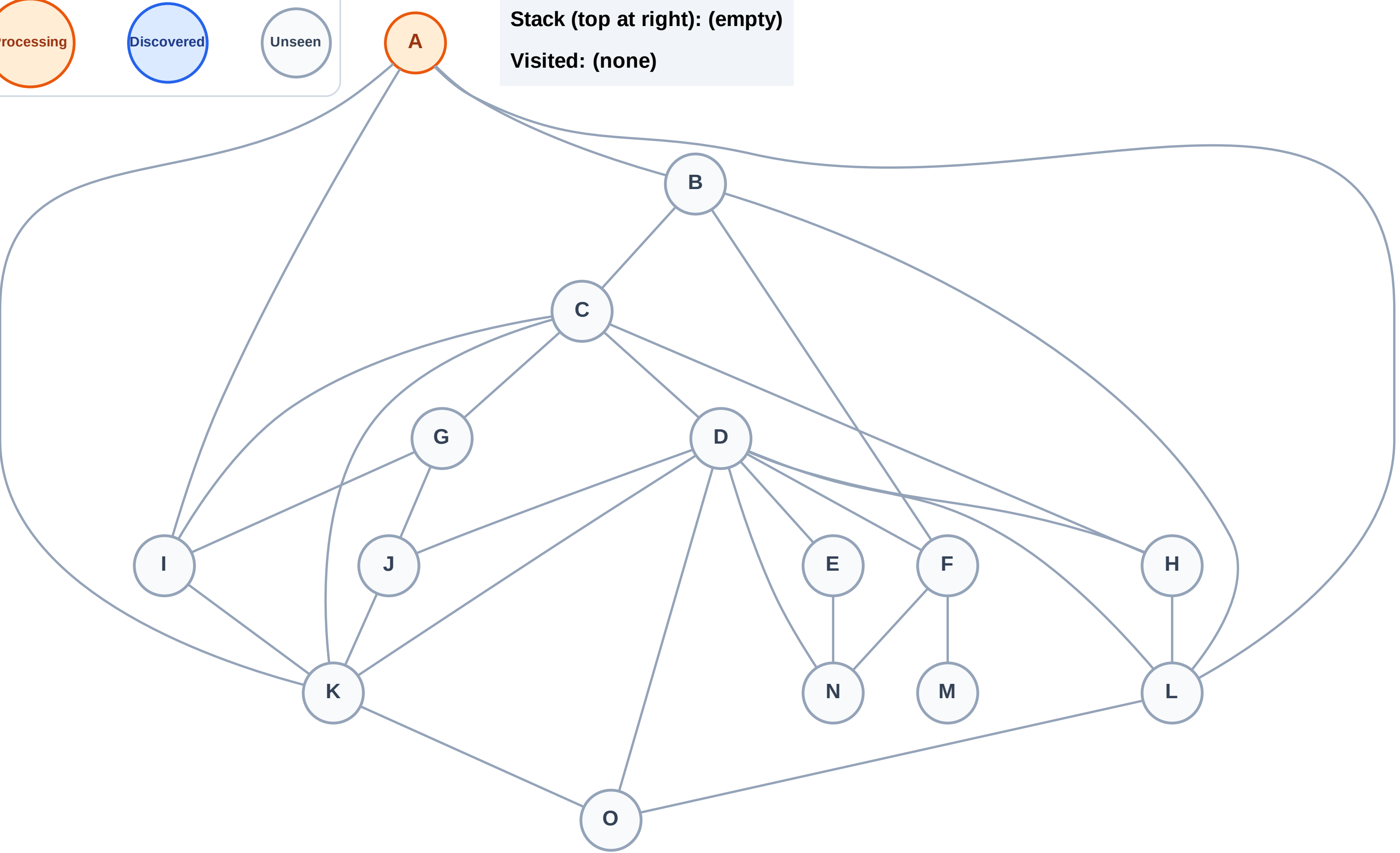
Complete

Processing

Discovered

Unseen

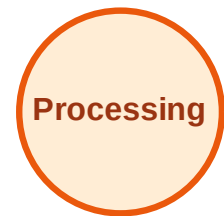
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

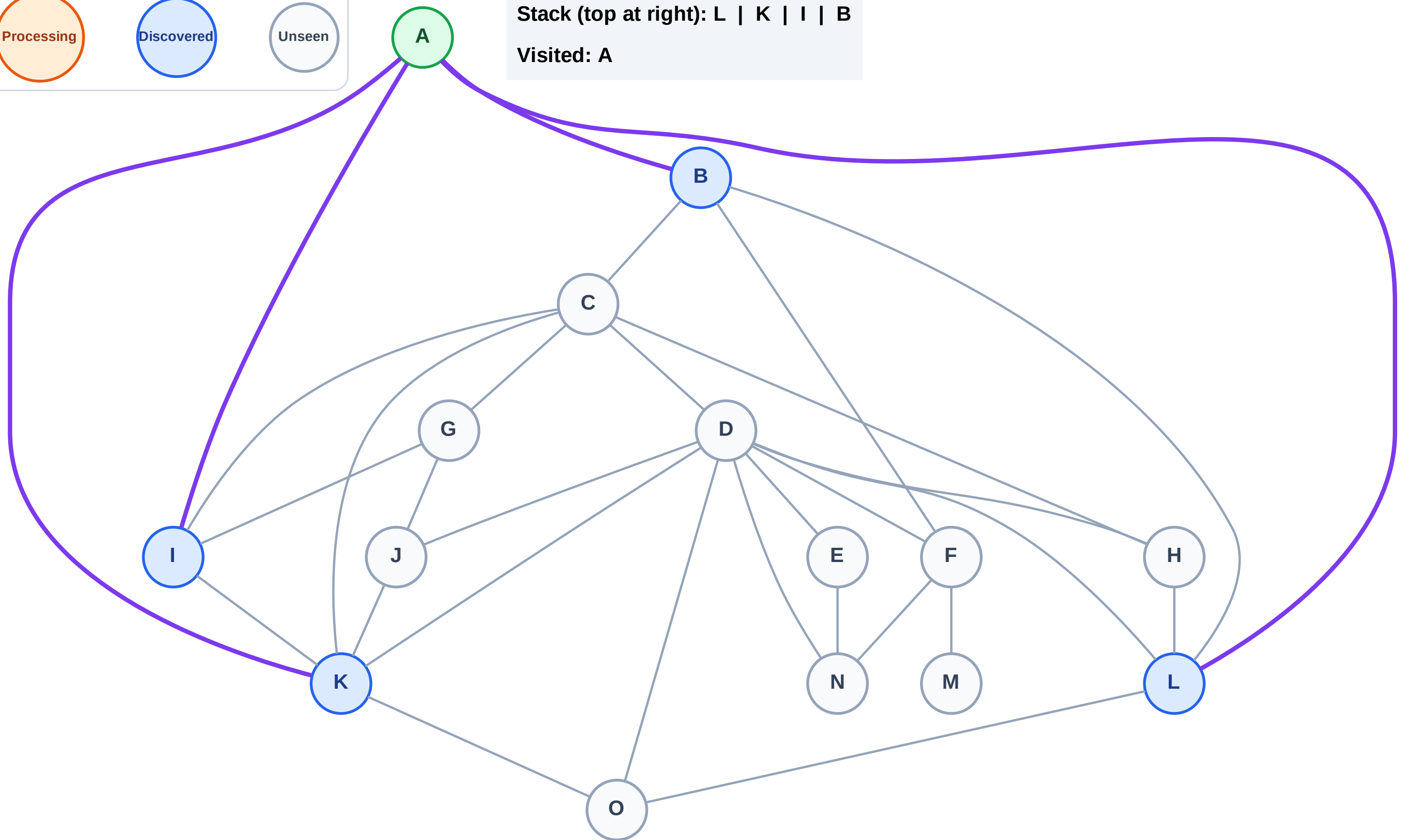
Step 3: Discover L, K, I, B from A

Legend



Stack (top at right): L | K | I | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

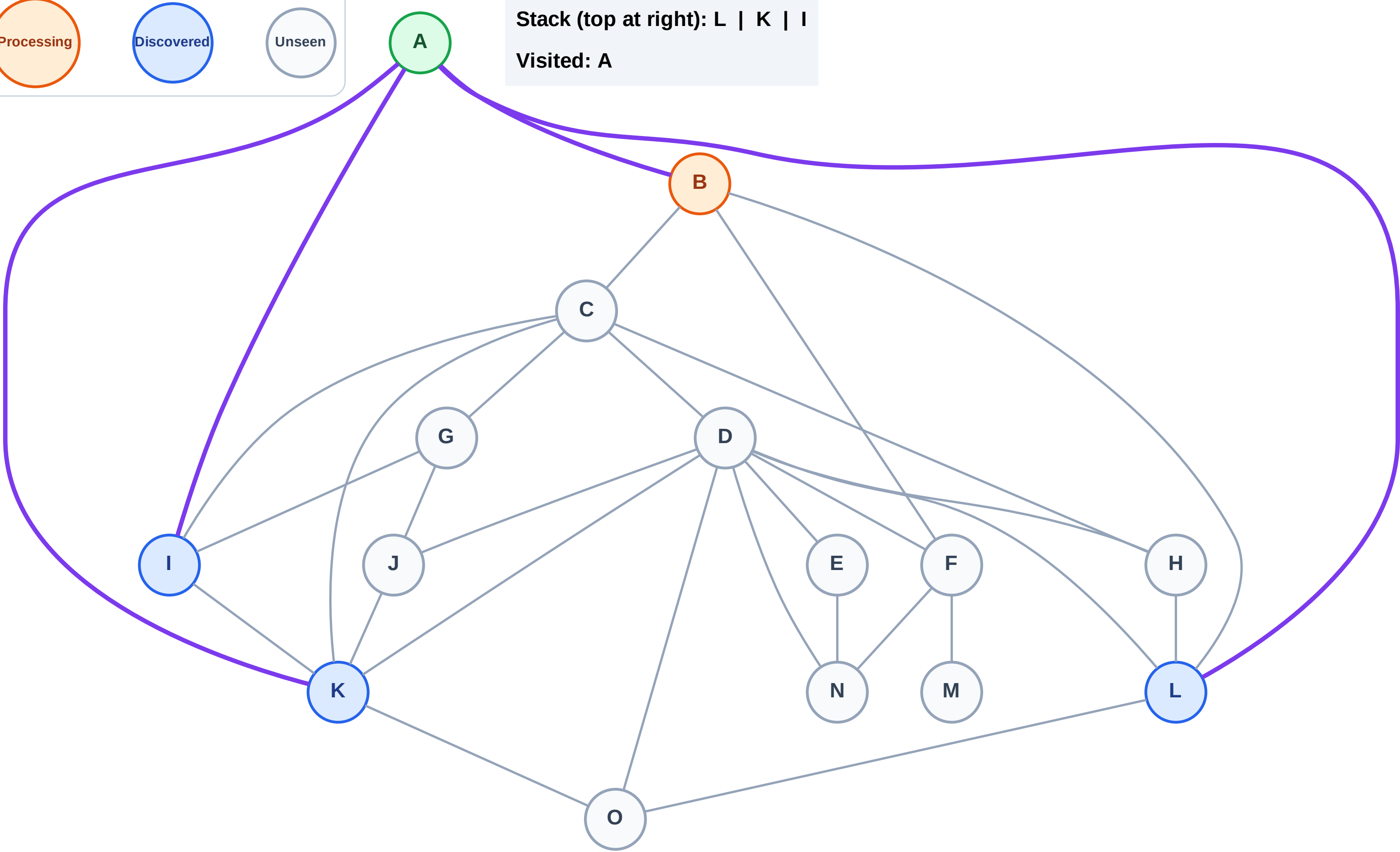
Complete

Processing

Discovered

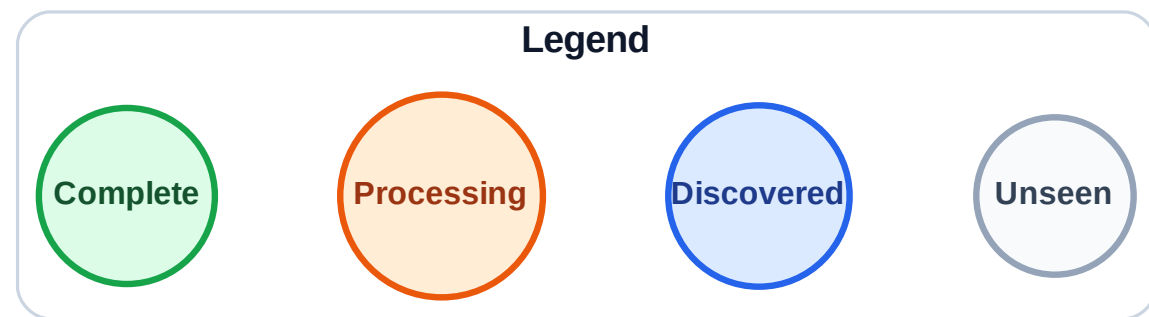
Unseen

Stack (top at right): L | K | I
Visited: A



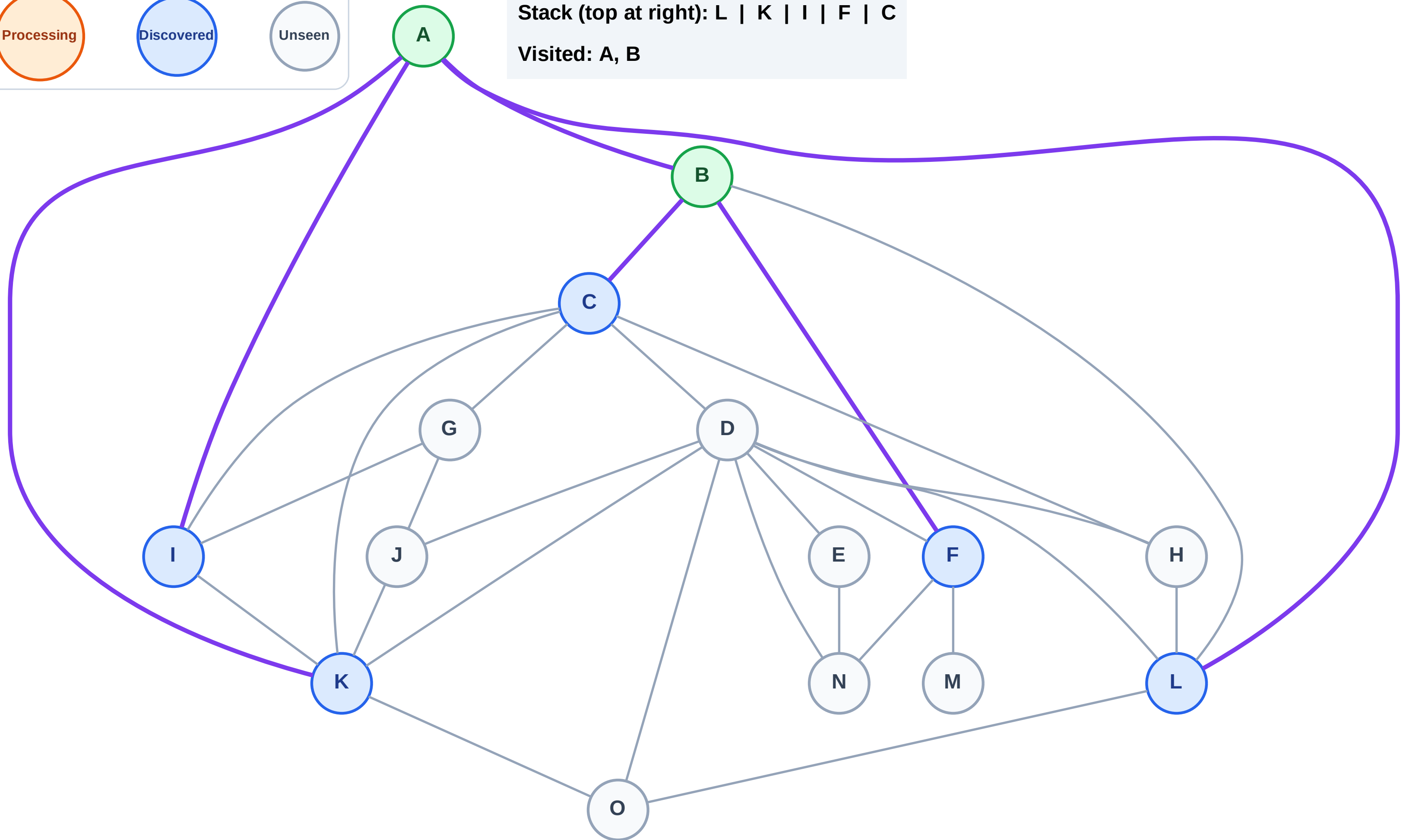
DFS Traversal

Step 5: Discover F, C from B



Stack (top at right): L | K | I | F | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

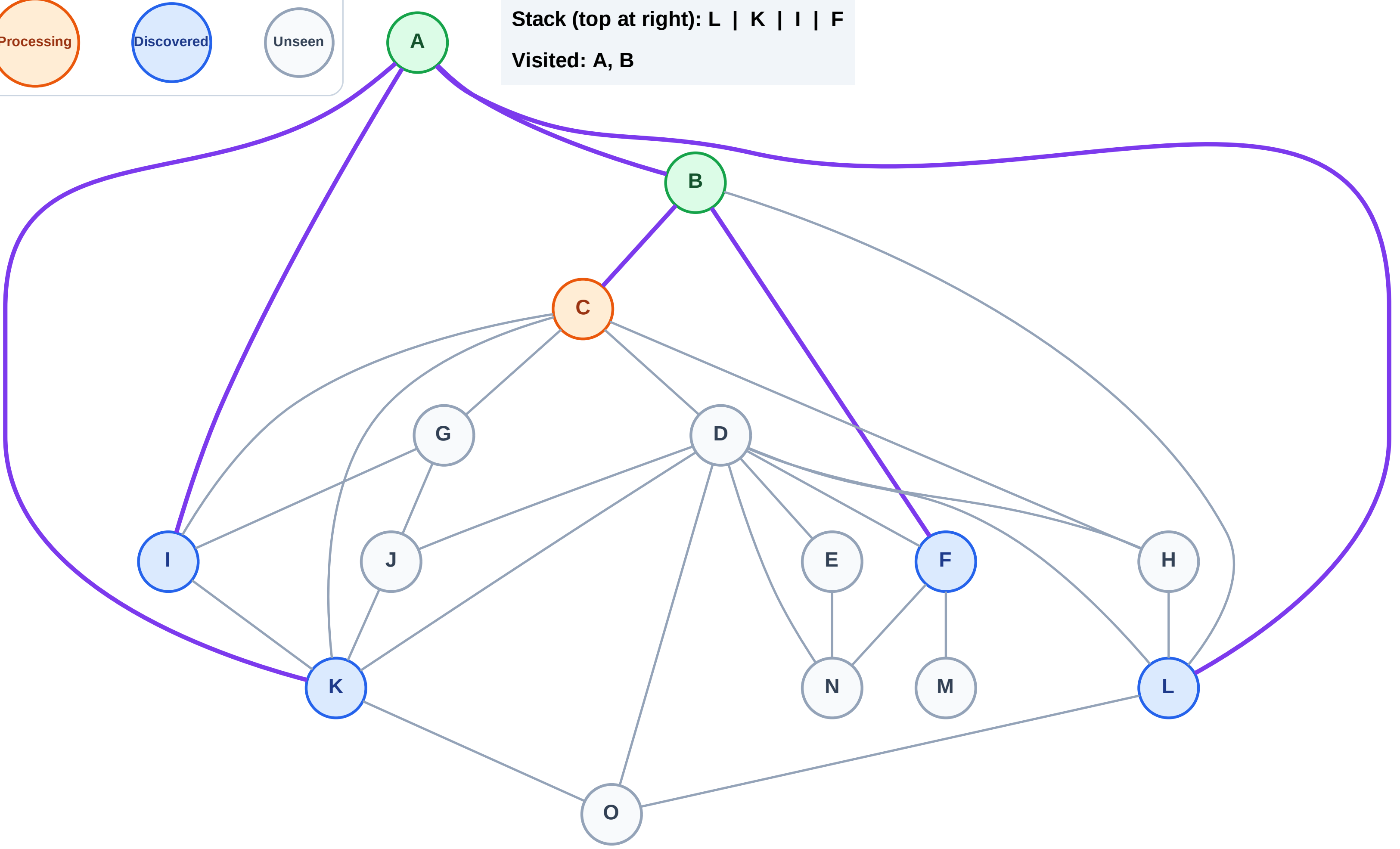
Complete

Processing

Discovered

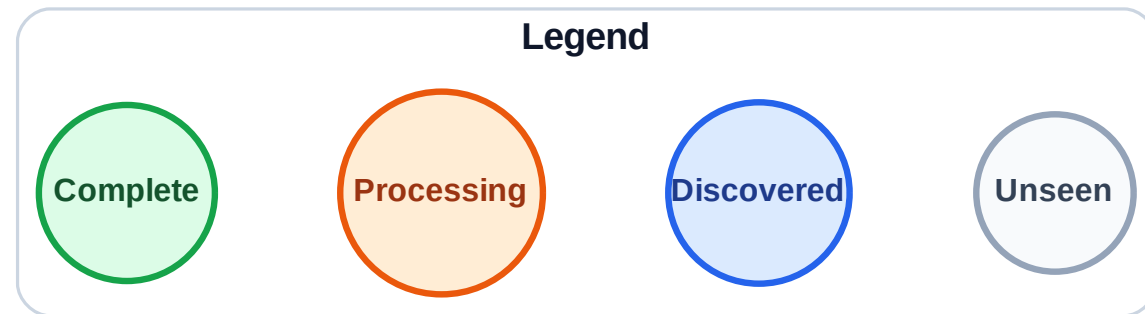
Unseen

Stack (top at right): L | K | I | F
Visited: A, B

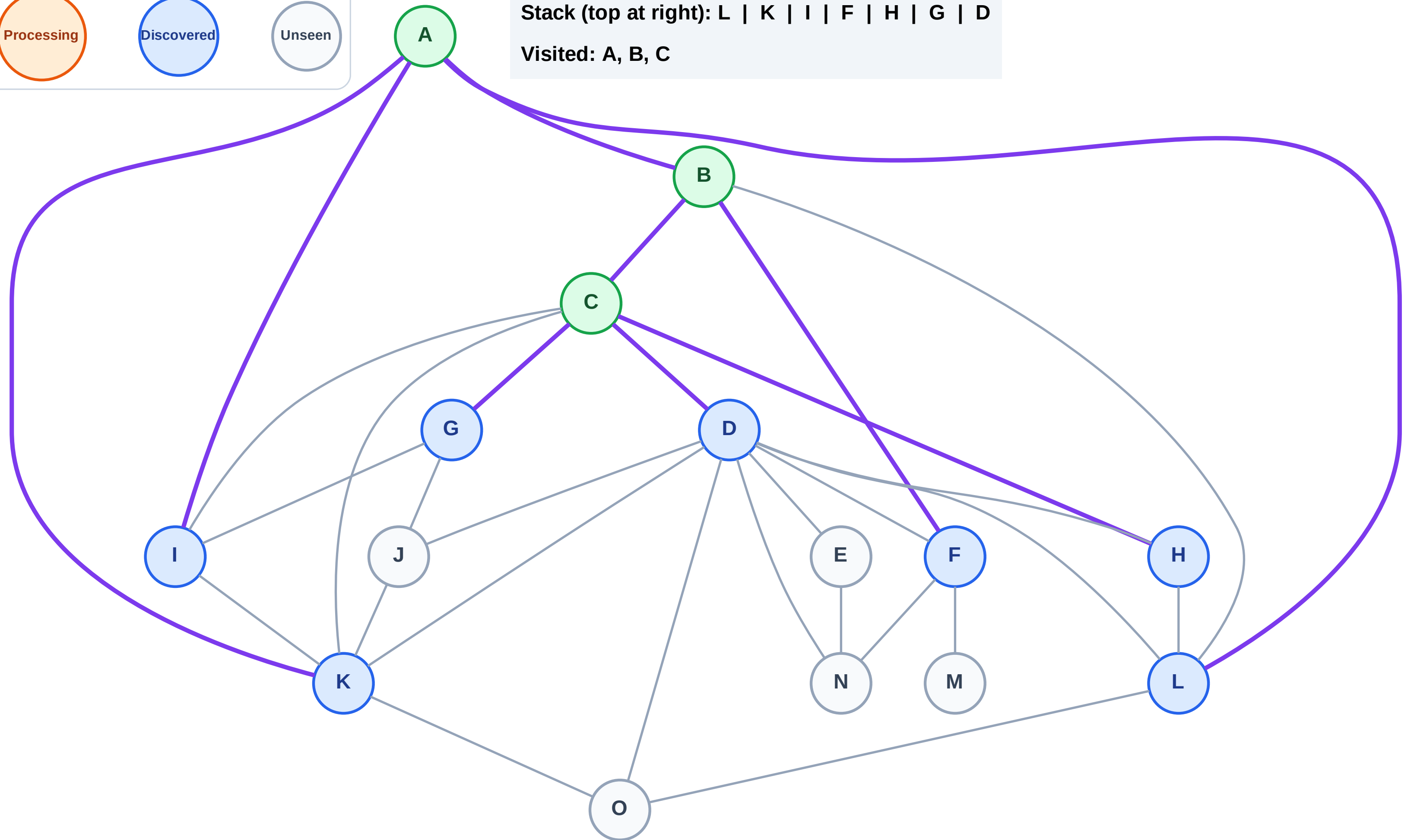


DFS Traversal

Step 7: Discover H, G, D from C



Stack (top at right): L | K | I | F | H | G | D
Visited: A, B, C



DFS Traversal

Step 8: Process node D

Legend

Complete

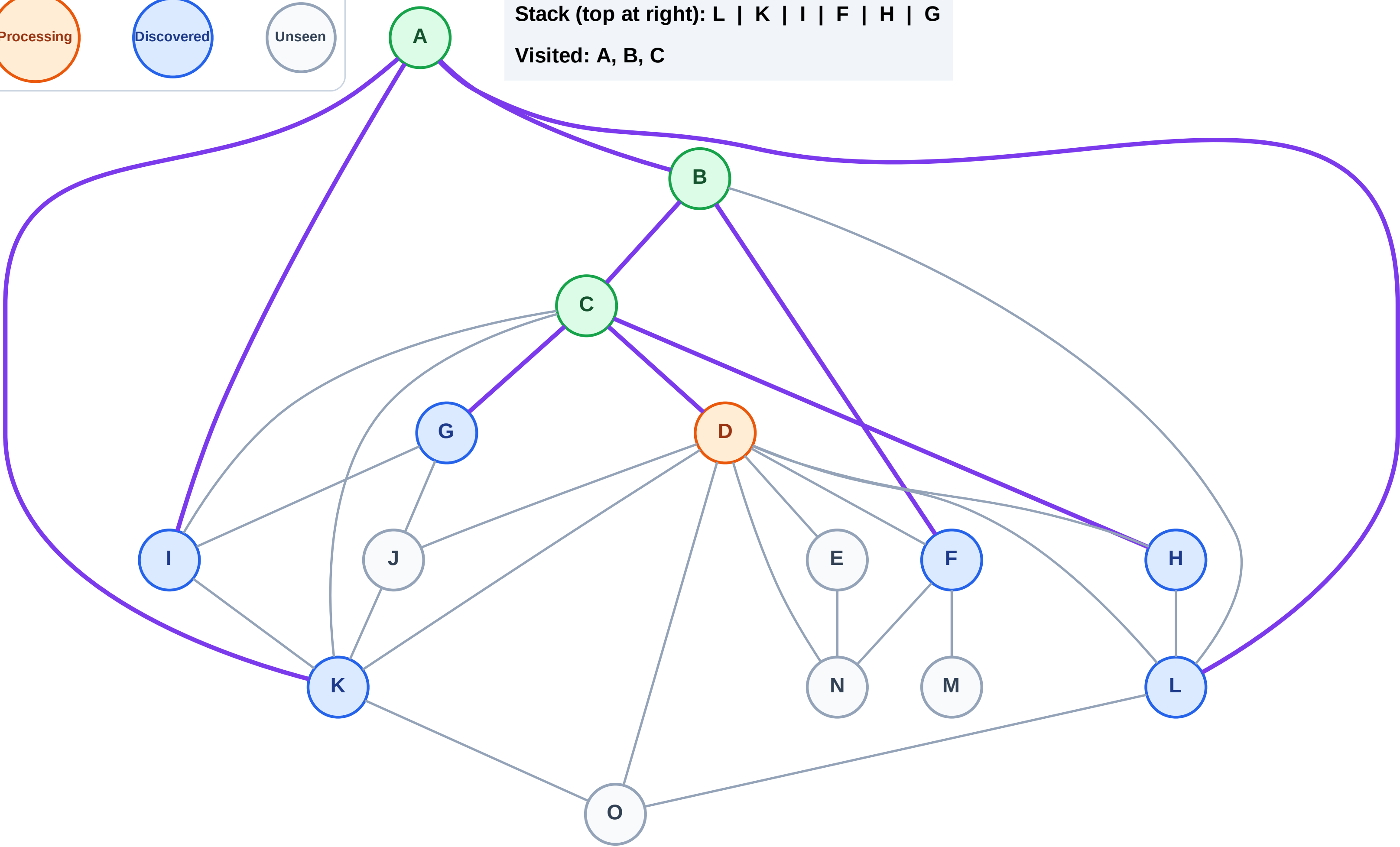
Processing

Discovered

Unseen

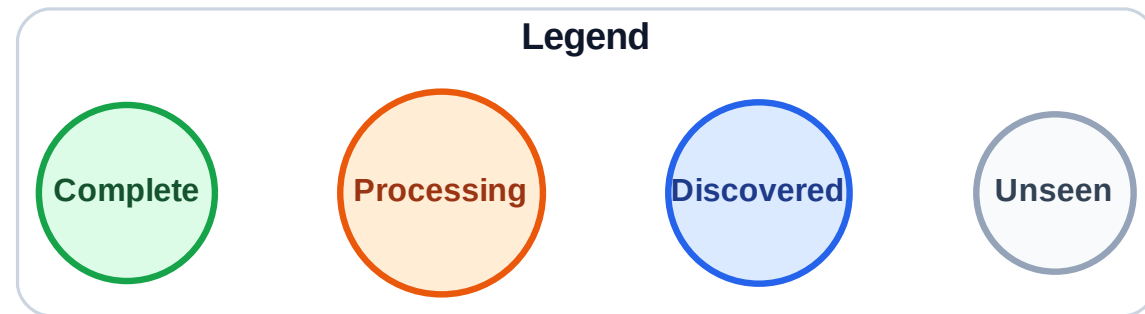
Stack (top at right): L | K | I | F | H | G

Visited: A, B, C



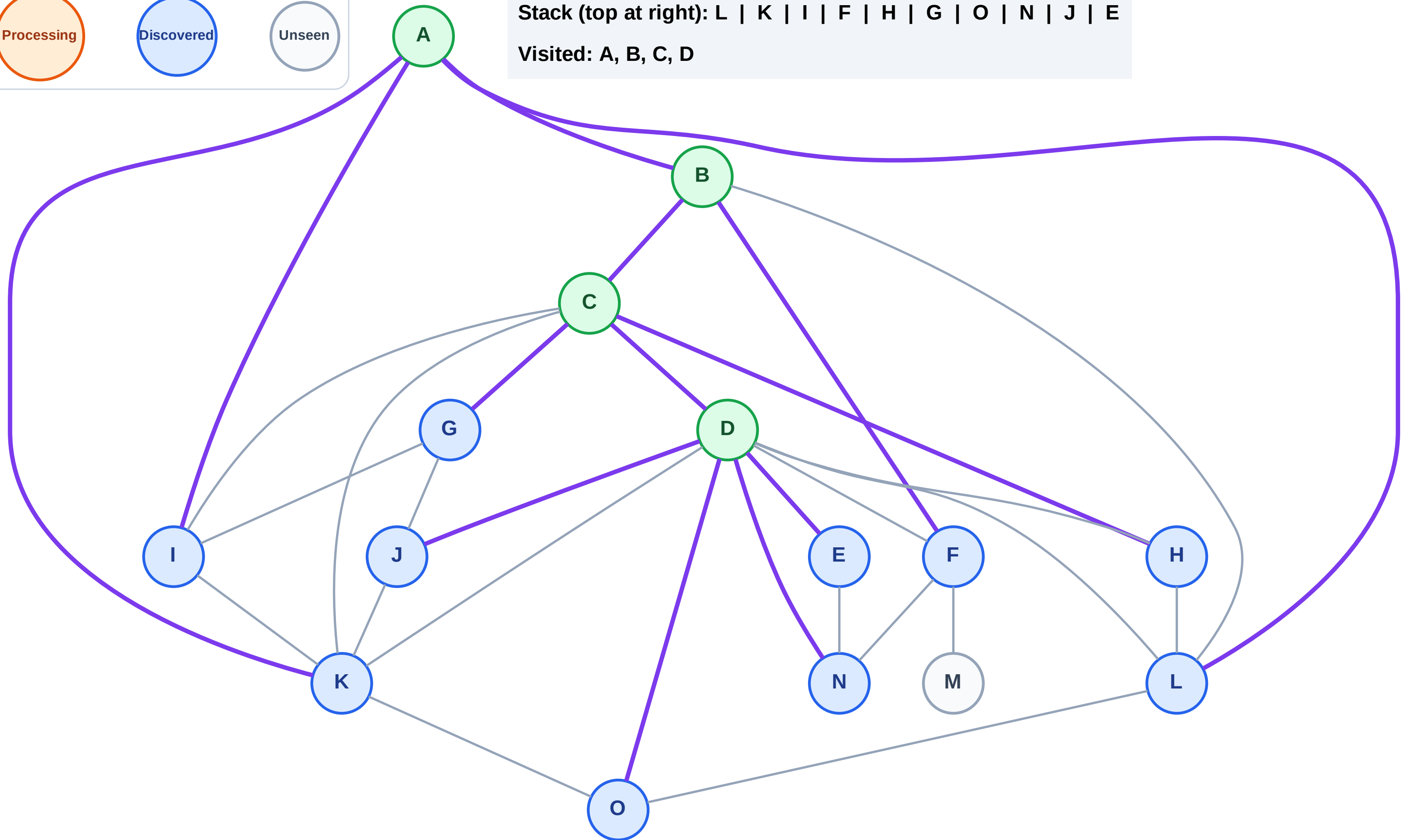
DFS Traversal

Step 9: Discover O, N, J, E from D

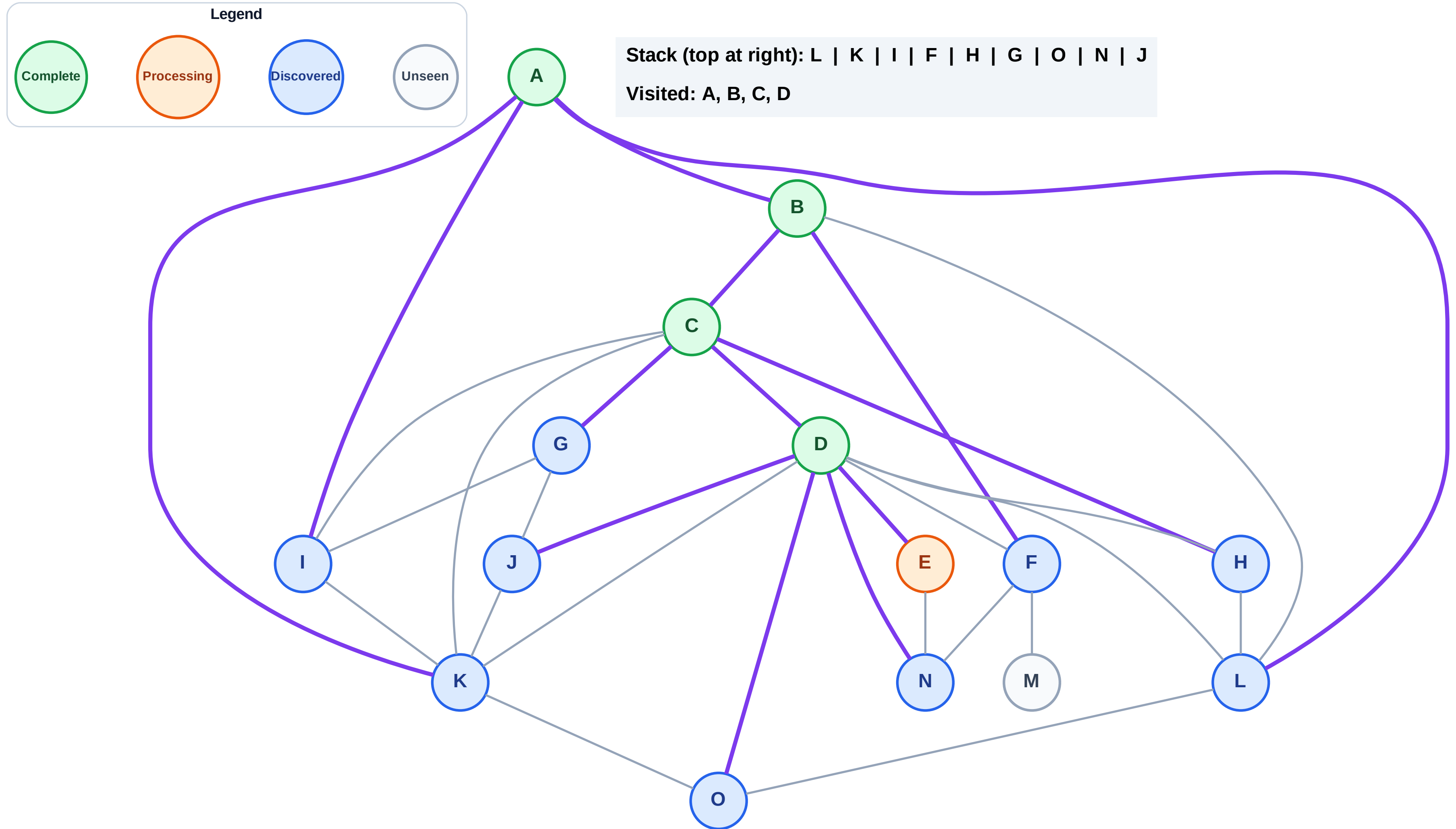


Stack (top at right): L | K | I | F | H | G | O | N | J | E

Visited: A, B, C, D

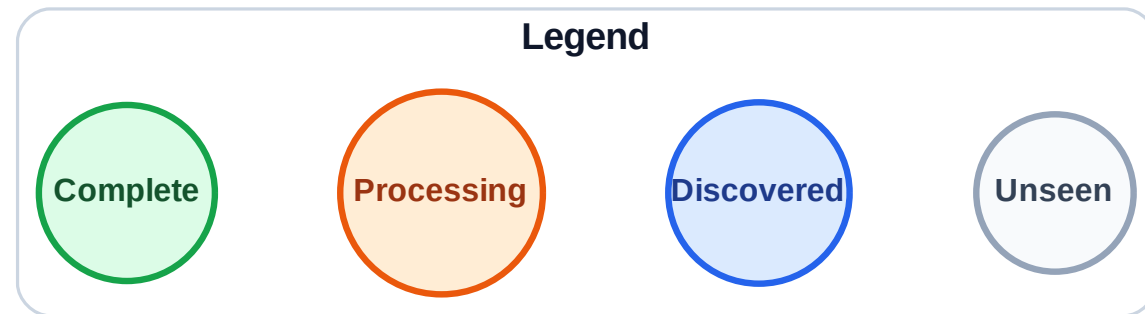


Step 10: Process node E



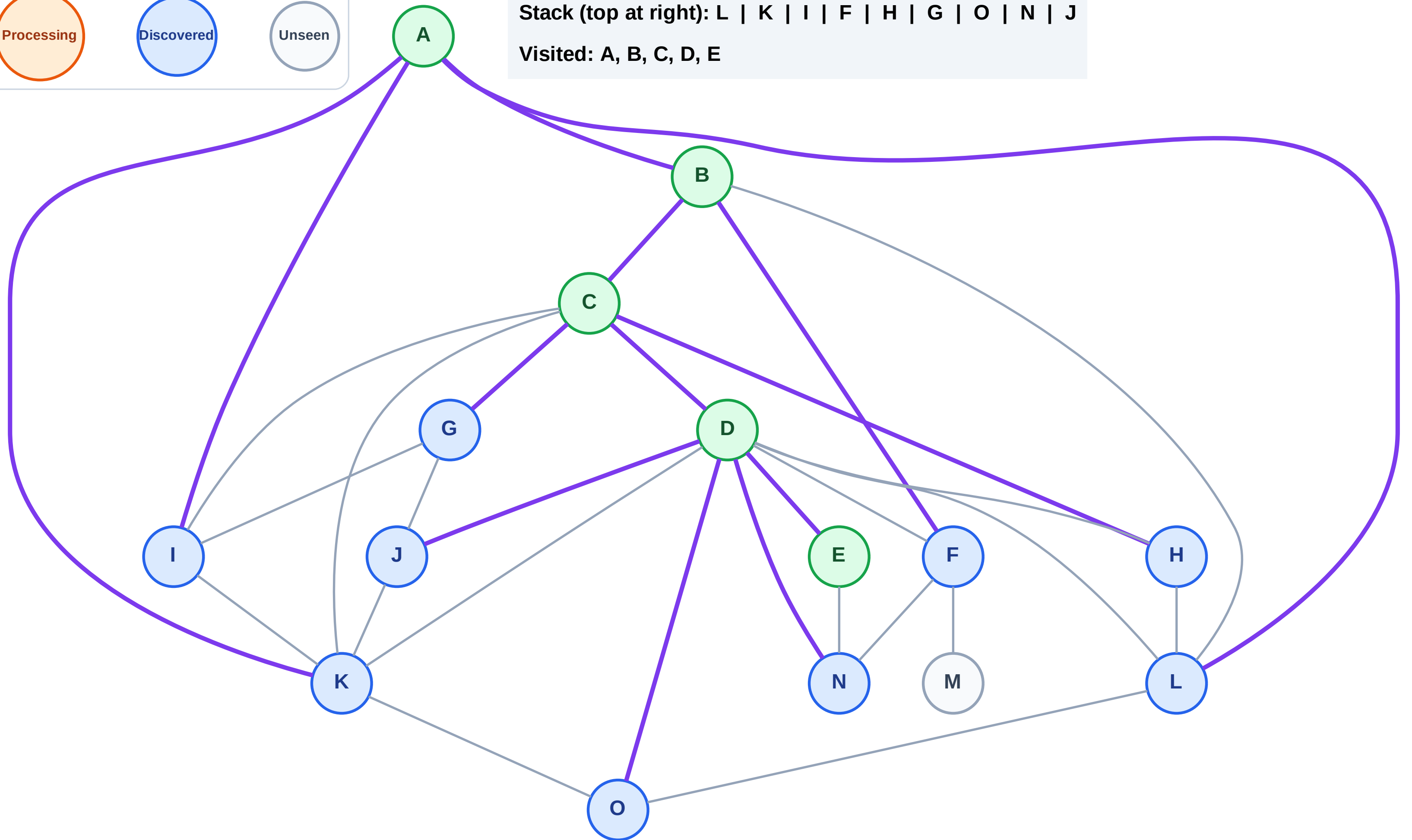
DFS Traversal

Step 11: No new neighbors from E

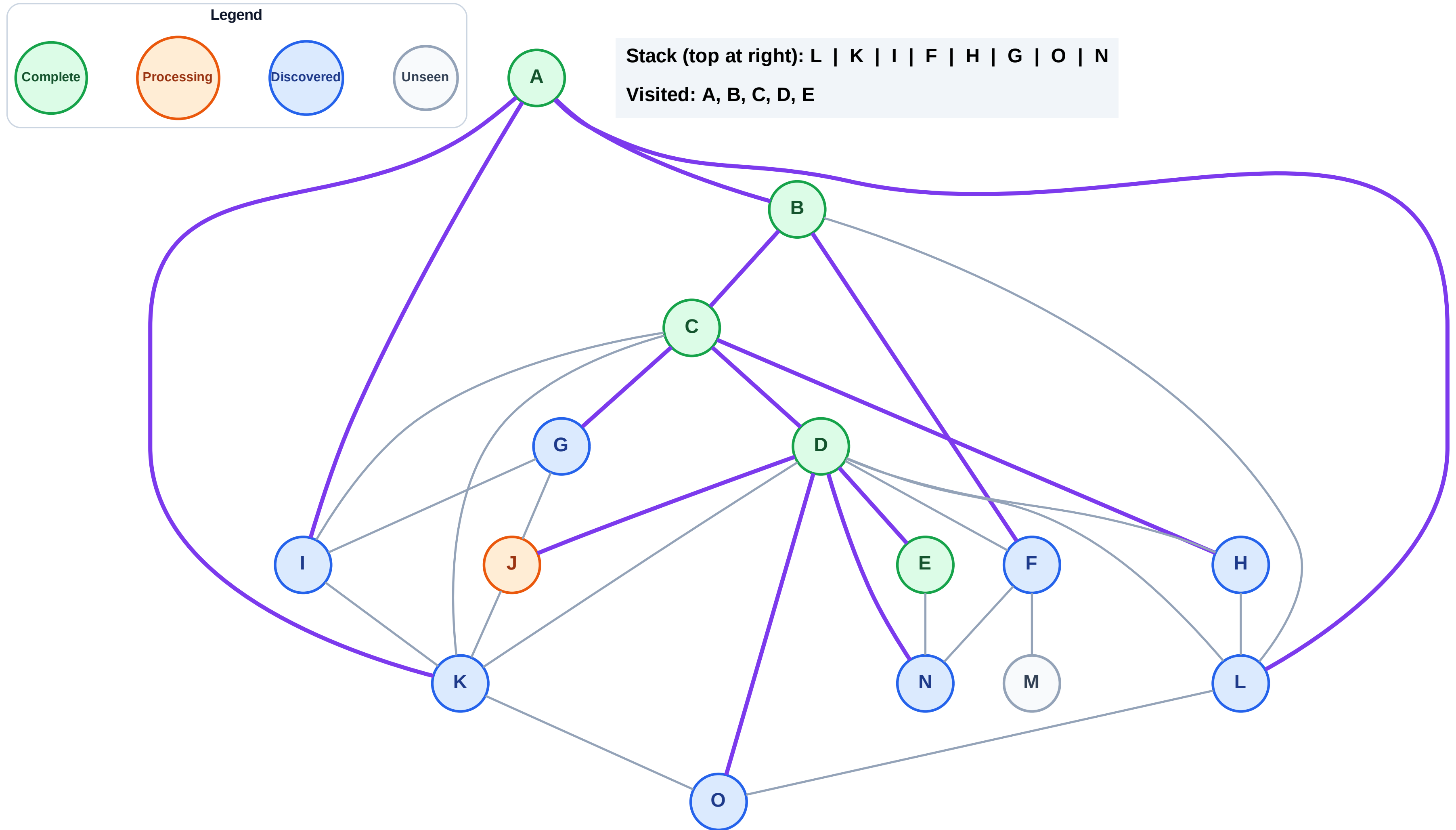


Stack (top at right): L | K | I | F | H | G | O | N | J

Visited: A, B, C, D, E



Step 12: Process node J

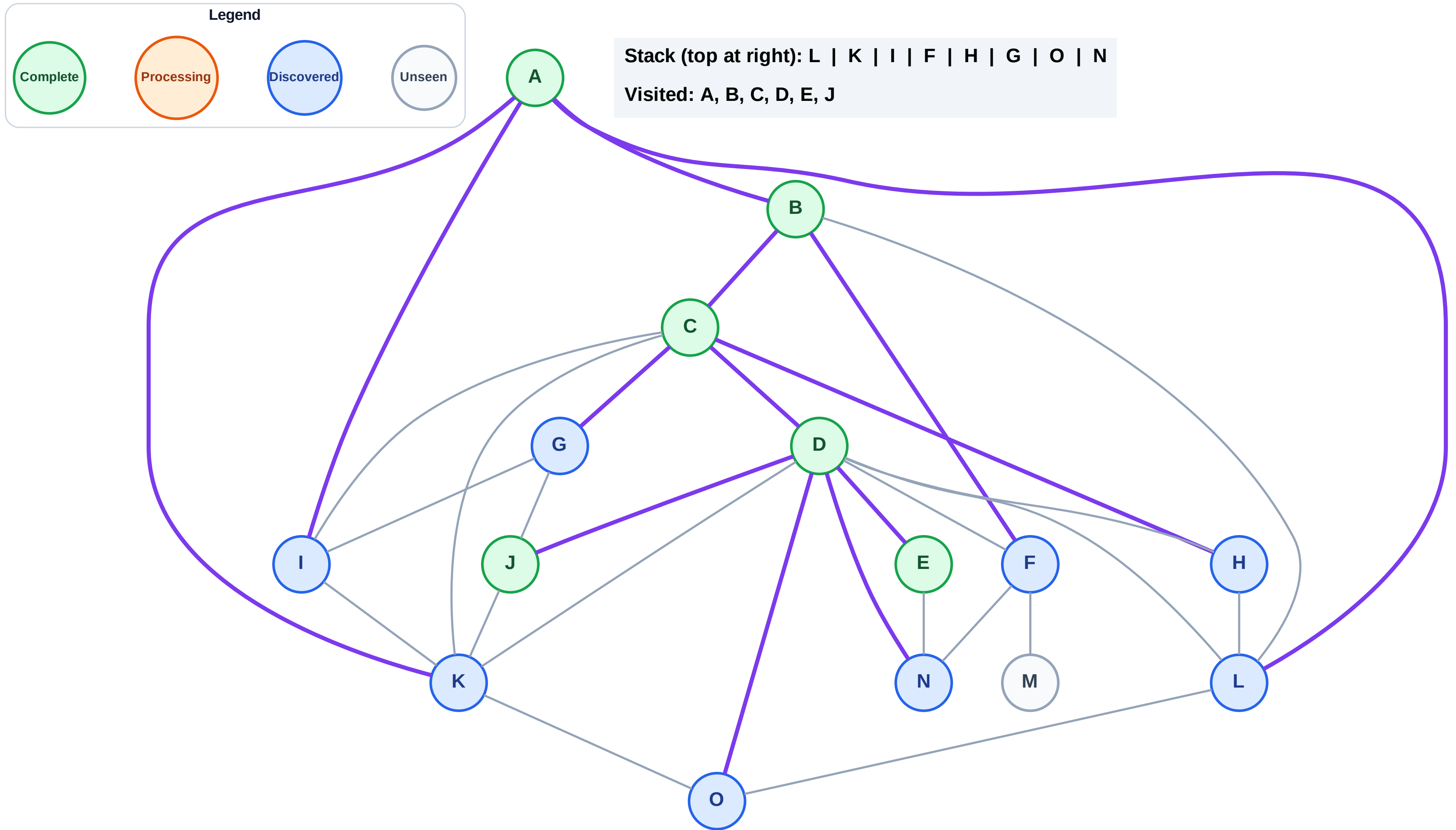


Step 13: No new neighbors from J

Stack (top at right): L | K | I | F | H | G | O | N

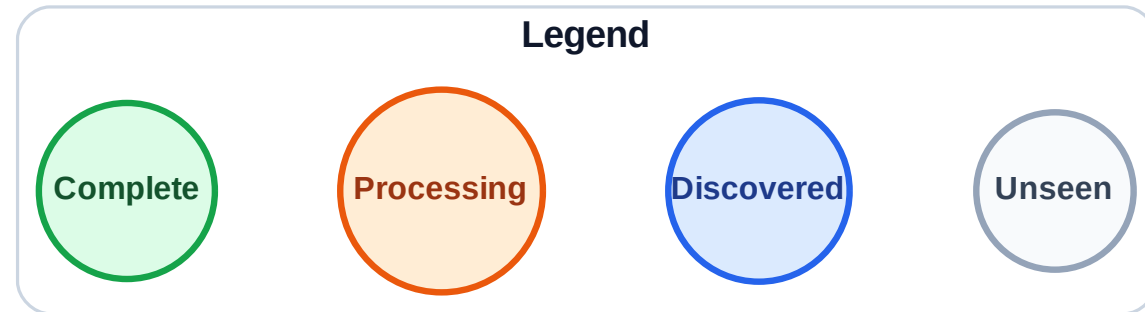
Stack (top at right): L | K | I | F | H | G | O | N

Visited: A, B, C, D, E, J



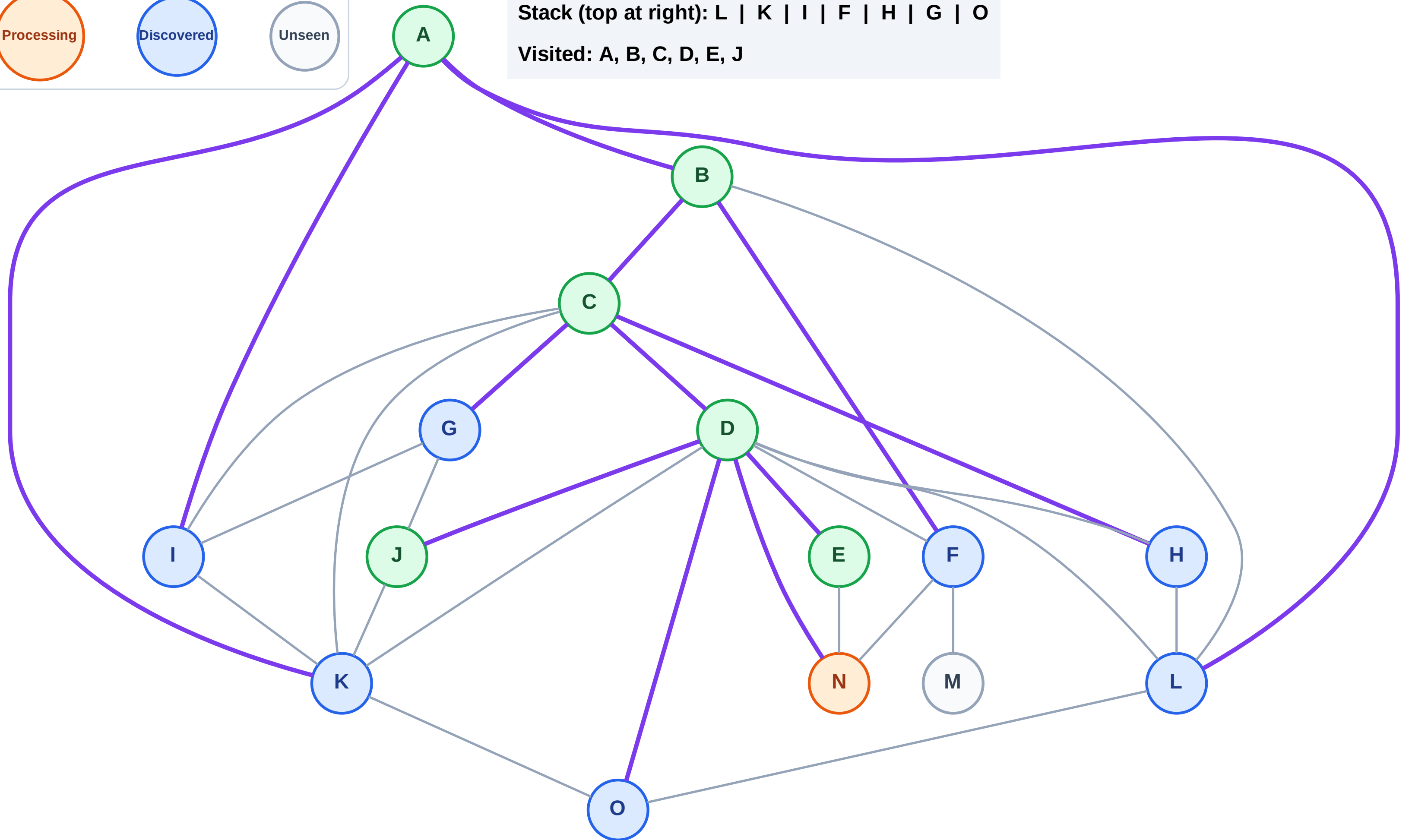
DFS Traversal

Step 14: Process node N



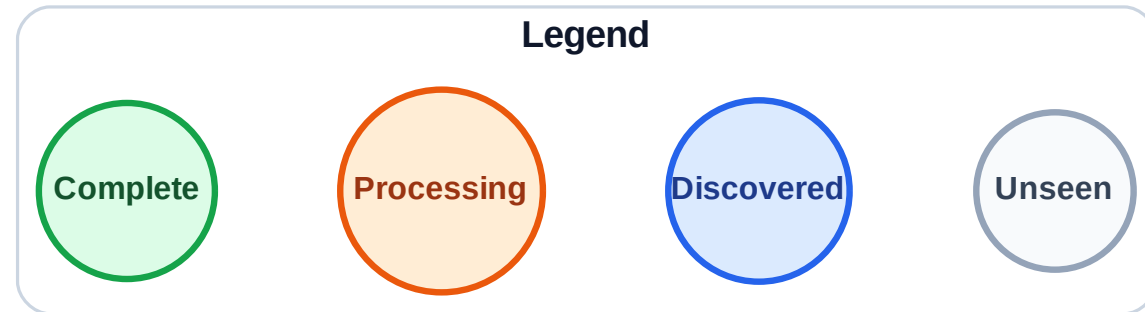
Stack (top at right): L | K | I | F | H | G | O

Visited: A, B, C, D, E, J



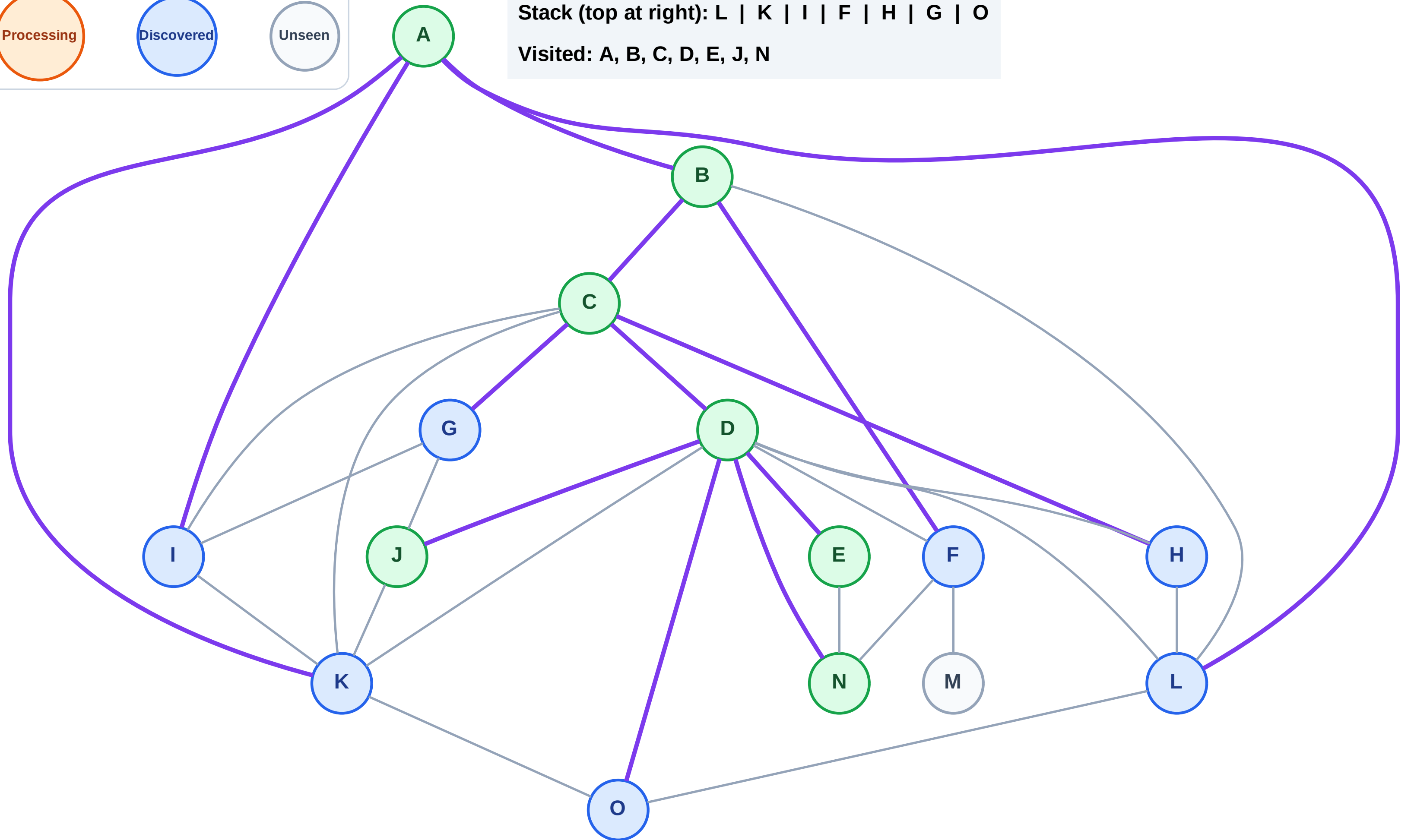
DFS Traversal

Step 15: No new neighbors from N



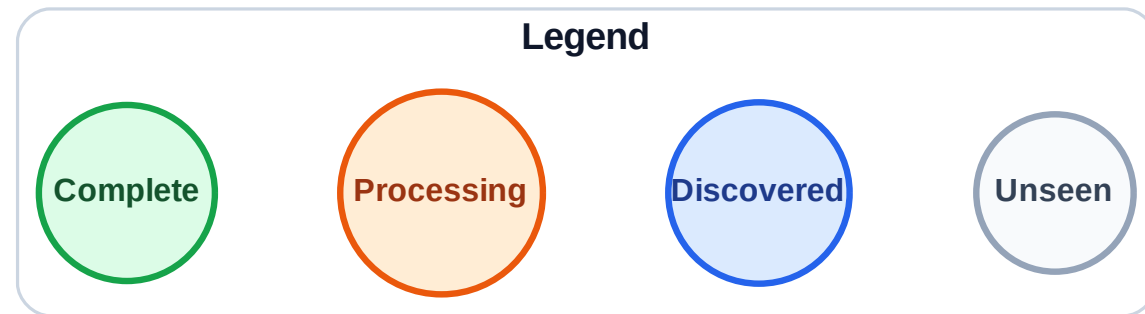
Stack (top at right): L | K | I | F | H | G | O

Visited: A, B, C, D, E, J, N



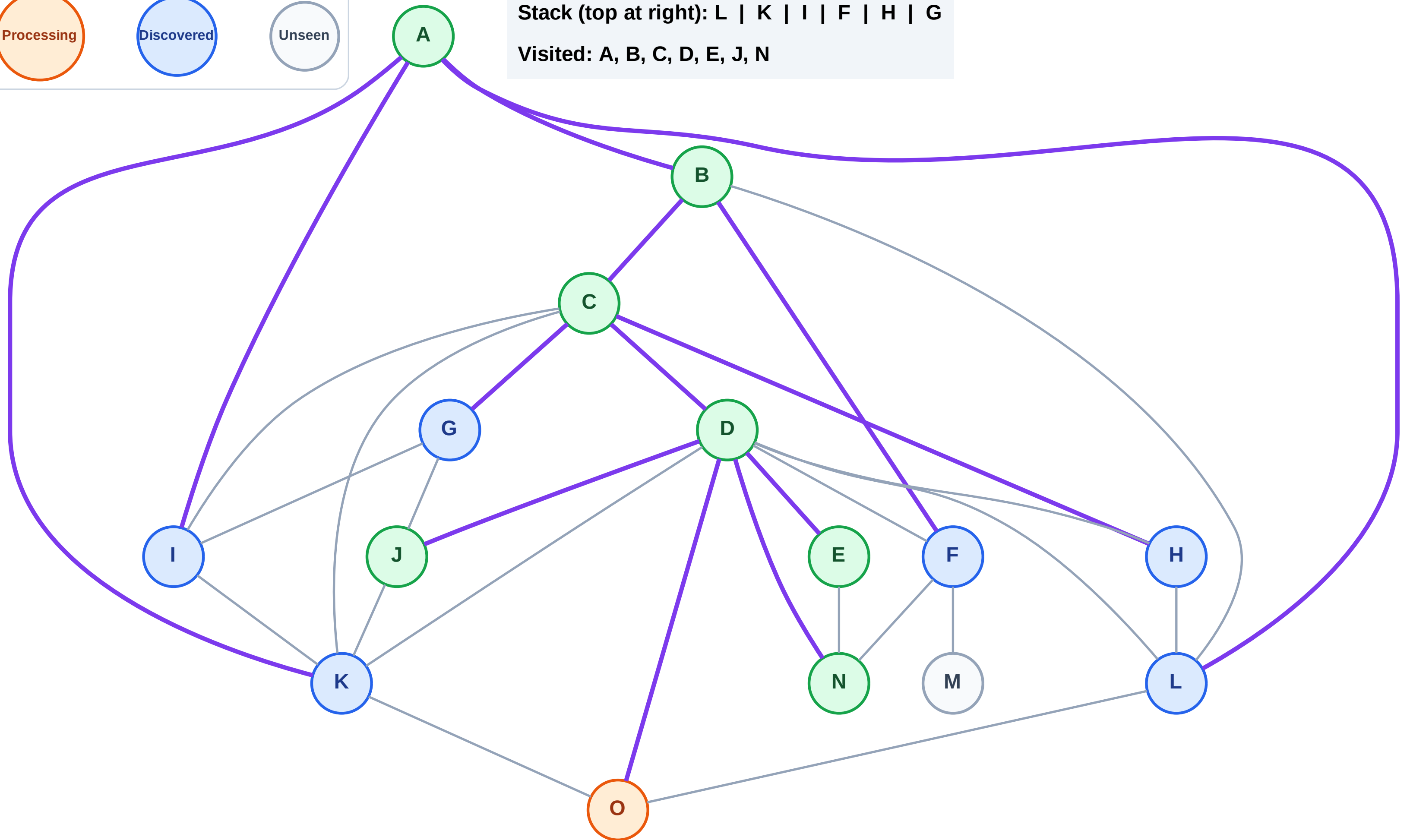
DFS Traversal

Step 16: Process node O



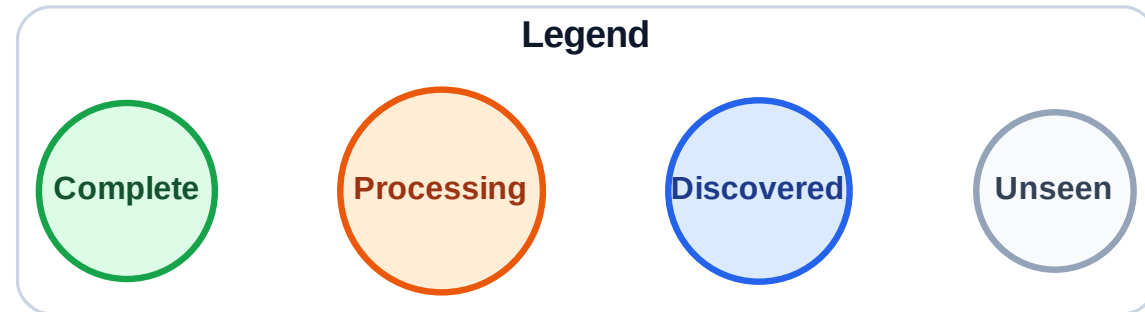
Stack (top at right): L | K | I | F | H | G

Visited: A, B, C, D, E, J, N



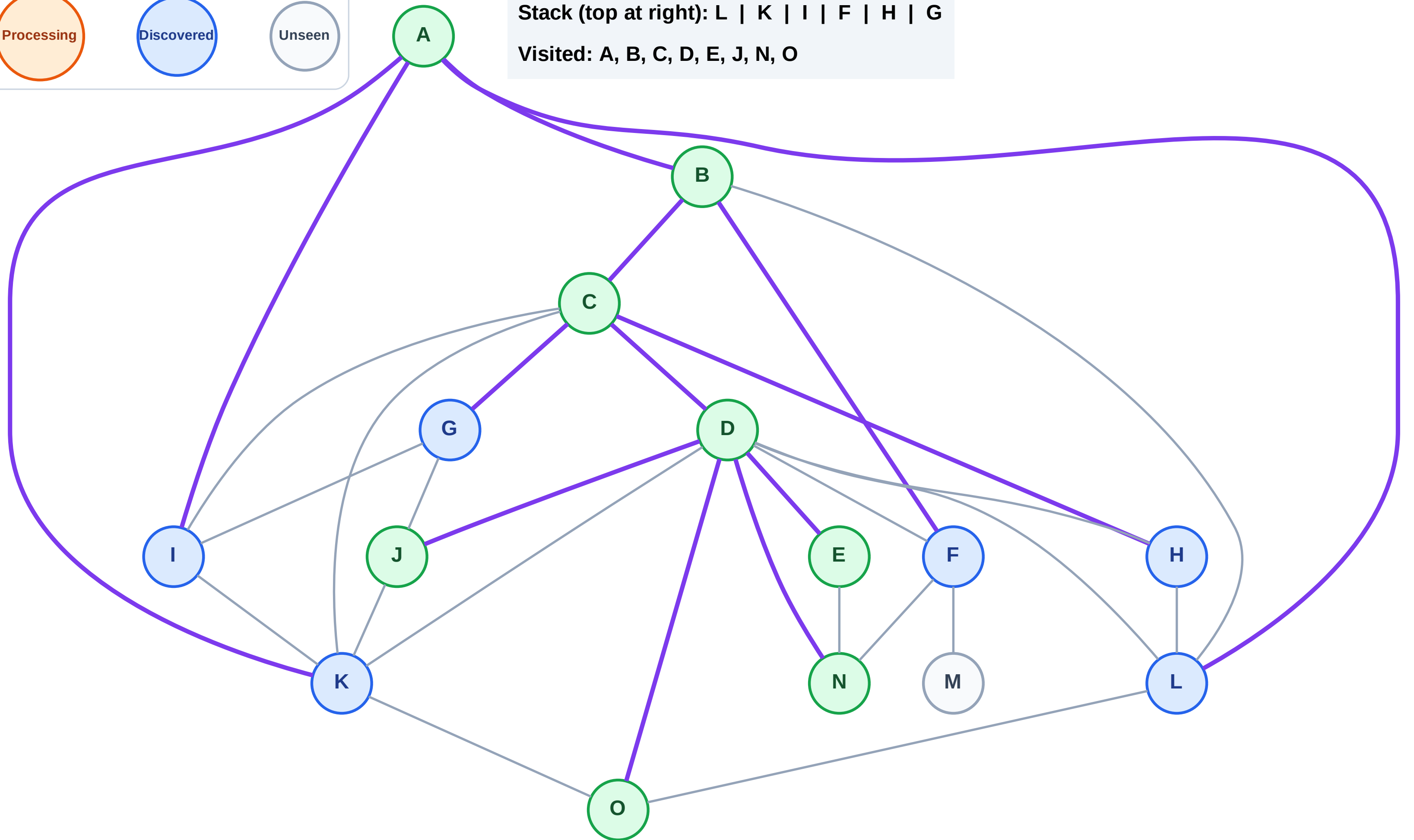
DFS Traversal

Step 17: No new neighbors from O



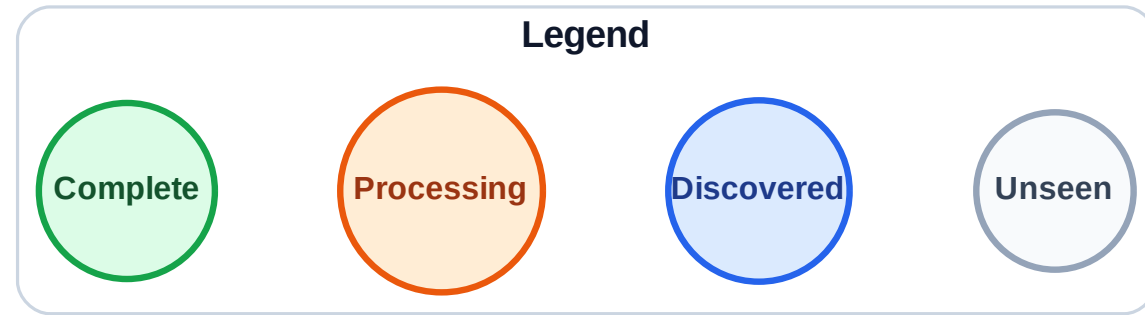
Stack (top at right): L | K | I | F | H | G

Visited: A, B, C, D, E, J, N, O



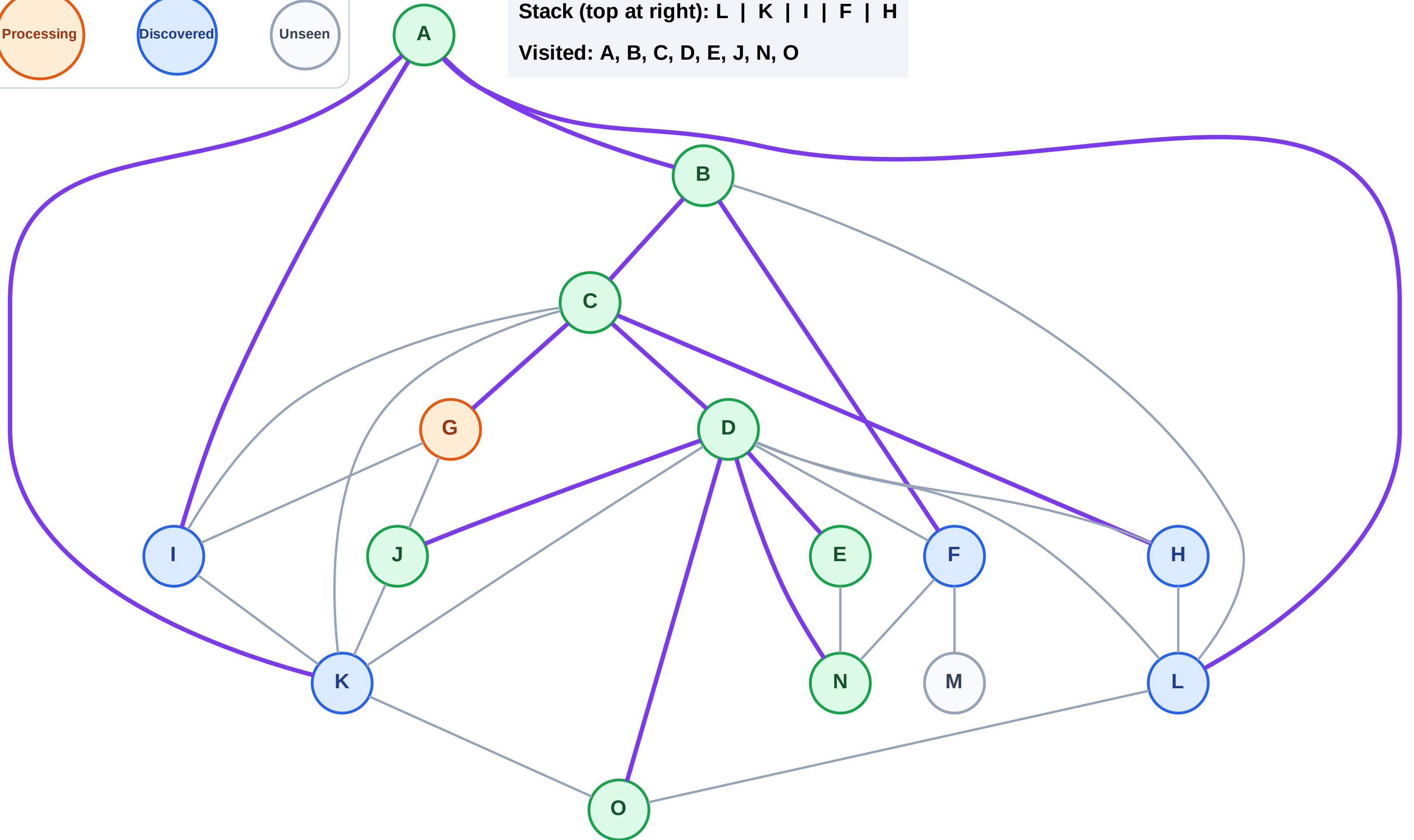
DFS Traversal

Step 18: Process node G



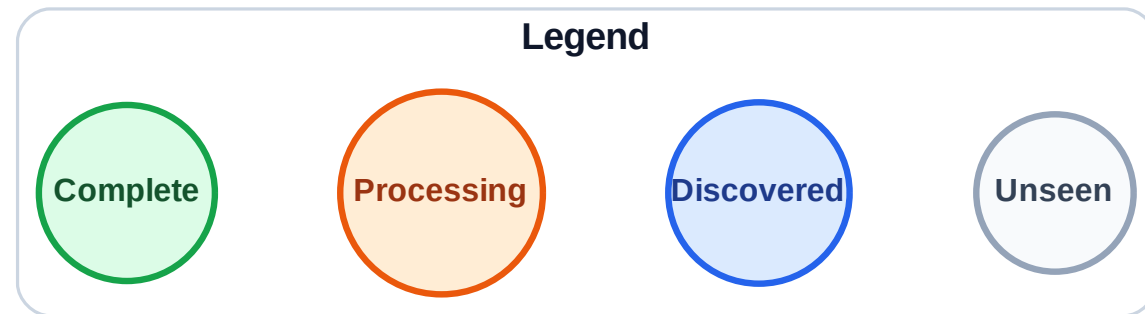
Stack (top at right): L | K | I | F | H

Visited: A, B, C, D, E, J, N, O



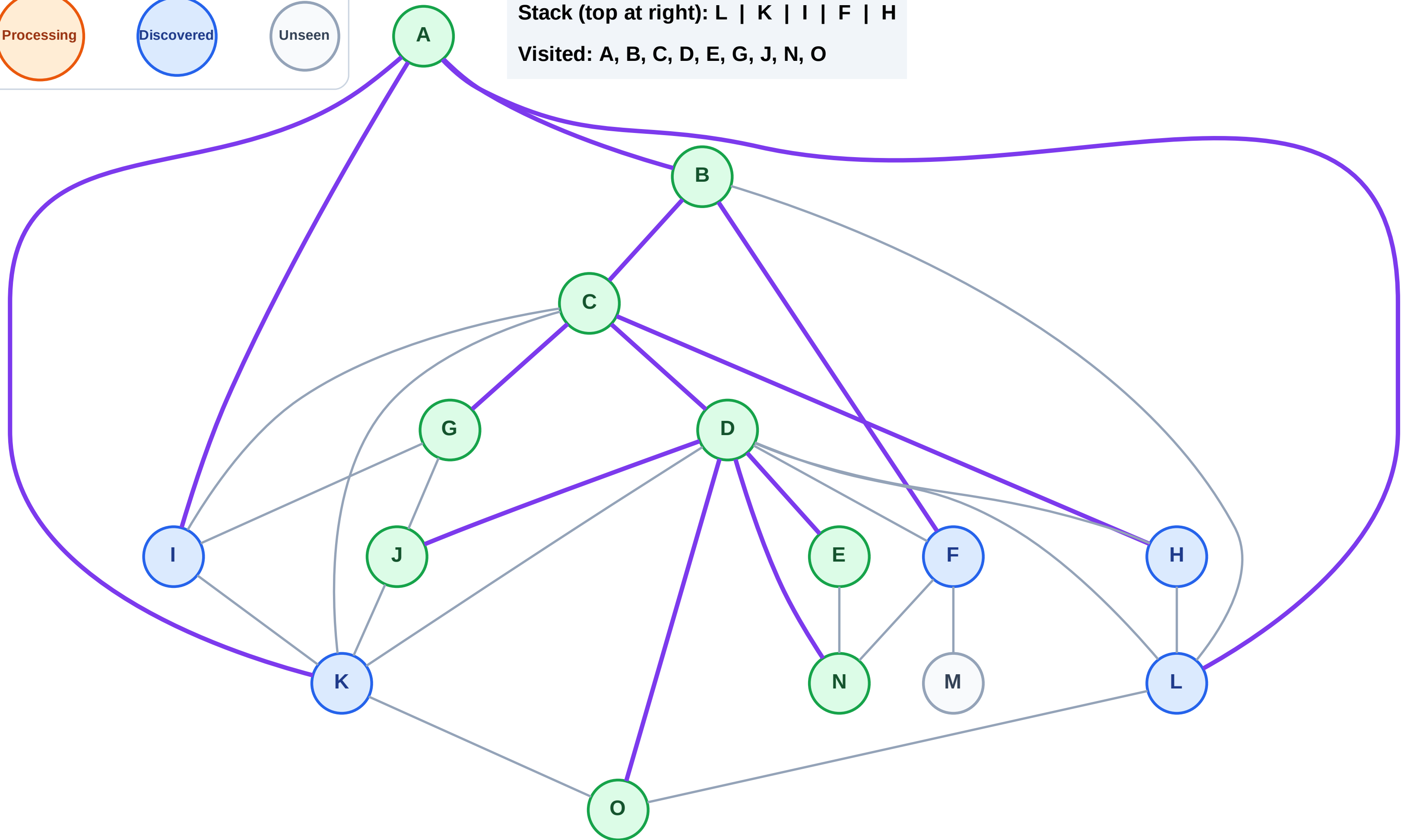
DFS Traversal

Step 19: No new neighbors from G



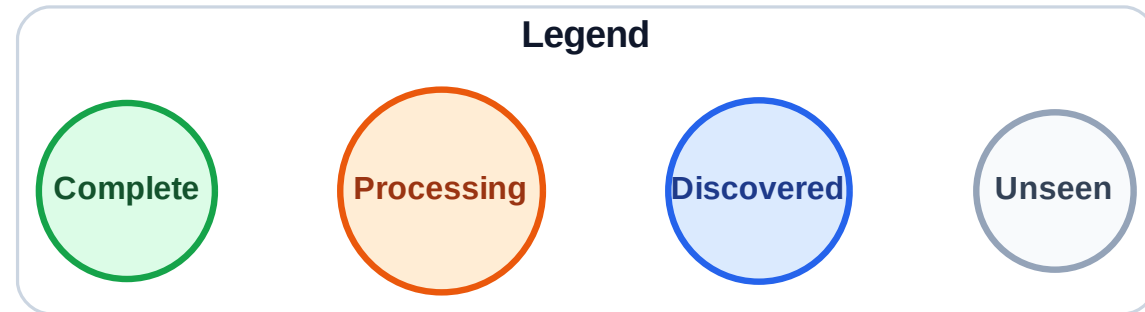
Stack (top at right): L | K | I | F | H

Visited: A, B, C, D, E, G, J, N, O



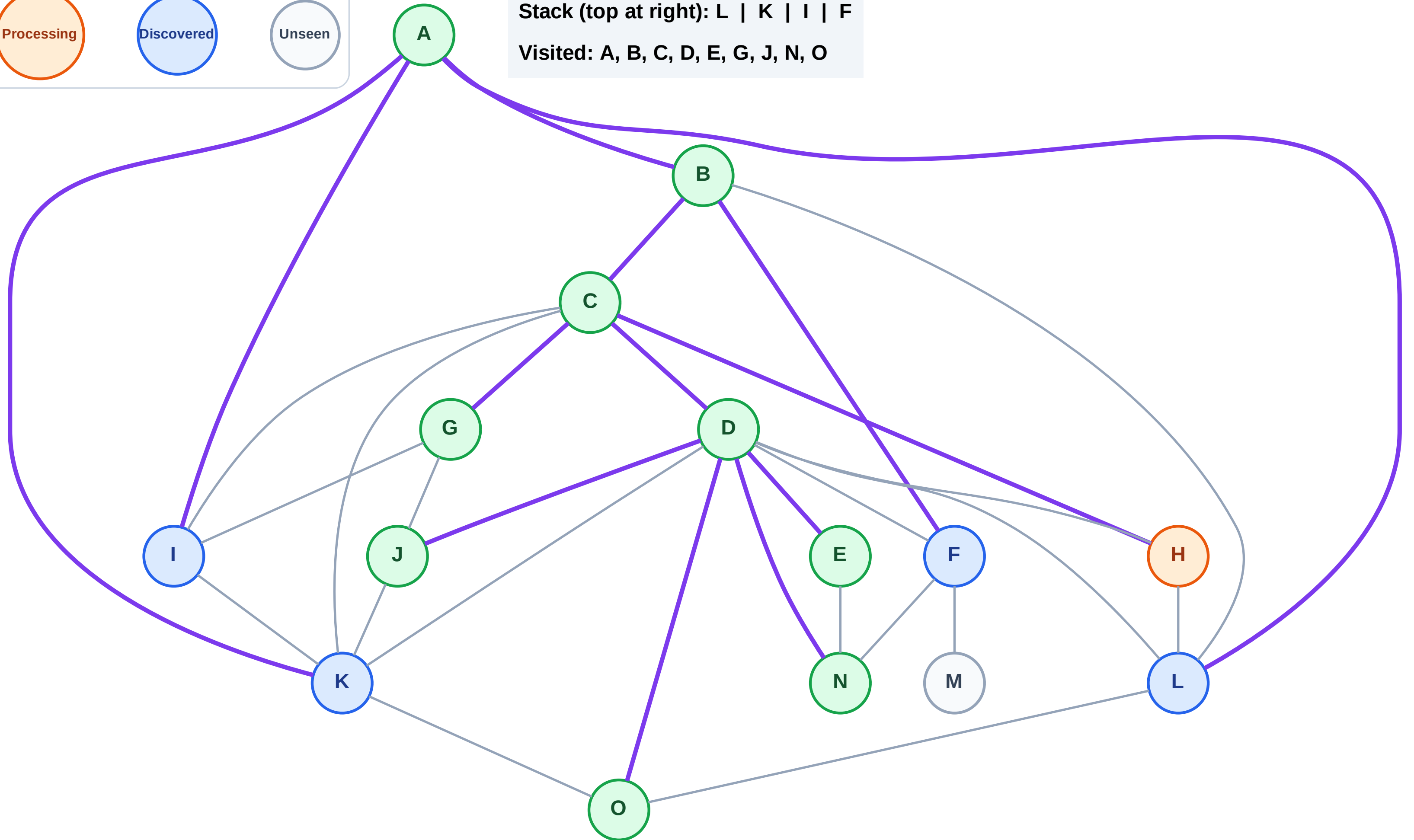
DFS Traversal

Step 20: Process node H



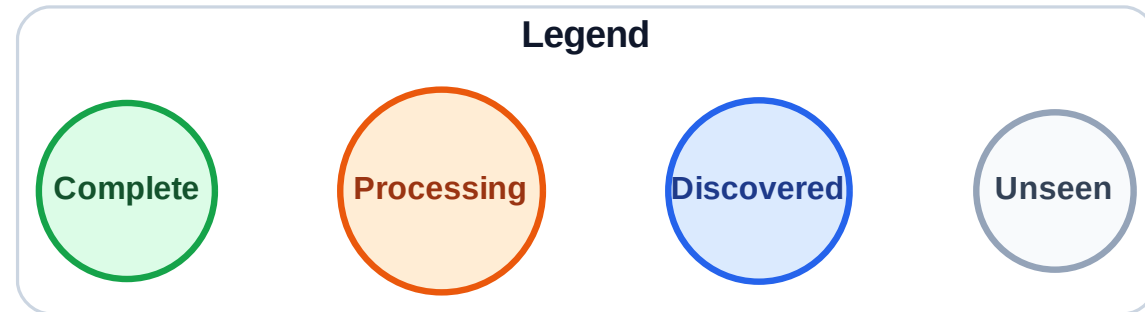
Stack (top at right): L | K | I | F

Visited: A, B, C, D, E, G, J, N, O



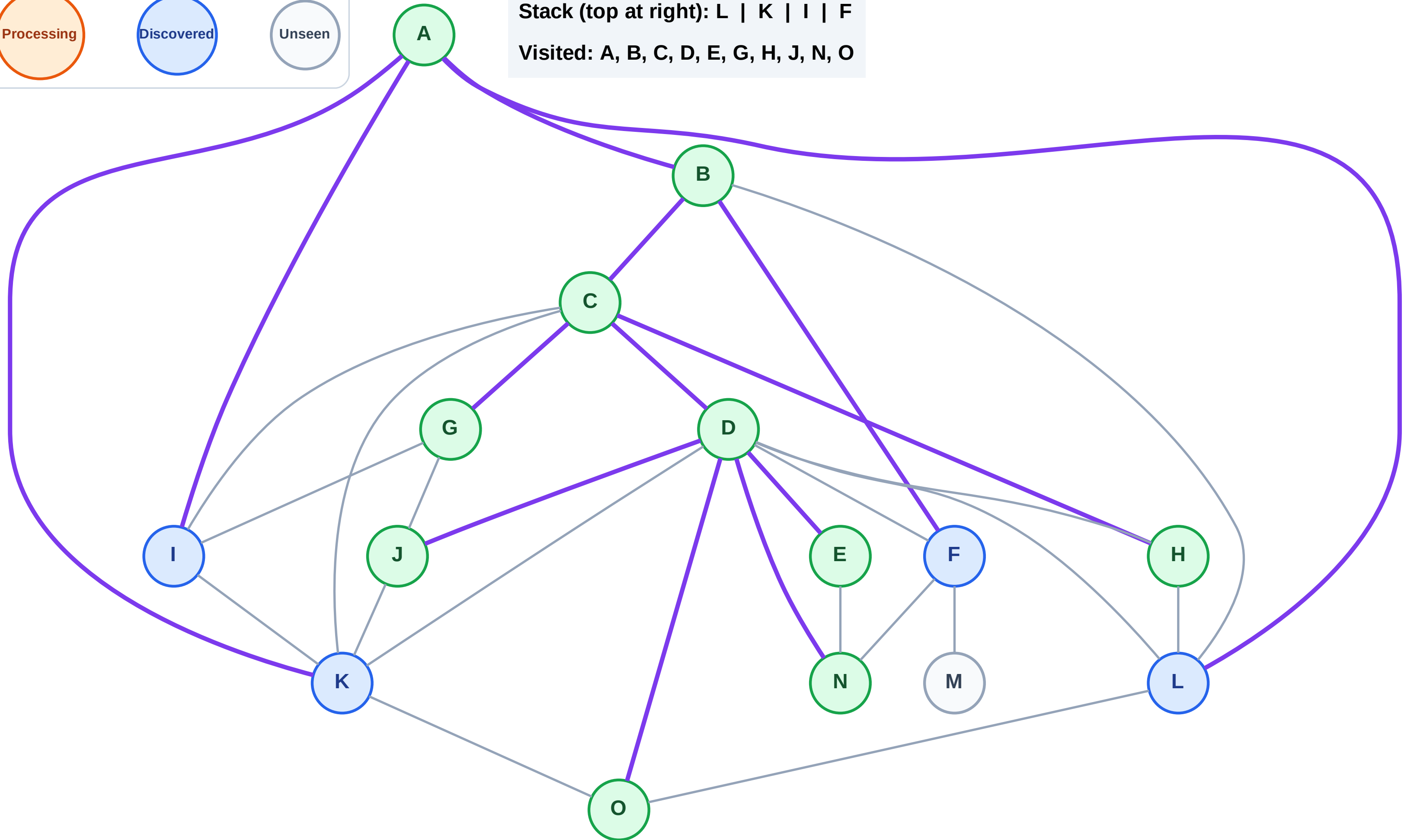
DFS Traversal

Step 21: No new neighbors from H



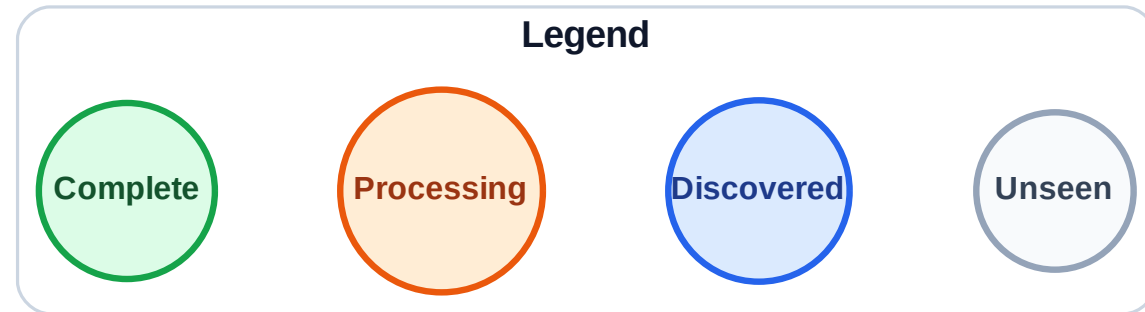
Stack (top at right): L | K | I | F

Visited: A, B, C, D, E, G, H, J, N, O



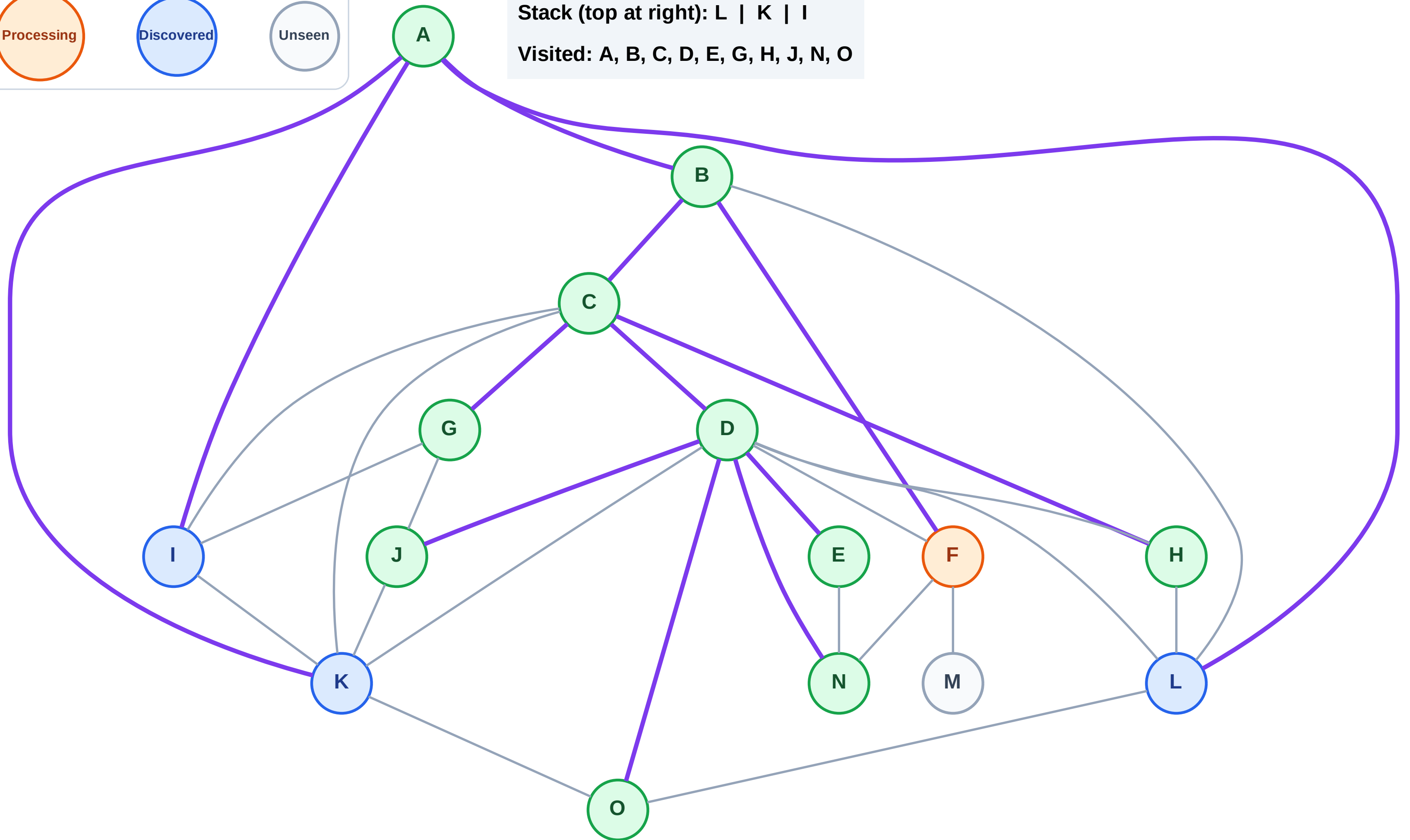
DFS Traversal

Step 22: Process node F



Stack (top at right): L | K | I

Visited: A, B, C, D, E, G, H, J, N, O



DFS Traversal

Step 23: Discover M from F

Stack (top at right): L | K | I | M
Visited: A, B, C, D, E, F, G, H, J, N, O

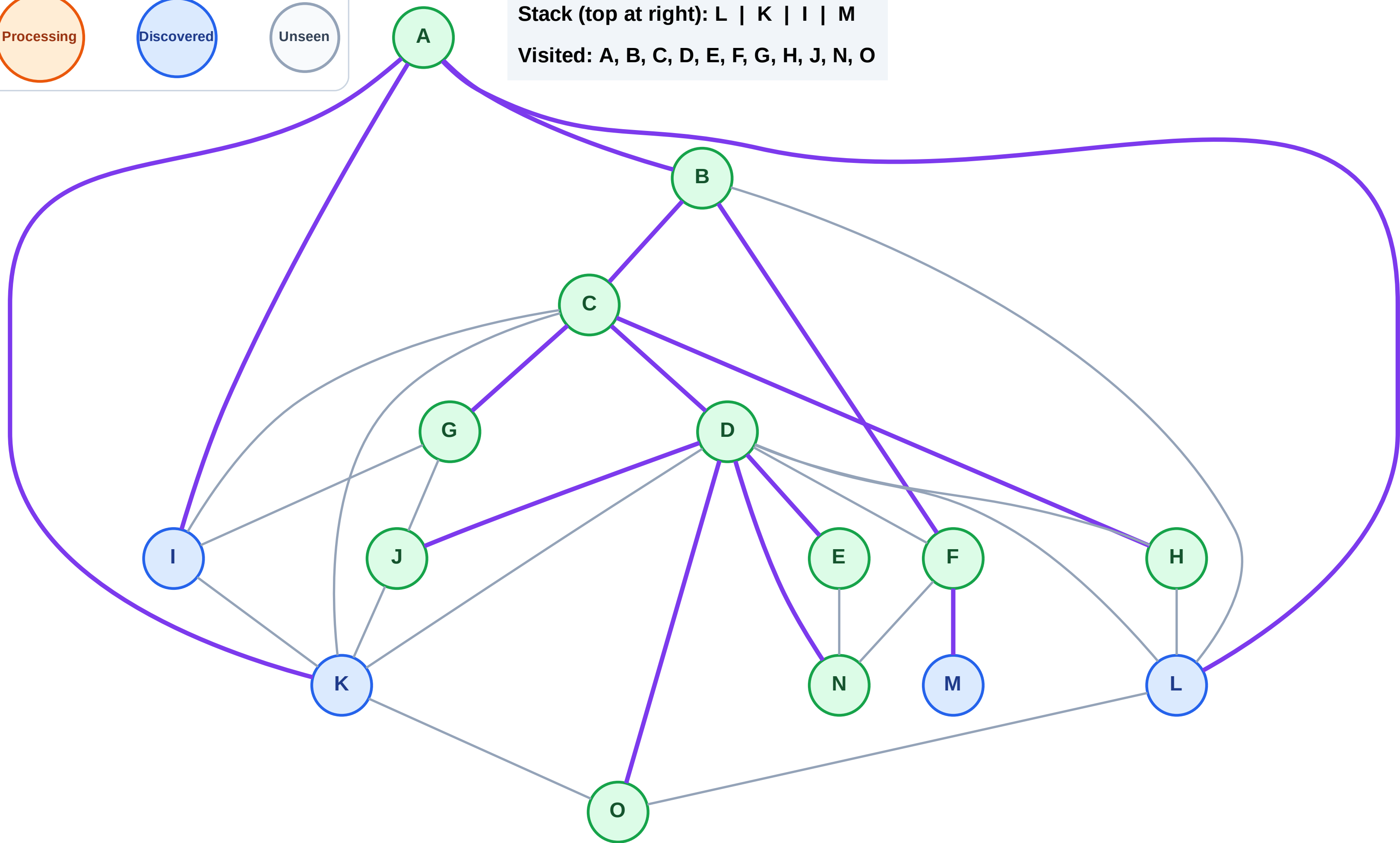
Legend

Complete

Processing

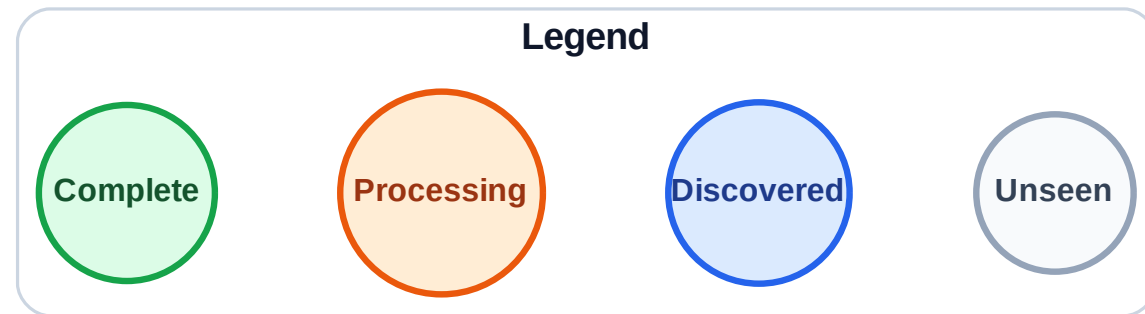
Discovered

Unseen



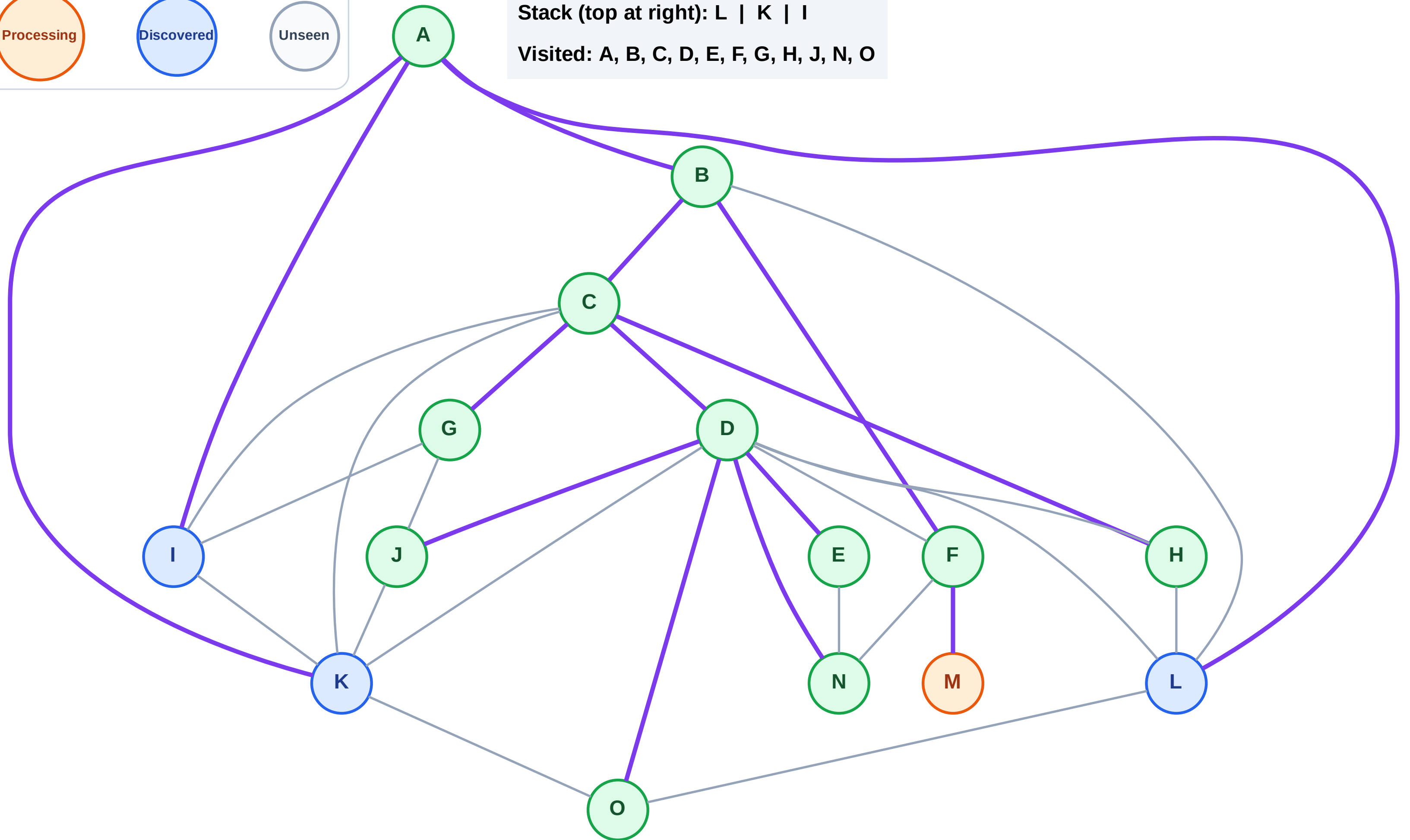
DFS Traversal

Step 24: Process node M

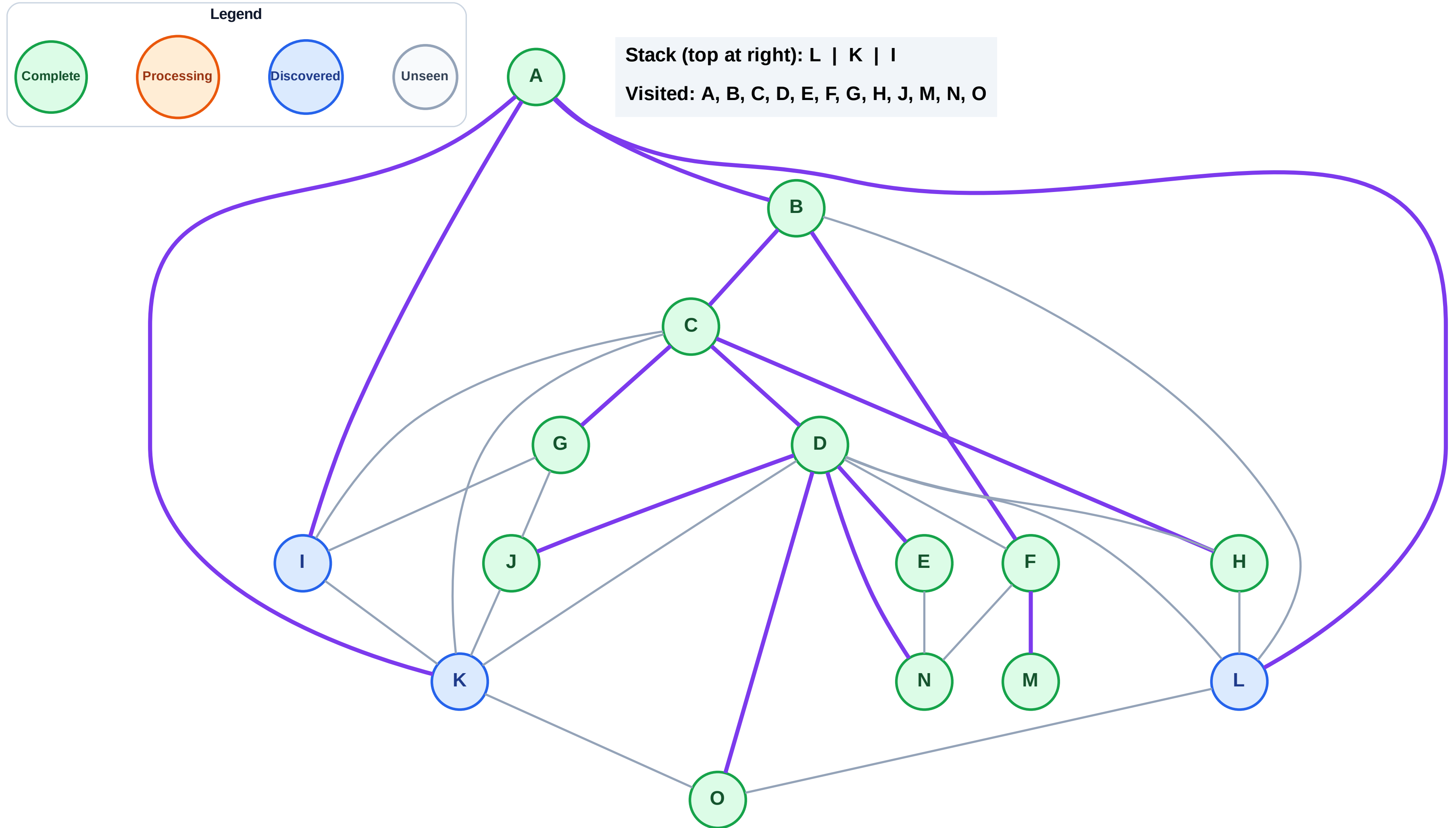


Stack (top at right): L | K | I

Visited: A, B, C, D, E, F, G, H, J, N, O



Step 25: No new neighbors from M



DFS Traversal

Step 26: Process node I

Legend

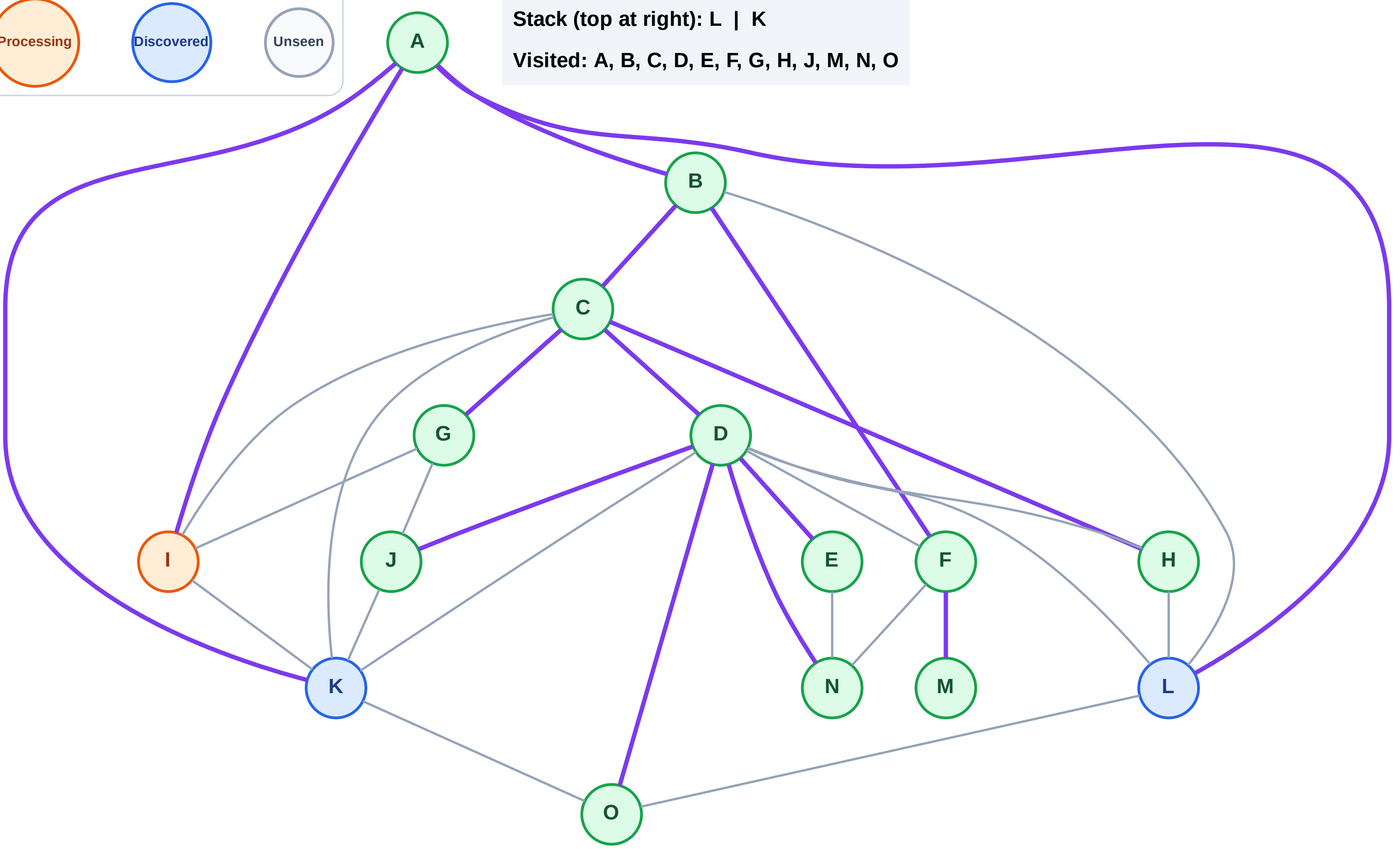
Complete

Processing

Discovered

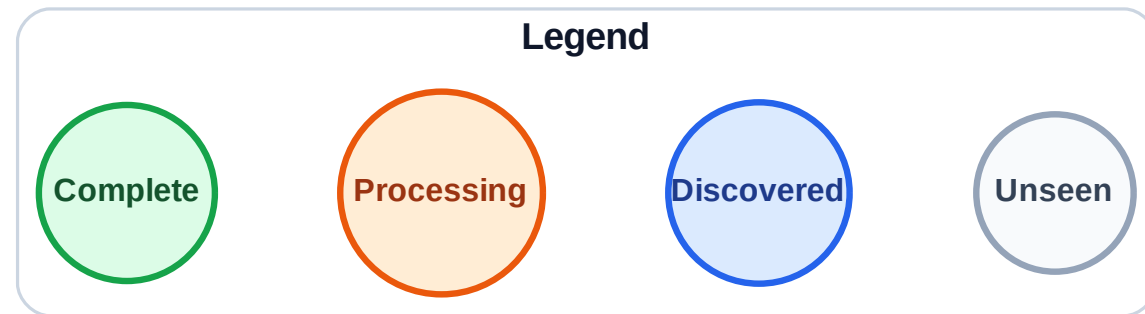
Unseen

Stack (top at right): L | K
Visited: A, B, C, D, E, F, G, H, J, M, N, O



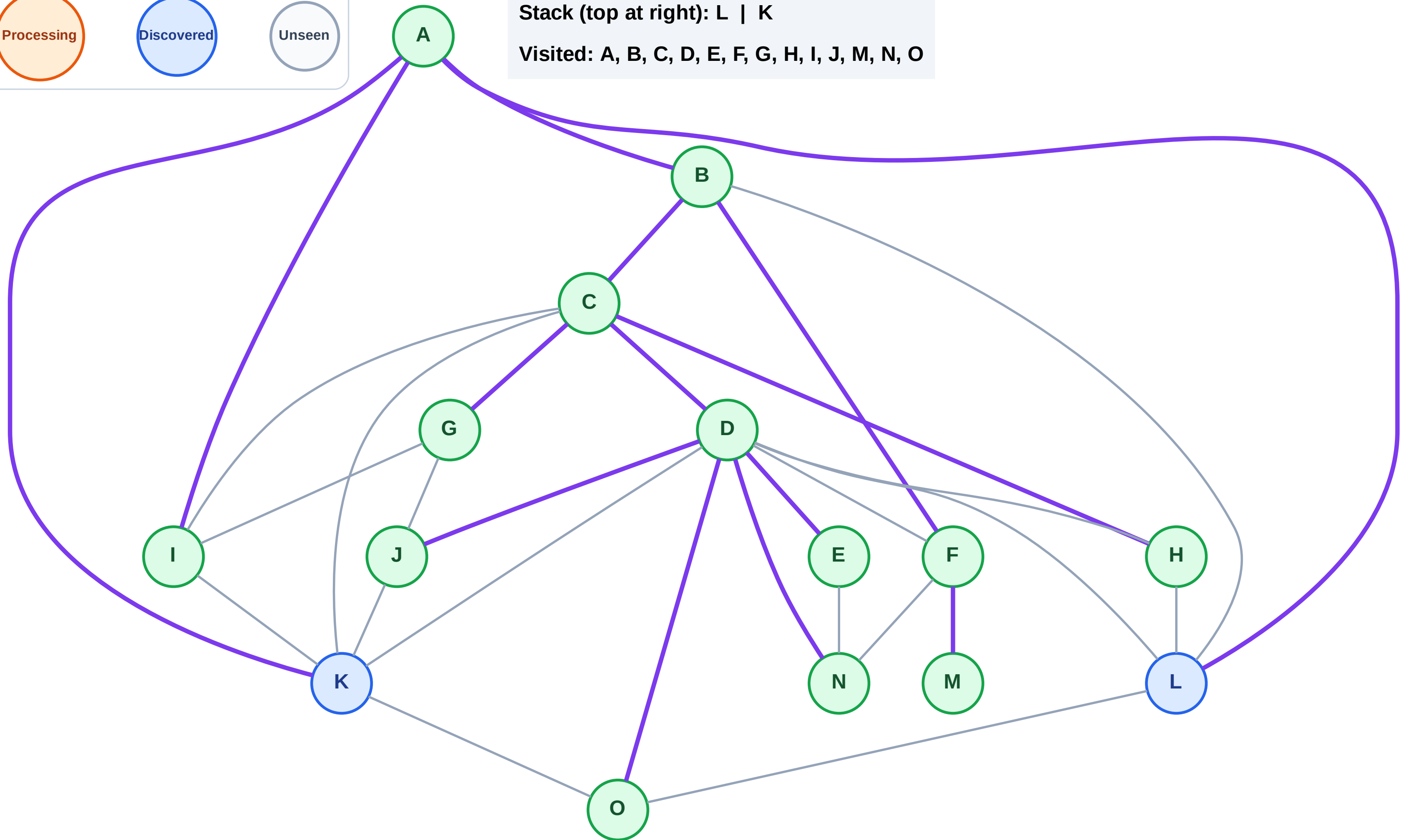
DFS Traversal

Step 27: No new neighbors from I



Stack (top at right): L | K

Visited: A, B, C, D, E, F, G, H, I, J, M, N, O



DFS Traversal

Step 28: Process node K

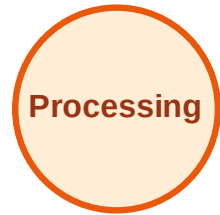
Stack (top at right): L

Visited: A, B, C, D, E, F, G, H, I, J, M, N, O

Legend



Complete



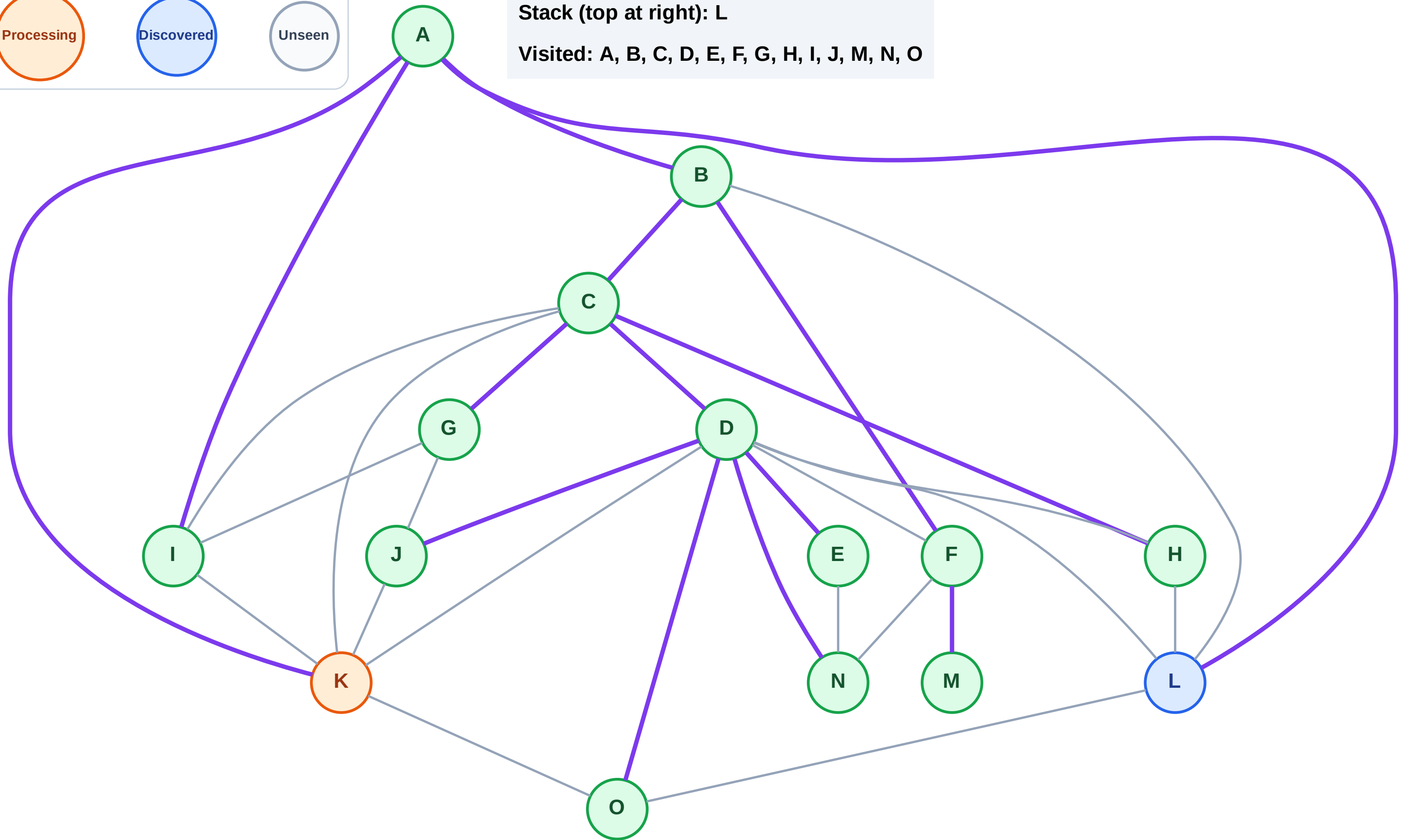
Processing



Discovered

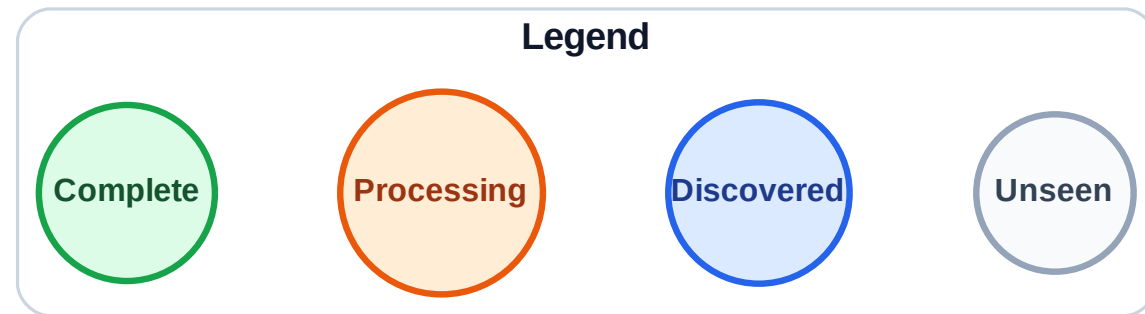


Unseen



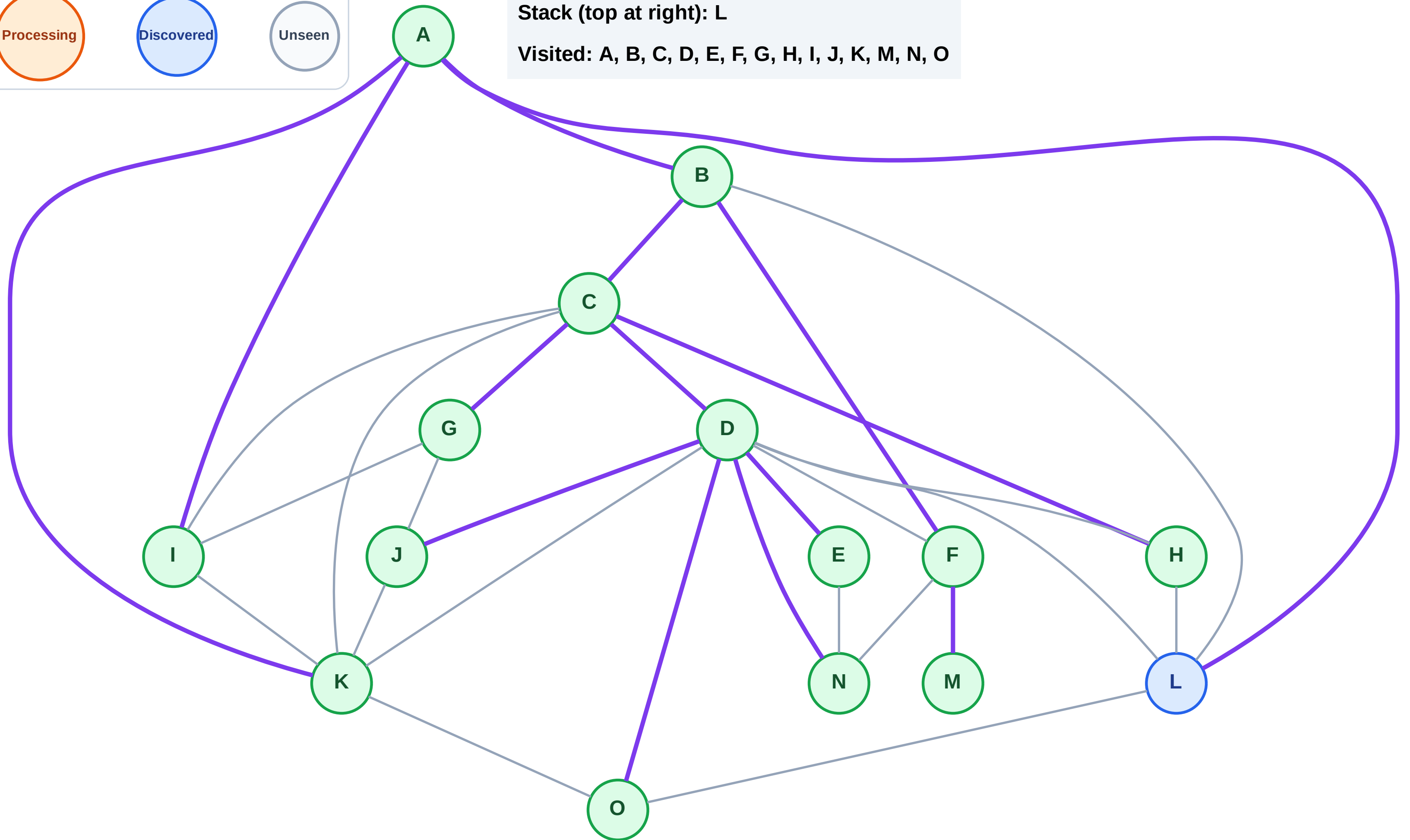
DFS Traversal

Step 29: No new neighbors from K



Stack (top at right): L

Visited: A, B, C, D, E, F, G, H, I, J, K, M, N, O



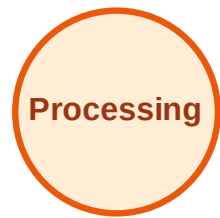
DFS Traversal

Step 30: Process node L

Legend



Complete



Processing



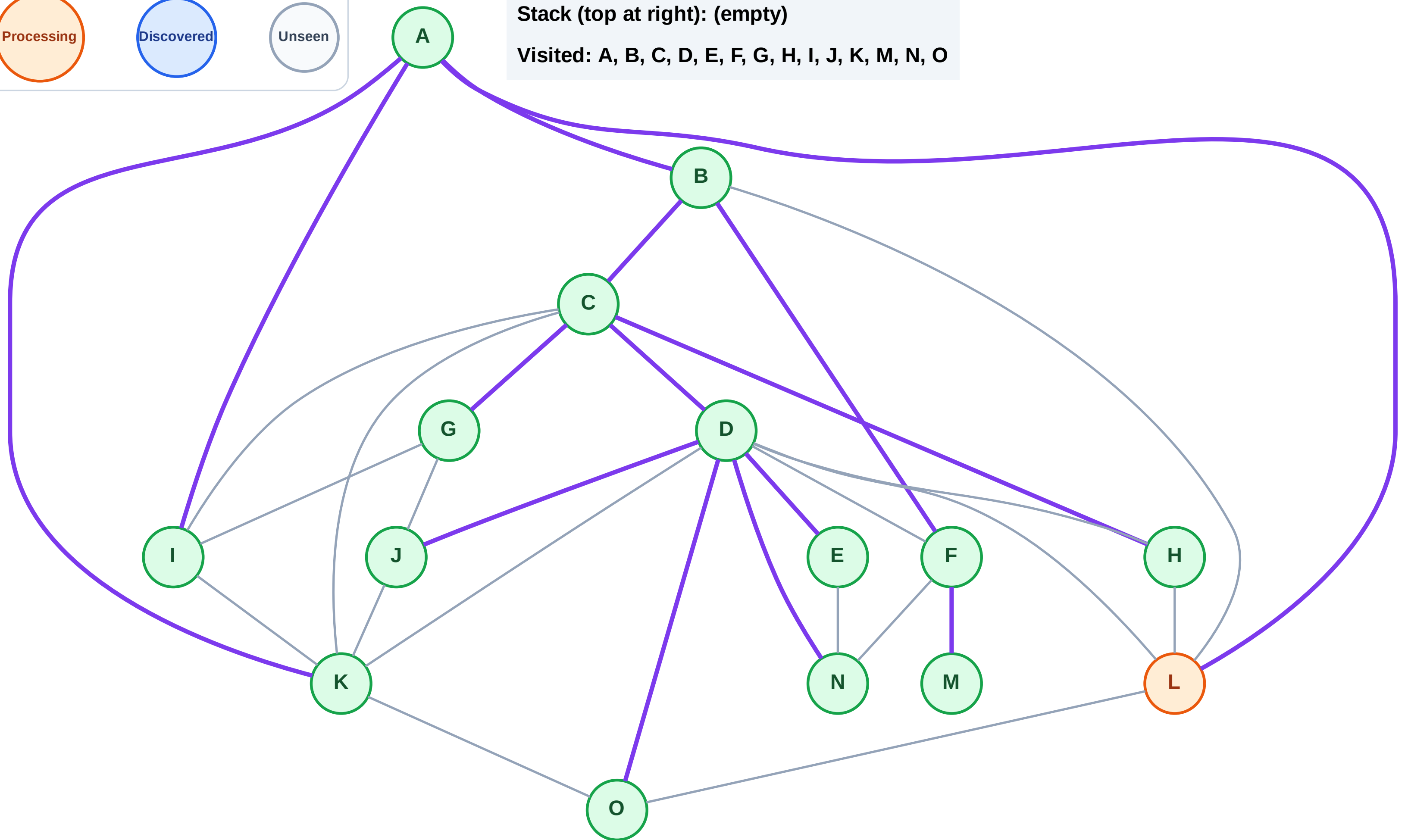
Discovered



Unseen

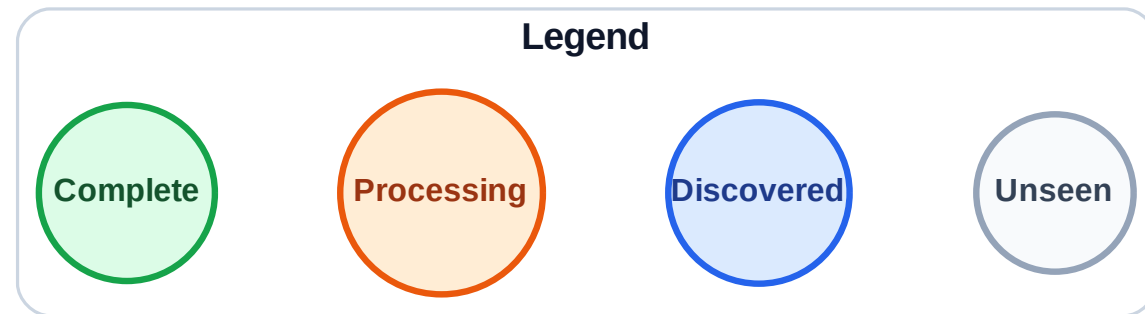
Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J, K, M, N, O



DFS Traversal

Step 31: No new neighbors from L



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N, O

